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Gocho et al.

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(54) **SEMICONDUCTOR DEVICE INCLUDING A FIELD EFFECT TRANSISTOR WITH NANOSTRUCTURE BASED LAMINATED CHANNEL STRUCTURE AND A FIELD EFFECT TRANSISTOR WITH A SINGLE CHANNEL STRUCTURE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/618,329**

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(65) **Prior Publication Data**

US 2024/0243173 A1 Jul. 18, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/943,337, filed on Sep. 13, 2022, now Pat. No. 11,961,885, which is a (Continued)

(30) **Foreign Application Priority Data**

Jul. 26, 2018 (JP) 2018-140217

(51) **Int. Cl.**

H10D 62/10 (2025.01)

H10D 30/43 (2025.01)

(Continued)

(52) **U.S. Cl.**

CPC **H10D 62/121** (2025.01); **H10D 30/43** (2025.01); **H10D 62/235** (2025.01); **H10D 84/83** (2025.01)

(58) **Field of Classification Search**

CPC H01L 29/0673; H01L 29/78696; H01L 29/66545; H01L 27/0922; H01L 29/42392;

(Continued)

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Primary Examiner — Changhyun Yi

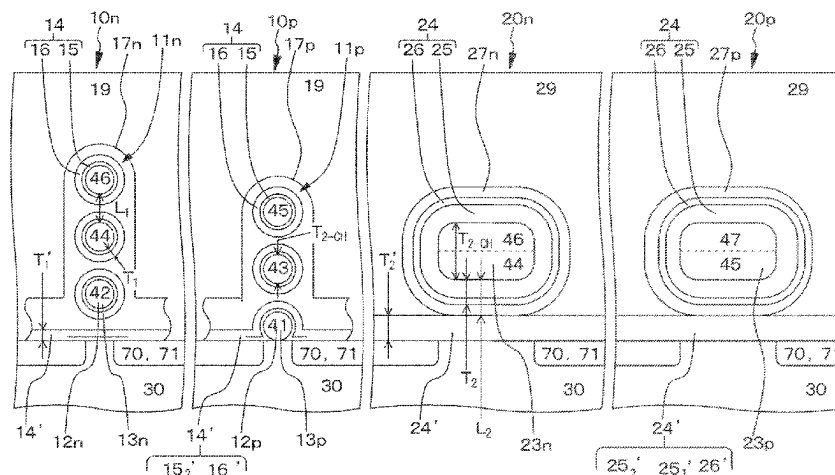
(74) *Attorney, Agent, or Firm* — Sheridan Ross PC

(57)

ABSTRACT

A semiconductor device includes a base, a first FET that includes at least two channel structure portions laminated, the channel structure portions each including a channel portion having a nanowire structure, a gate insulation film, and a gate electrode, and a second FET that includes a channel forming layer, a gate insulation layer, and a gate electrode. The first FET and the second FET are provided above the base. The channel portions of the first FET are disposed apart from each other in a laminating direction of

(Continued)



the channel structure portions. Assuming that each of a distance between the channel portions of the first FET is a distance L1 and that a thickness of the gate insulation layer of the second FET is a thickness T2, $T2 \geq (L1/2)$ is satisfied.

15 Claims, 50 Drawing Sheets

Related U.S. Application Data

continuation of application No. 17/261,226, filed as application No. PCT/JP2019/024075 on Jun. 18, 2019, now Pat. No. 11,476,329.

(51) **Int. Cl.**

H10D 62/17 (2025.01)

H10D 84/83 (2025.01)

(58) **Field of Classification Search**

CPC H01L 2029/7858; H01L 27/088; H01L 29/775; H01L 21/823462; H01L 29/1033; H01L 21/823857; H01L 29/66439; H10D 62/121; H10D 30/6735; H10D 62/235; H10D 84/83

See application file for complete search history.

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FIG. 1

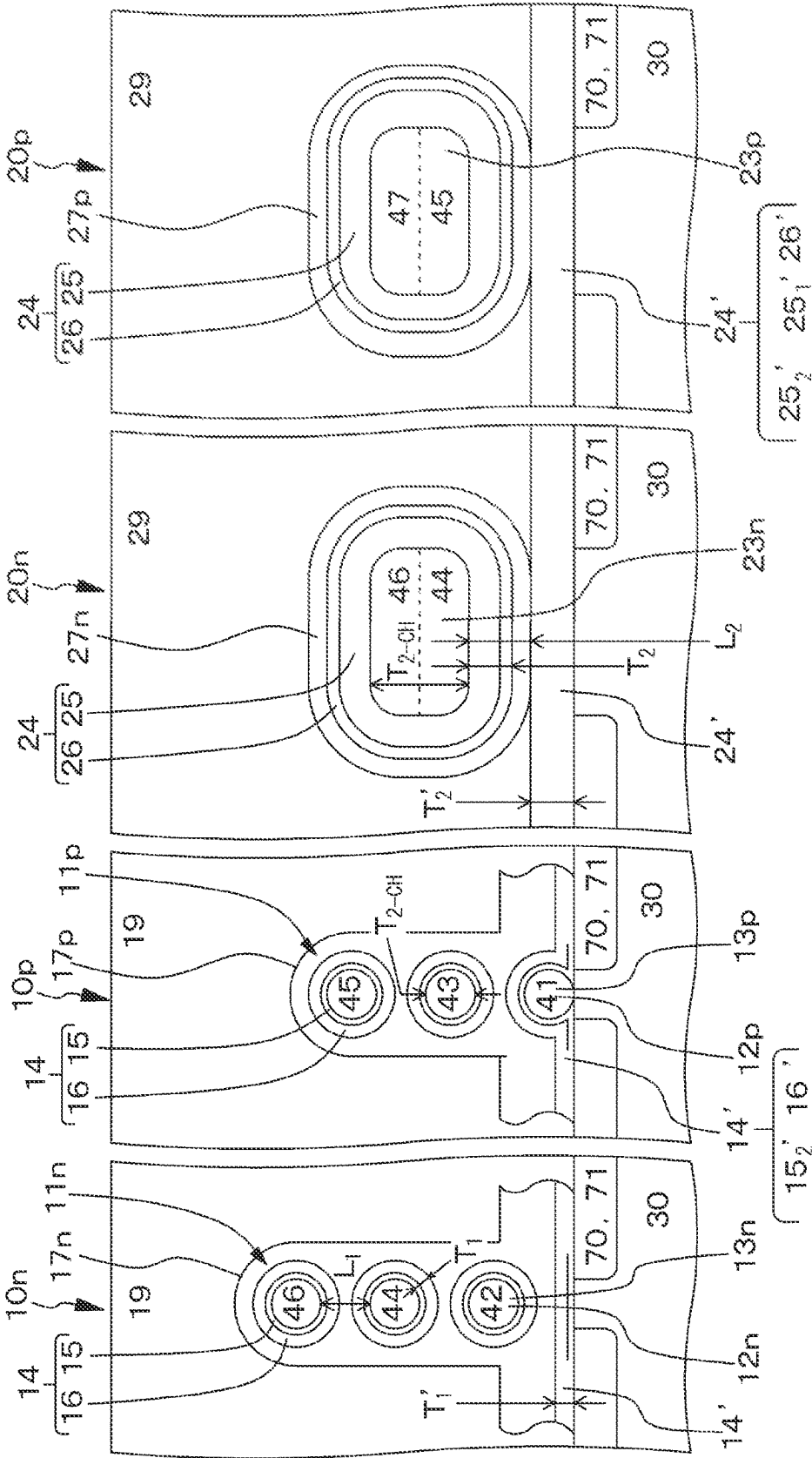


FIG. 2A

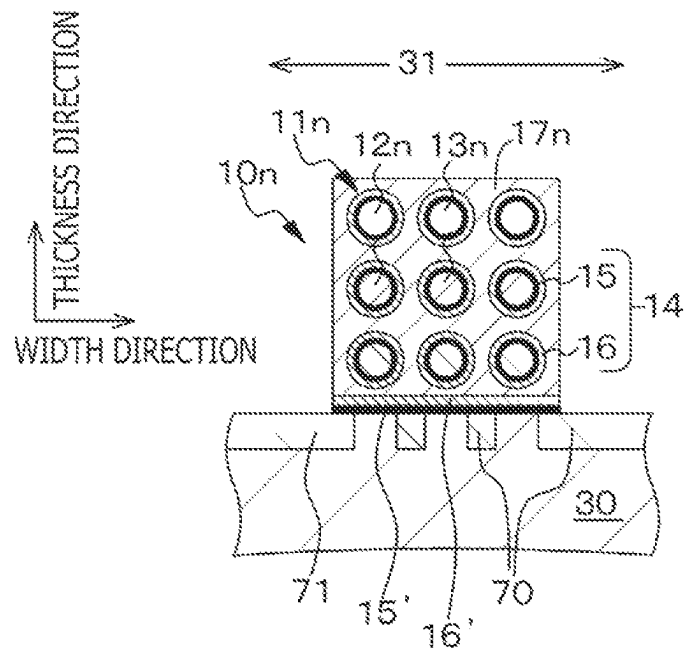
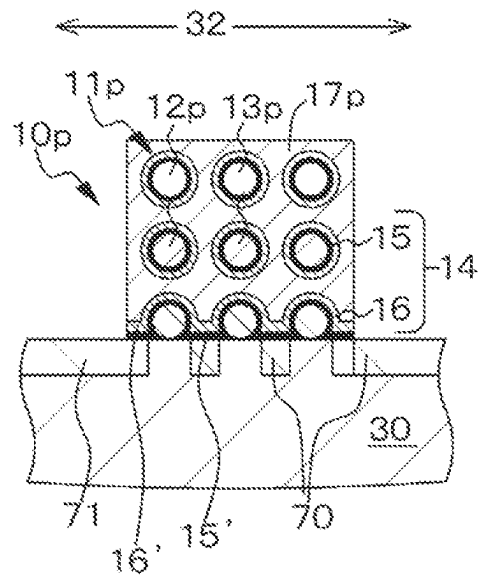


FIG. 2B



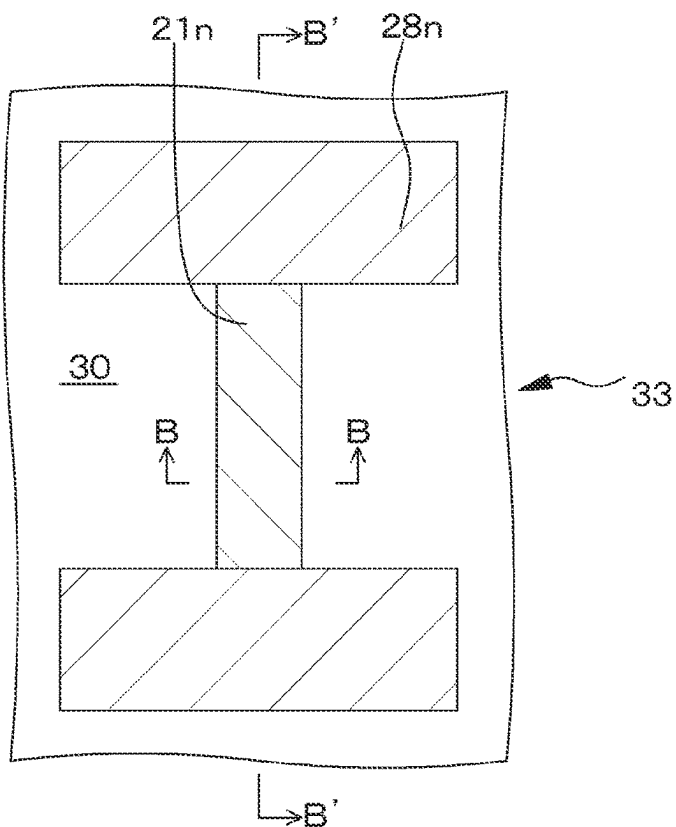


FIG. 4

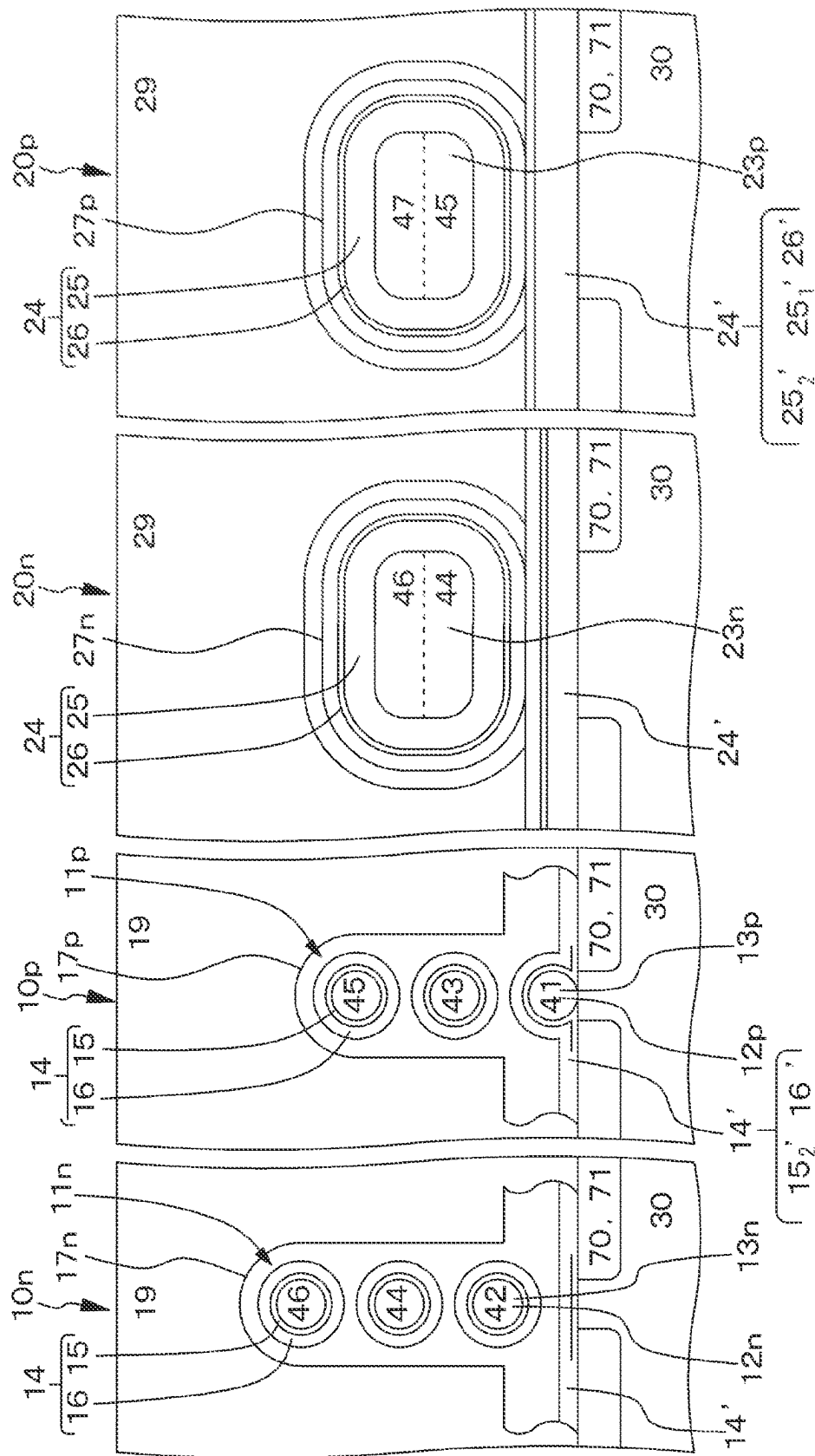
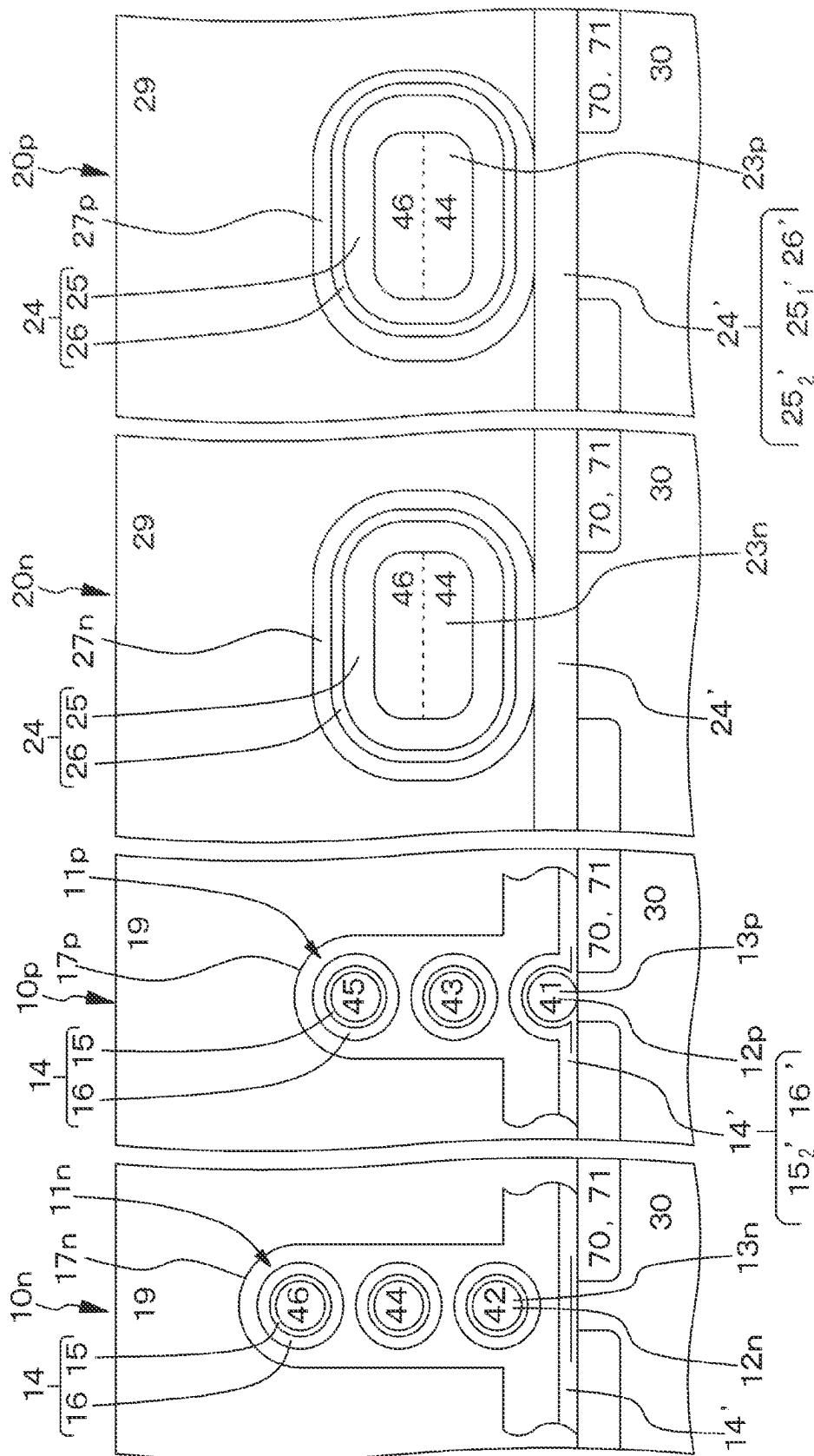
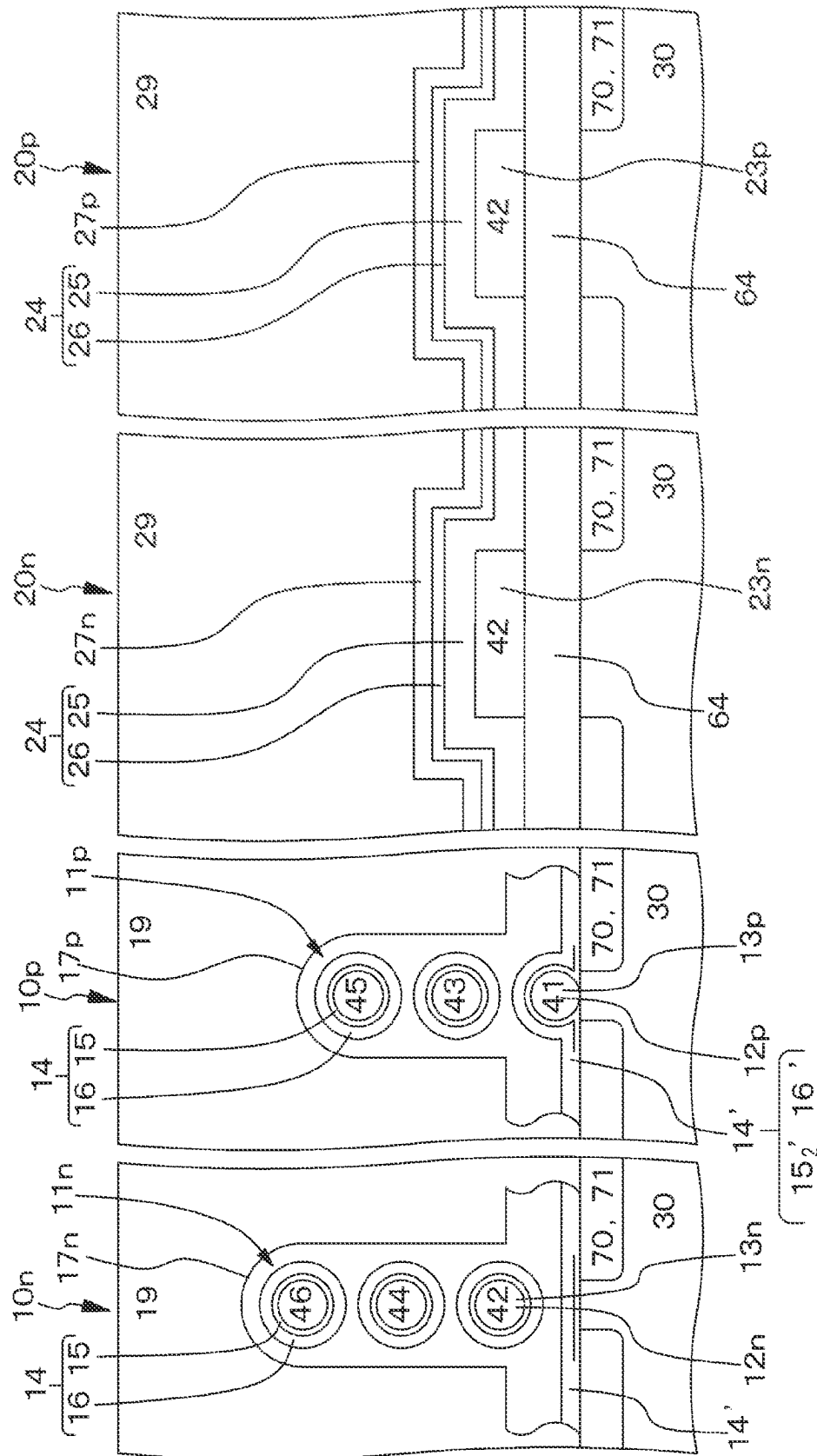


FIG. 5







7. GIL

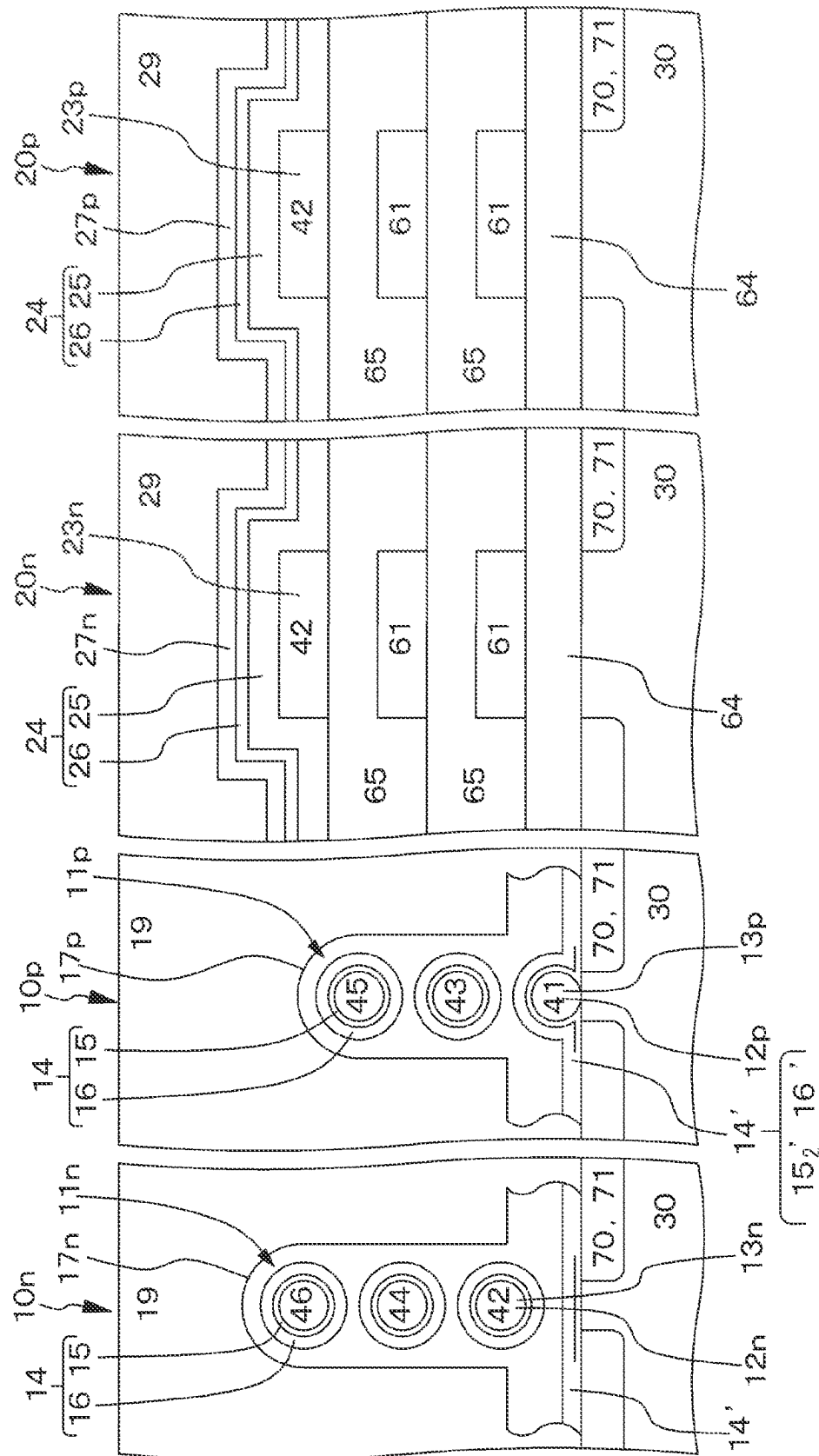


FIG. 8A

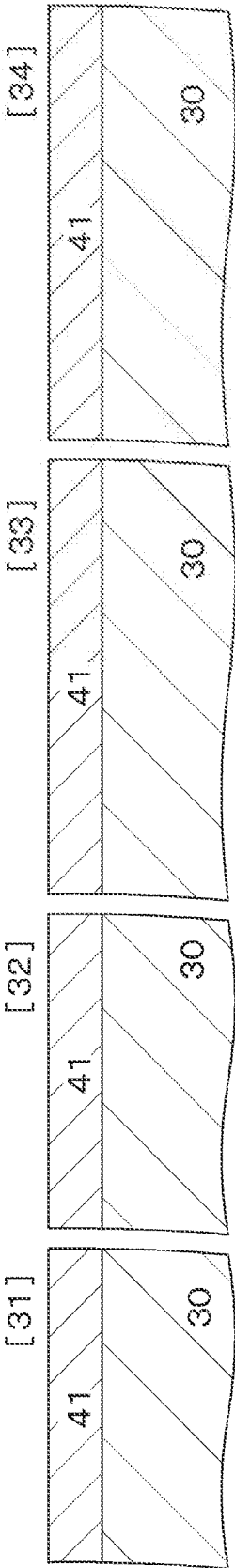


FIG. 8B

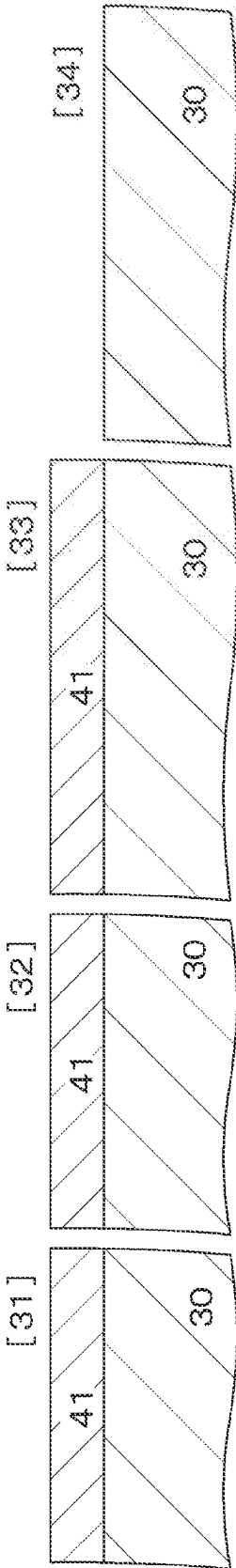


FIG. 8C

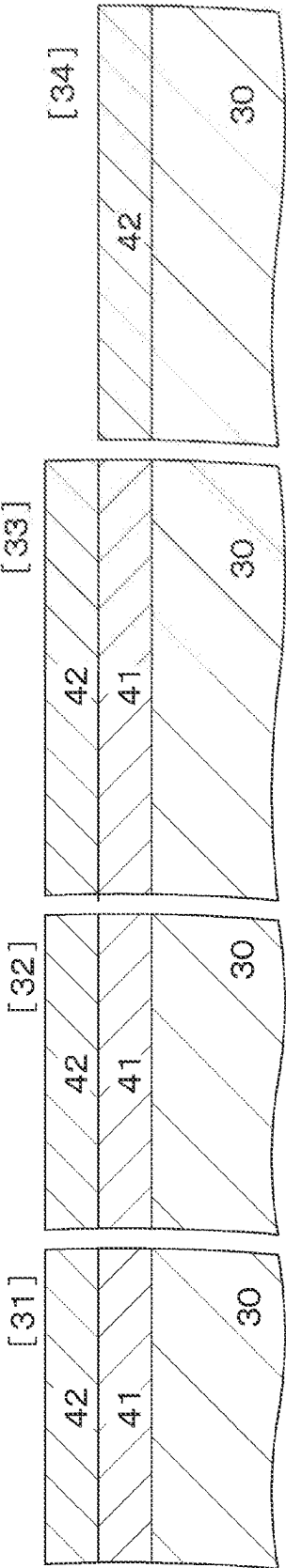


FIG. 9A

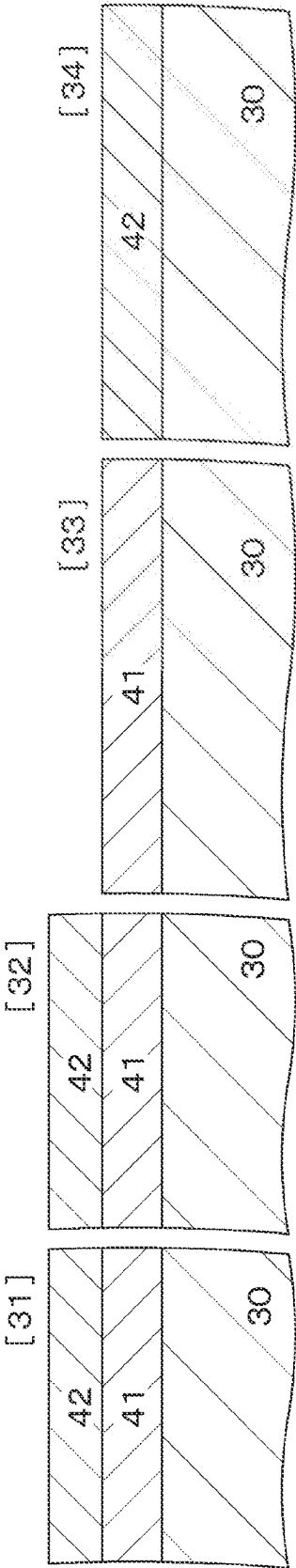


FIG. 9B

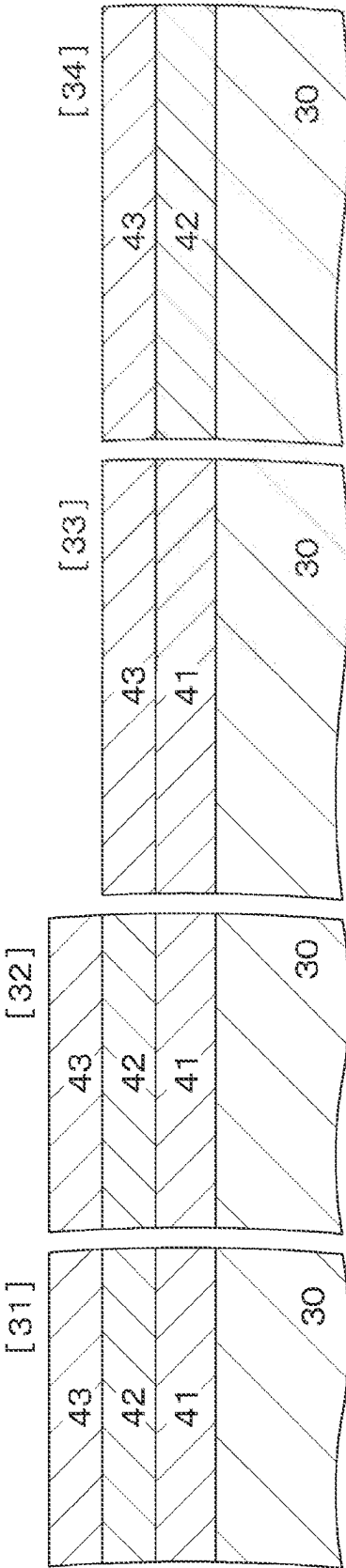


FIG. 10A

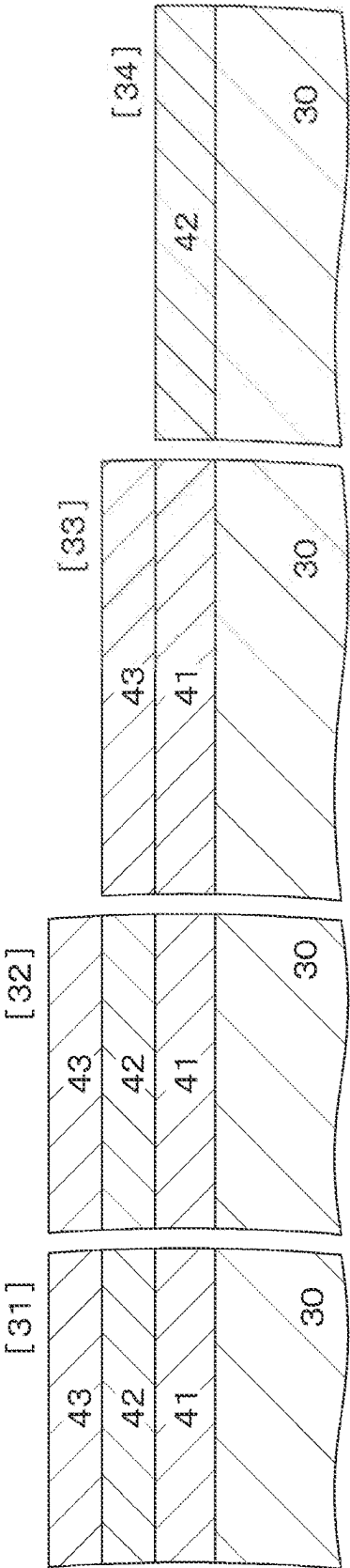


FIG. 10B

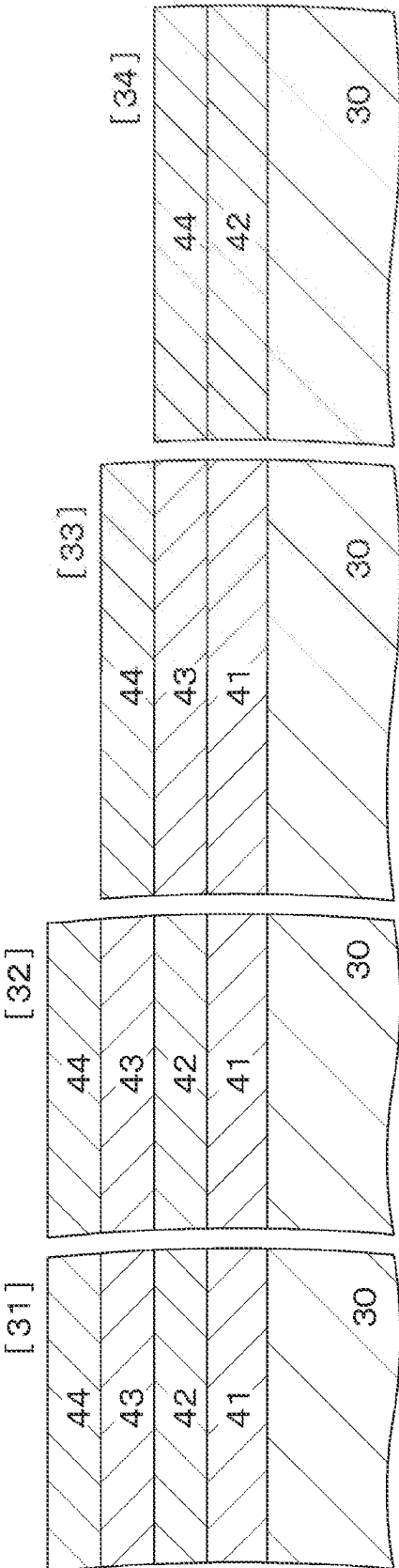


FIG. 11A

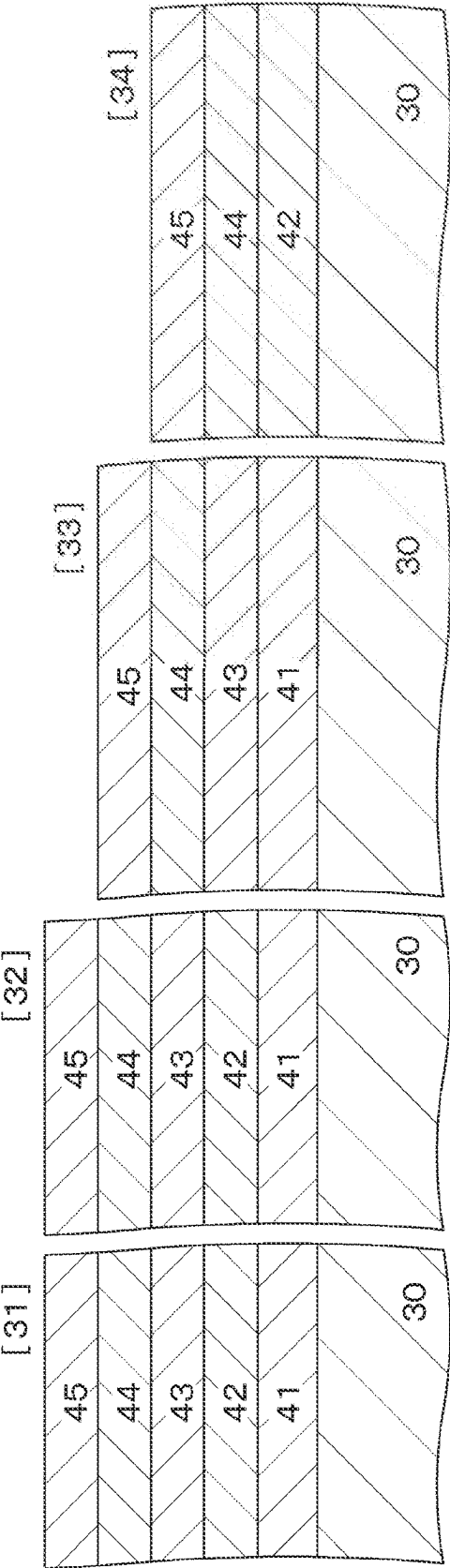


FIG. 11B

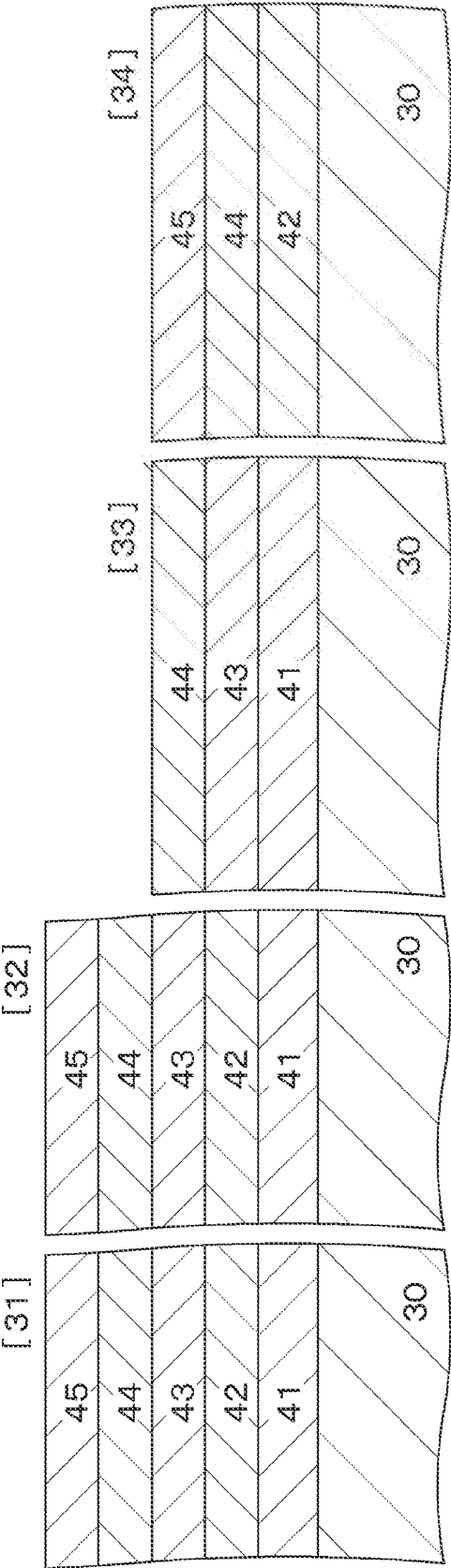
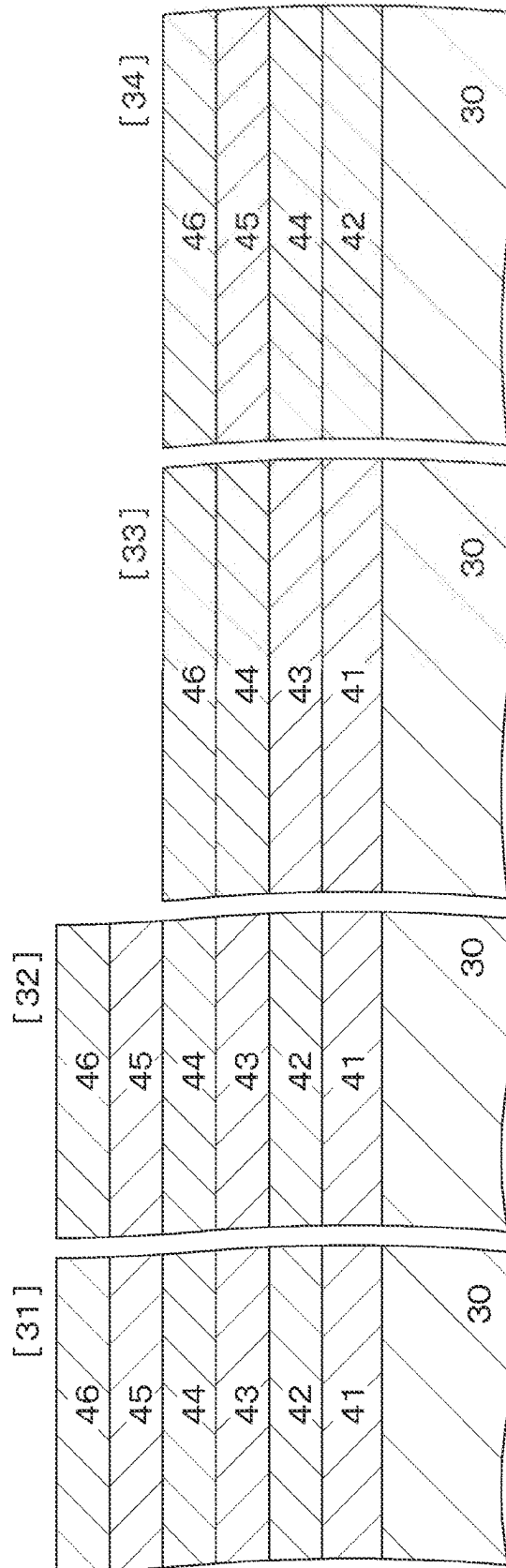


FIG. 12



31.6 FIFTEEN

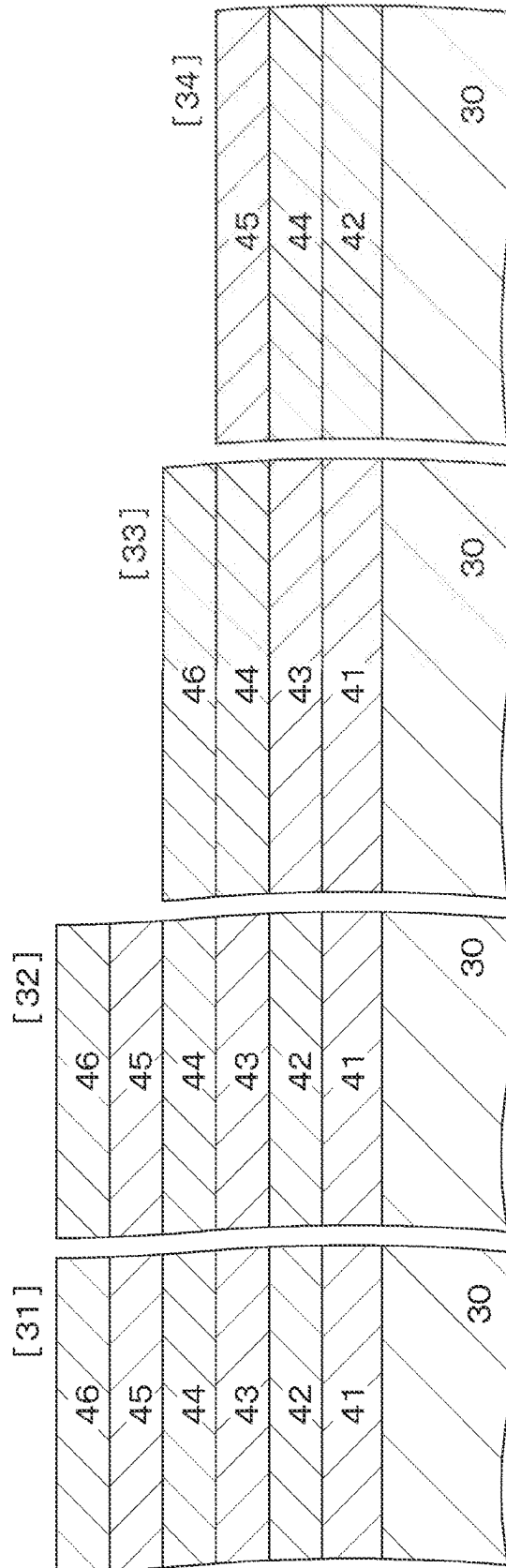


FIG. 14

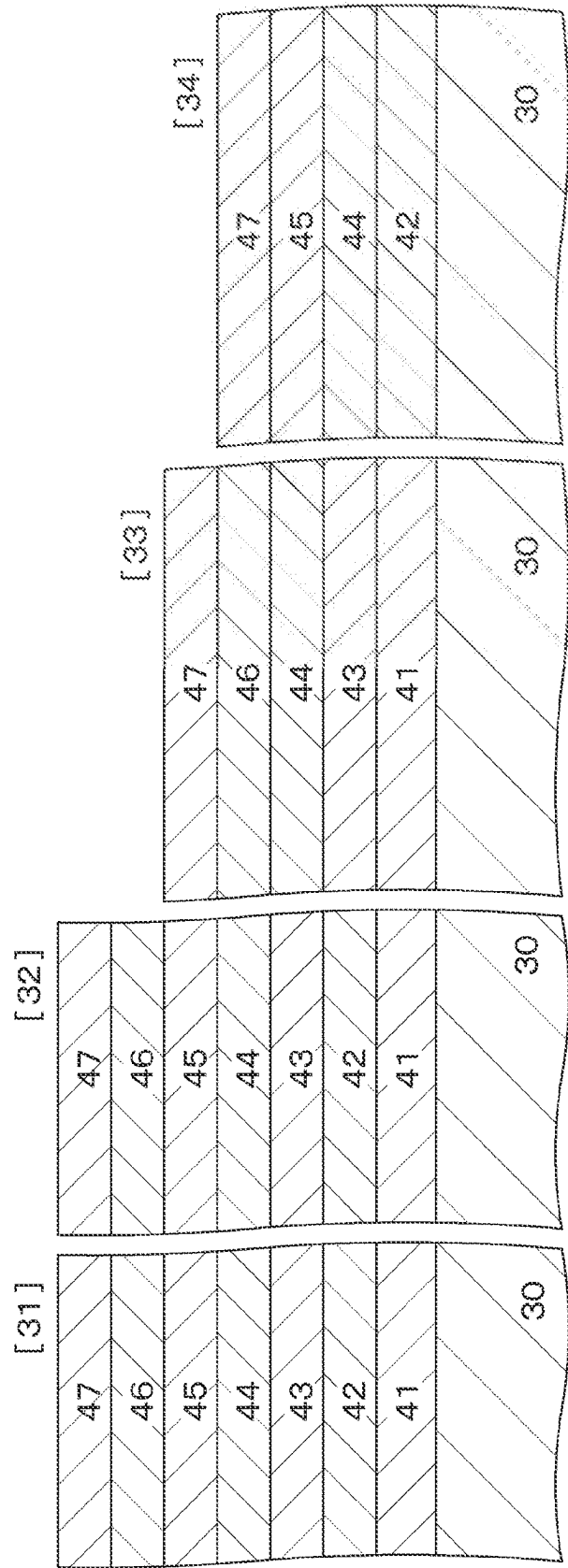


FIG. 15

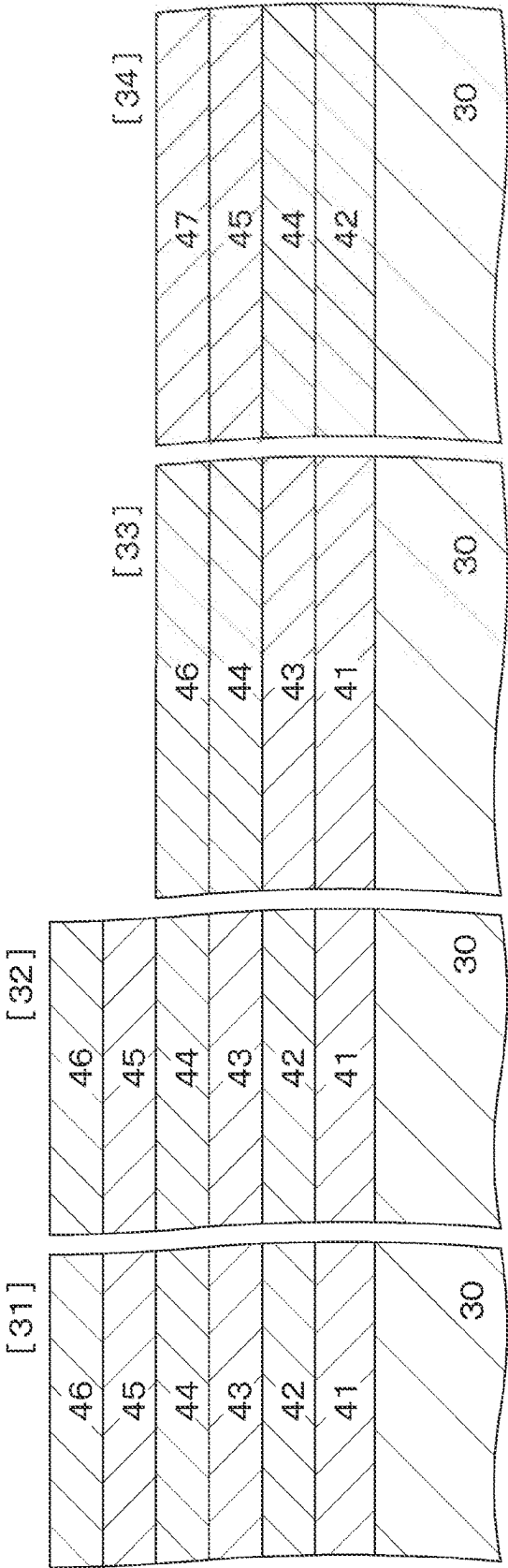


FIG. 16

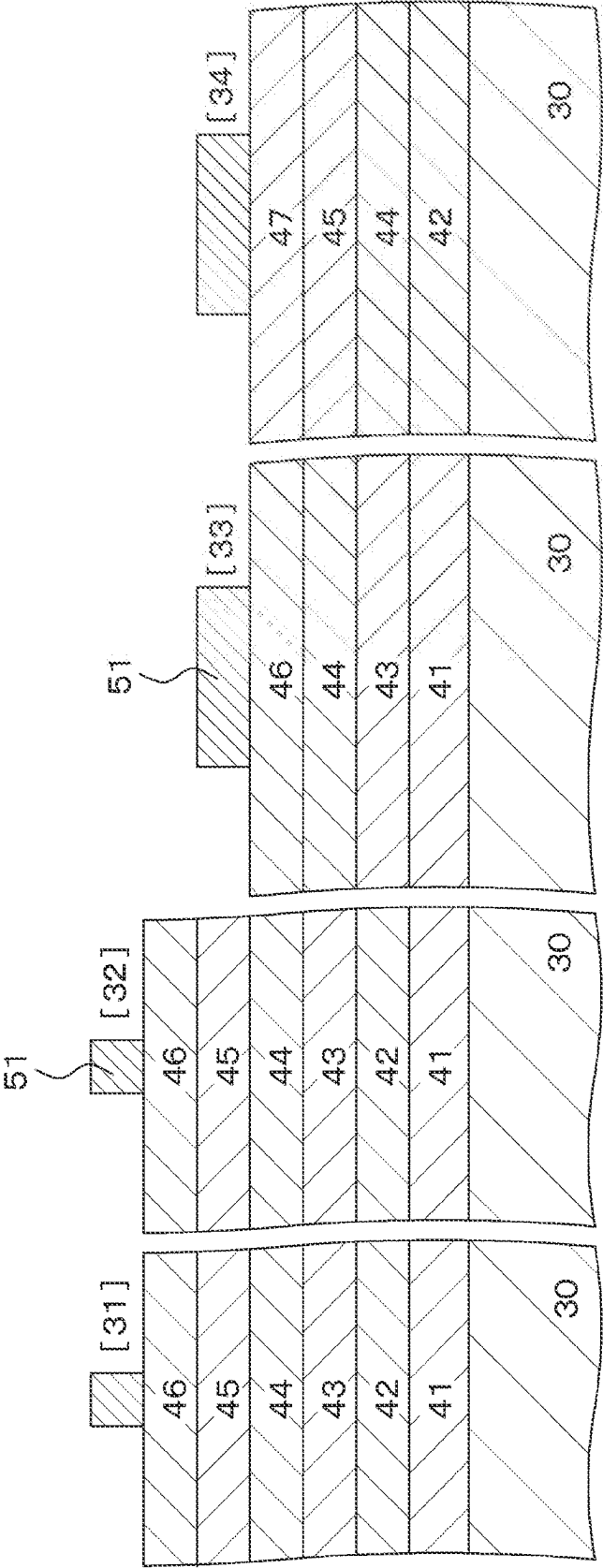


FIG. 17

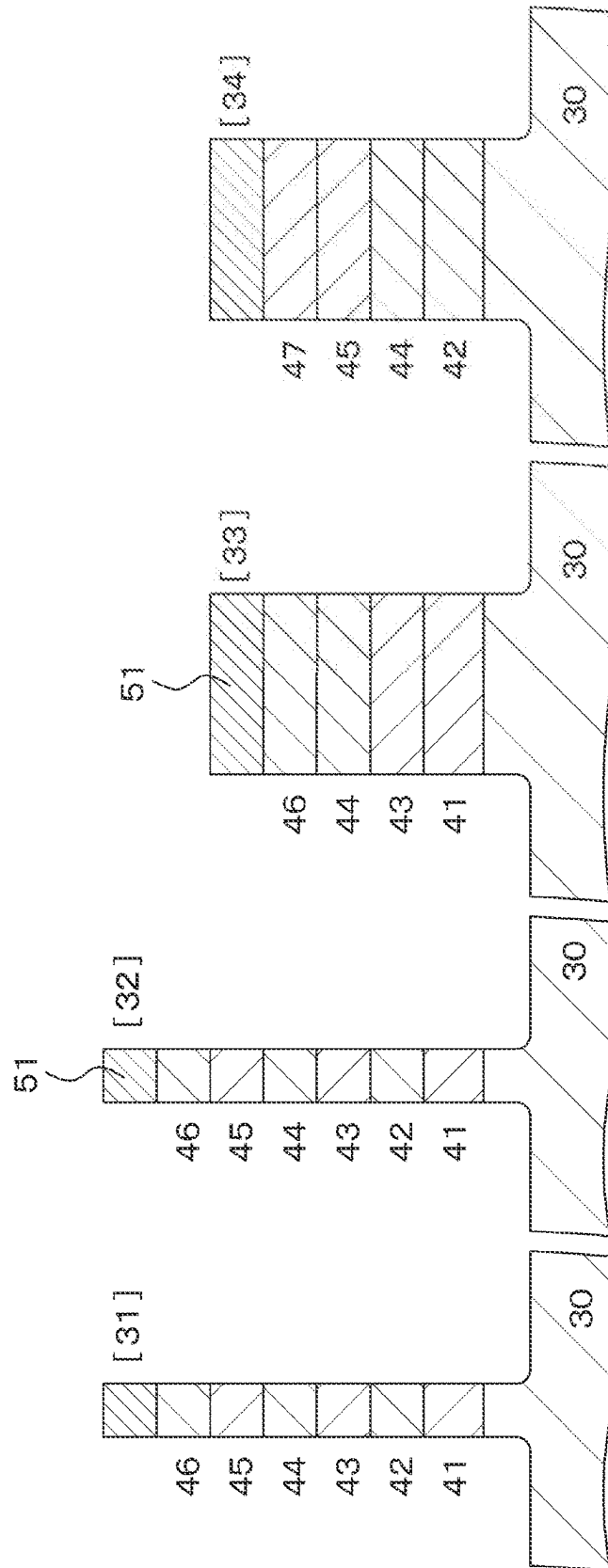


FIG. 18

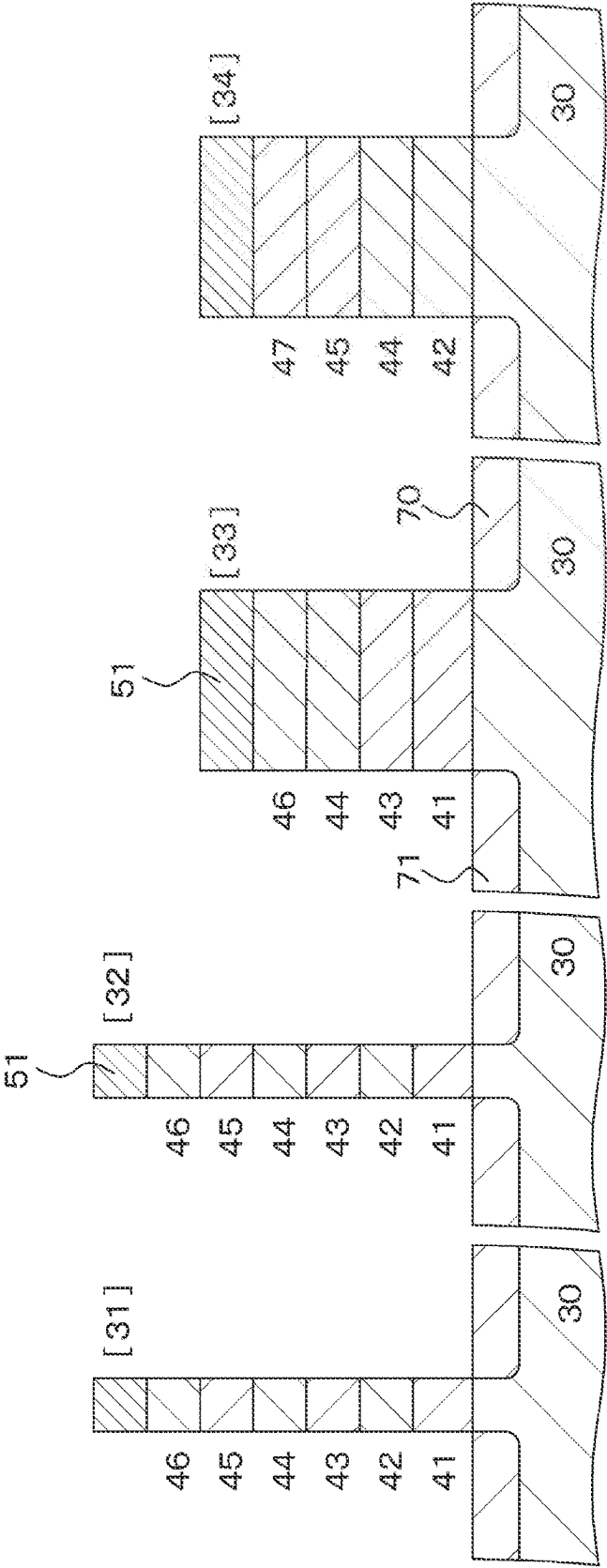


FIG. 19

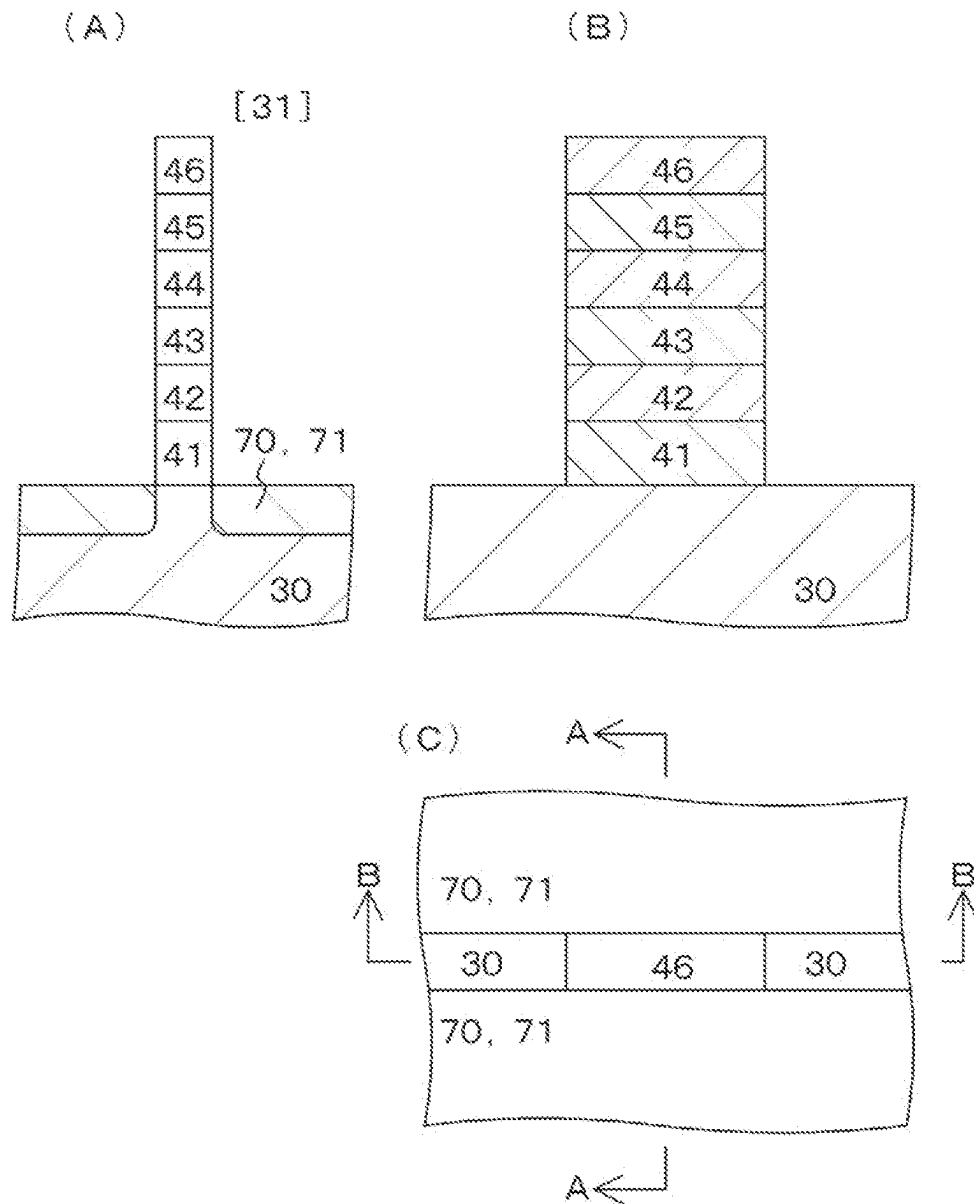


FIG. 20

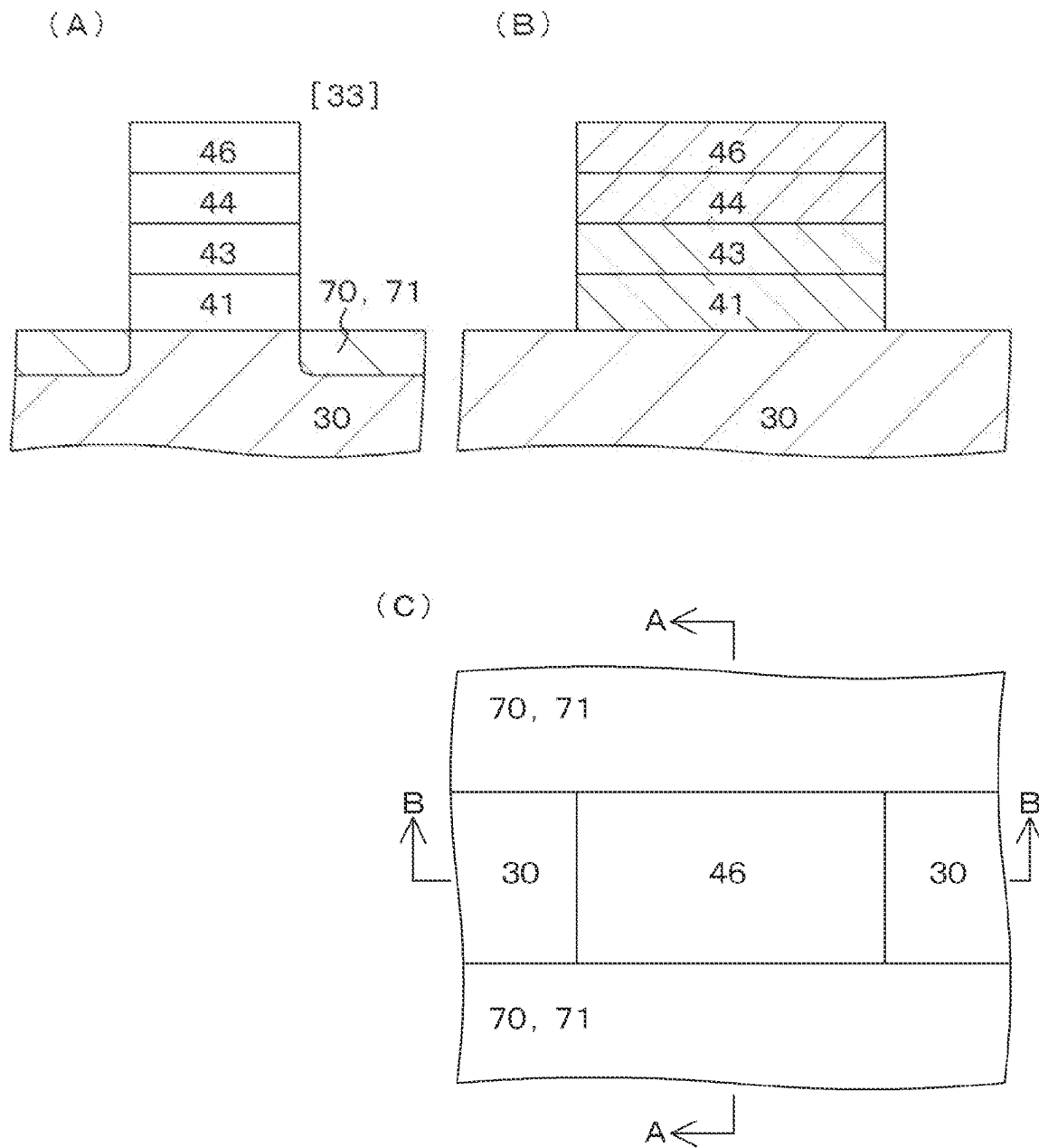


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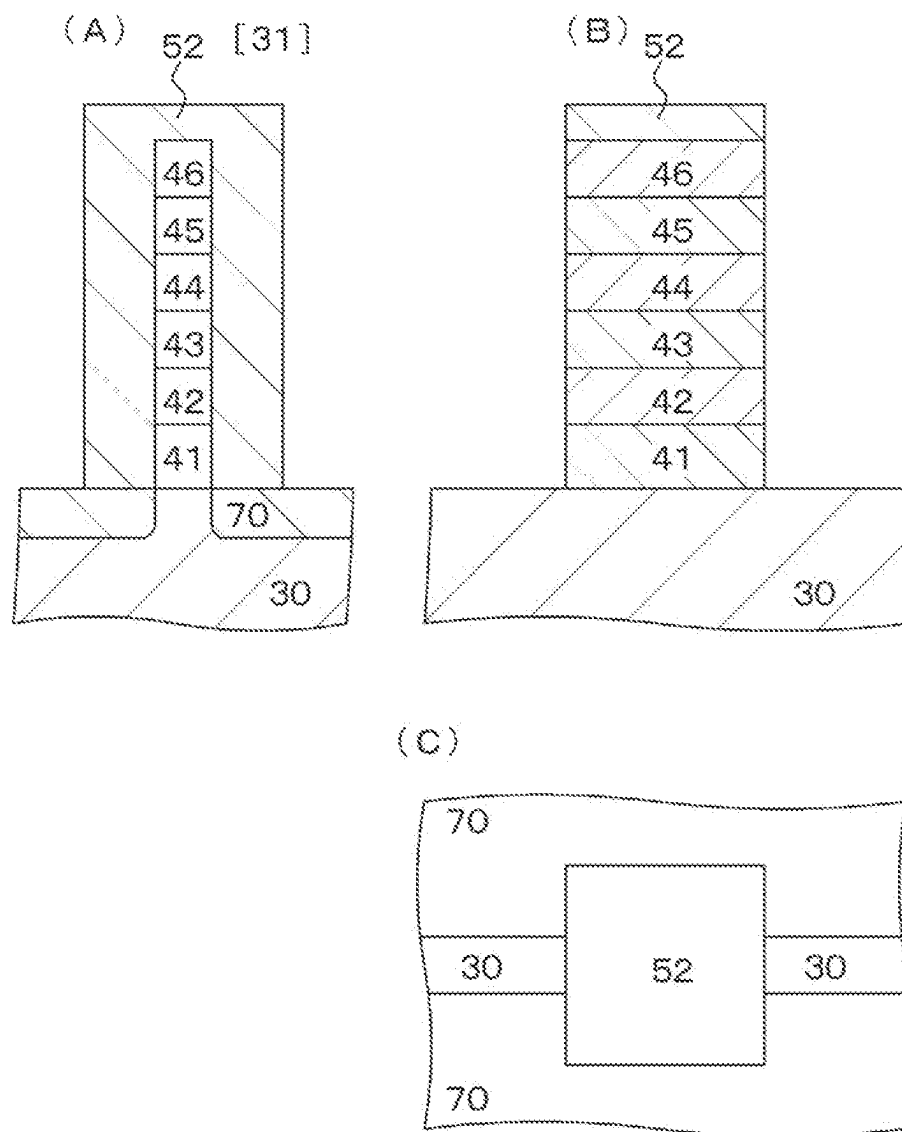


FIG. 22

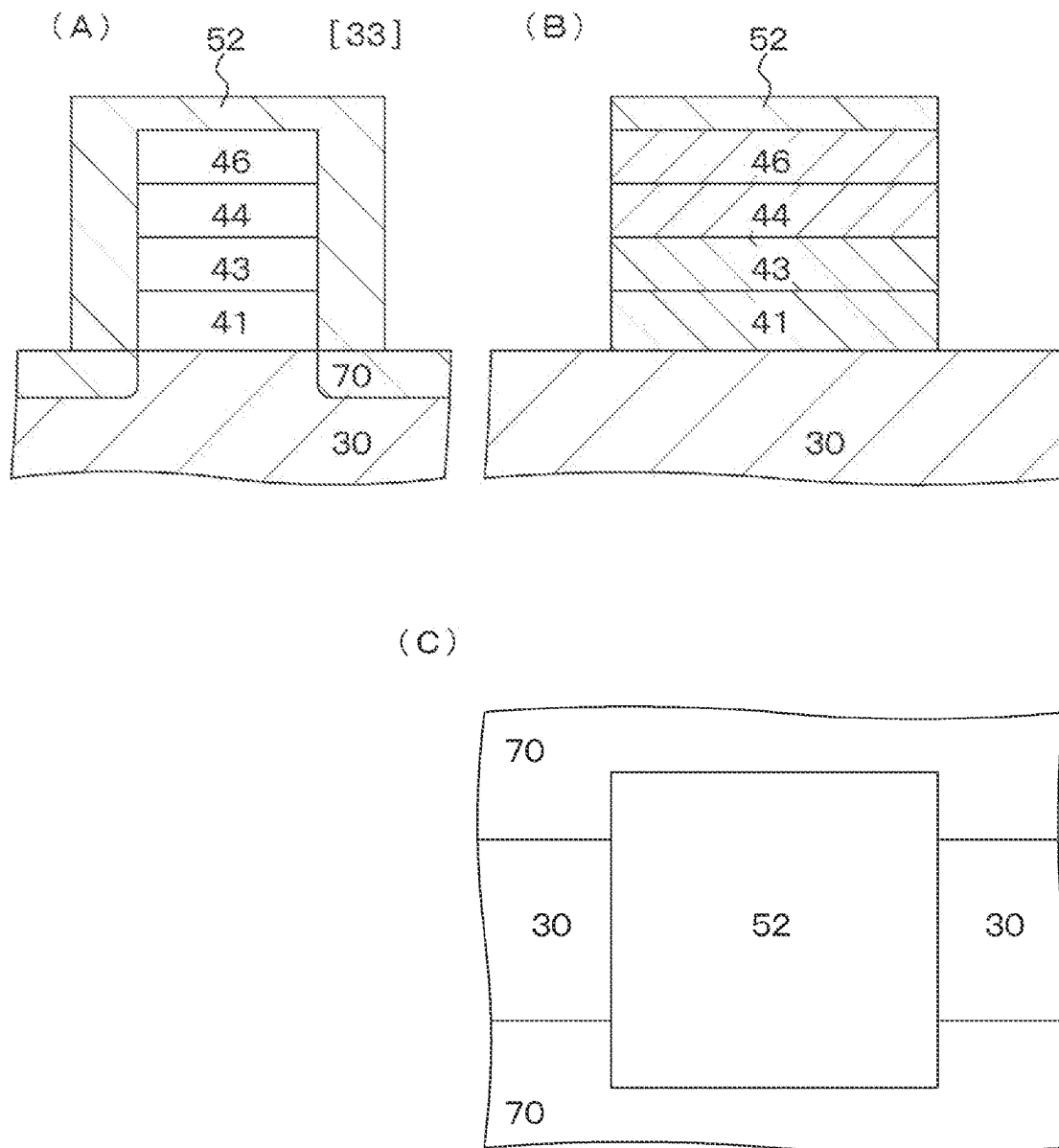


FIG. 23

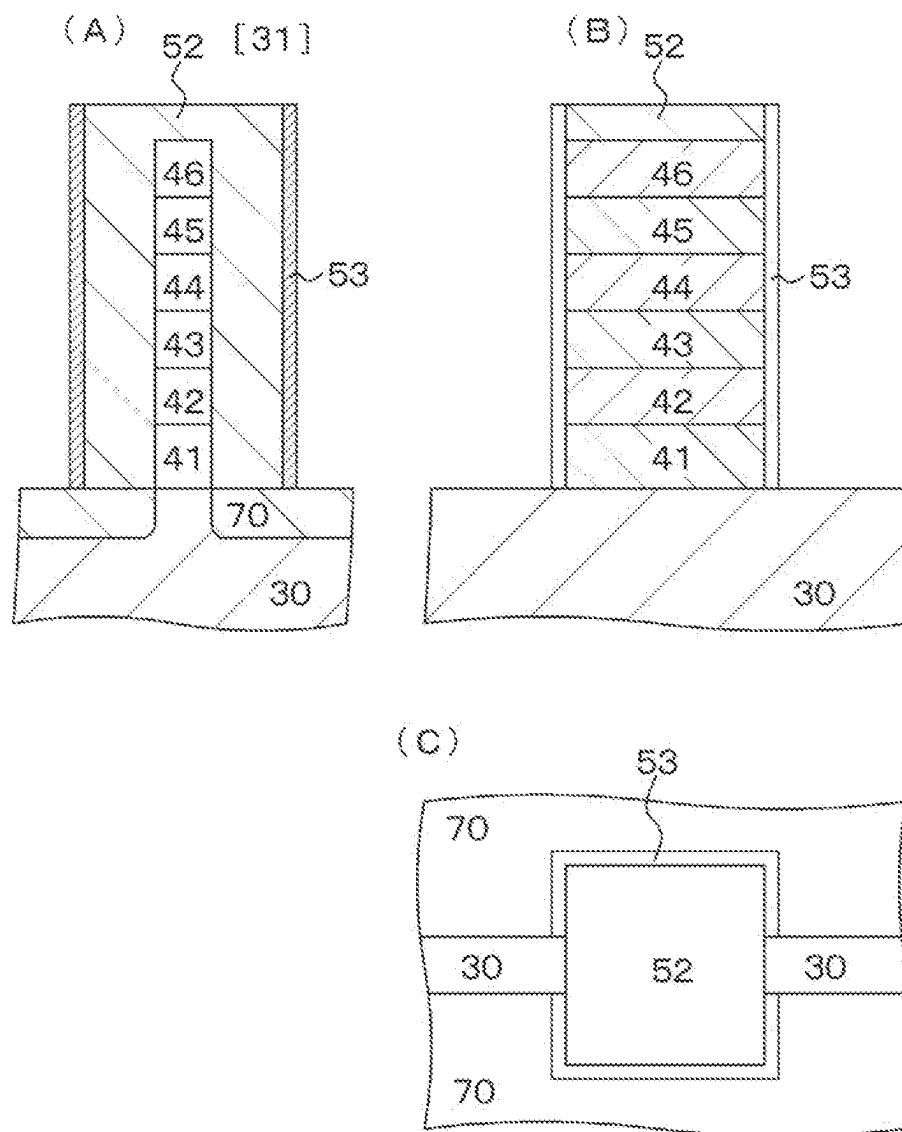


FIG. 24

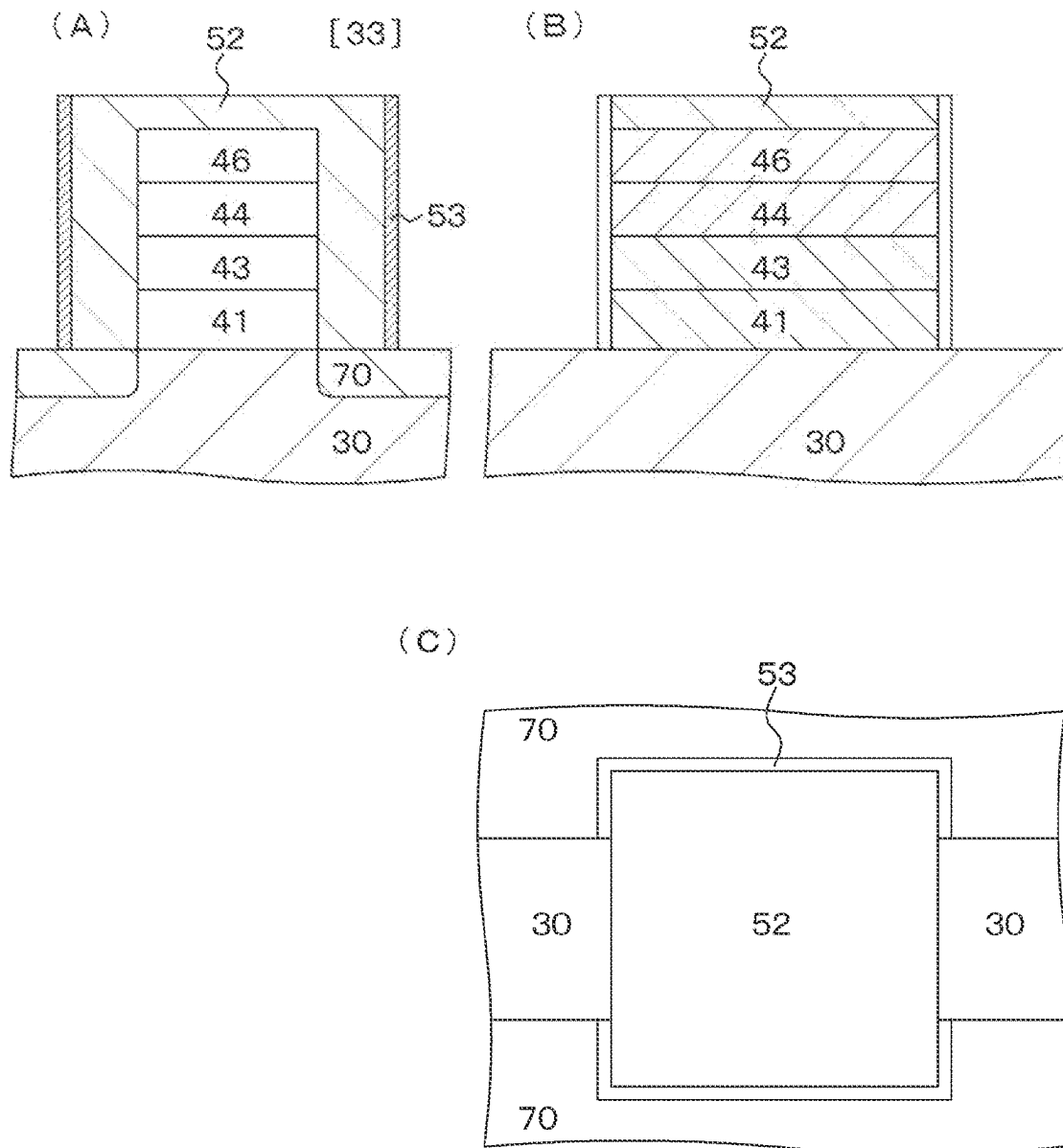


FIG. 25

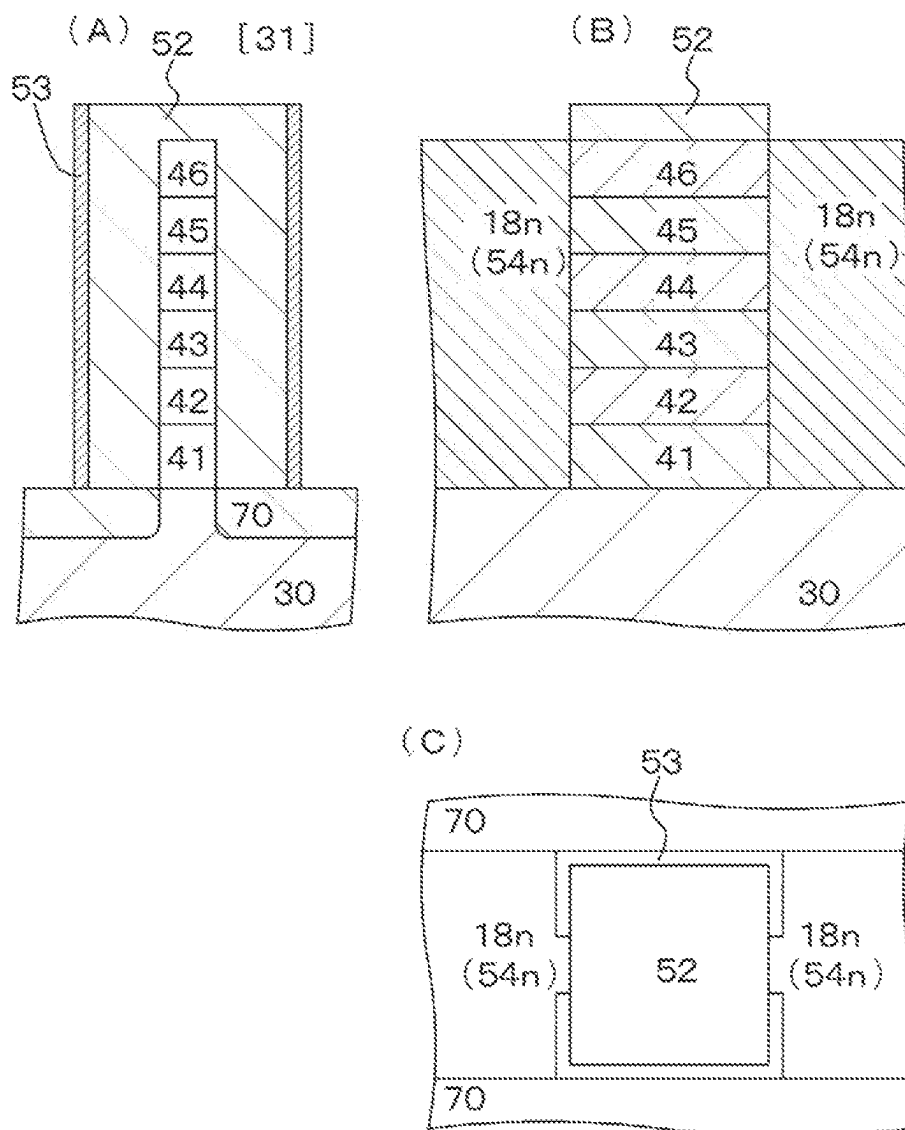


FIG. 26

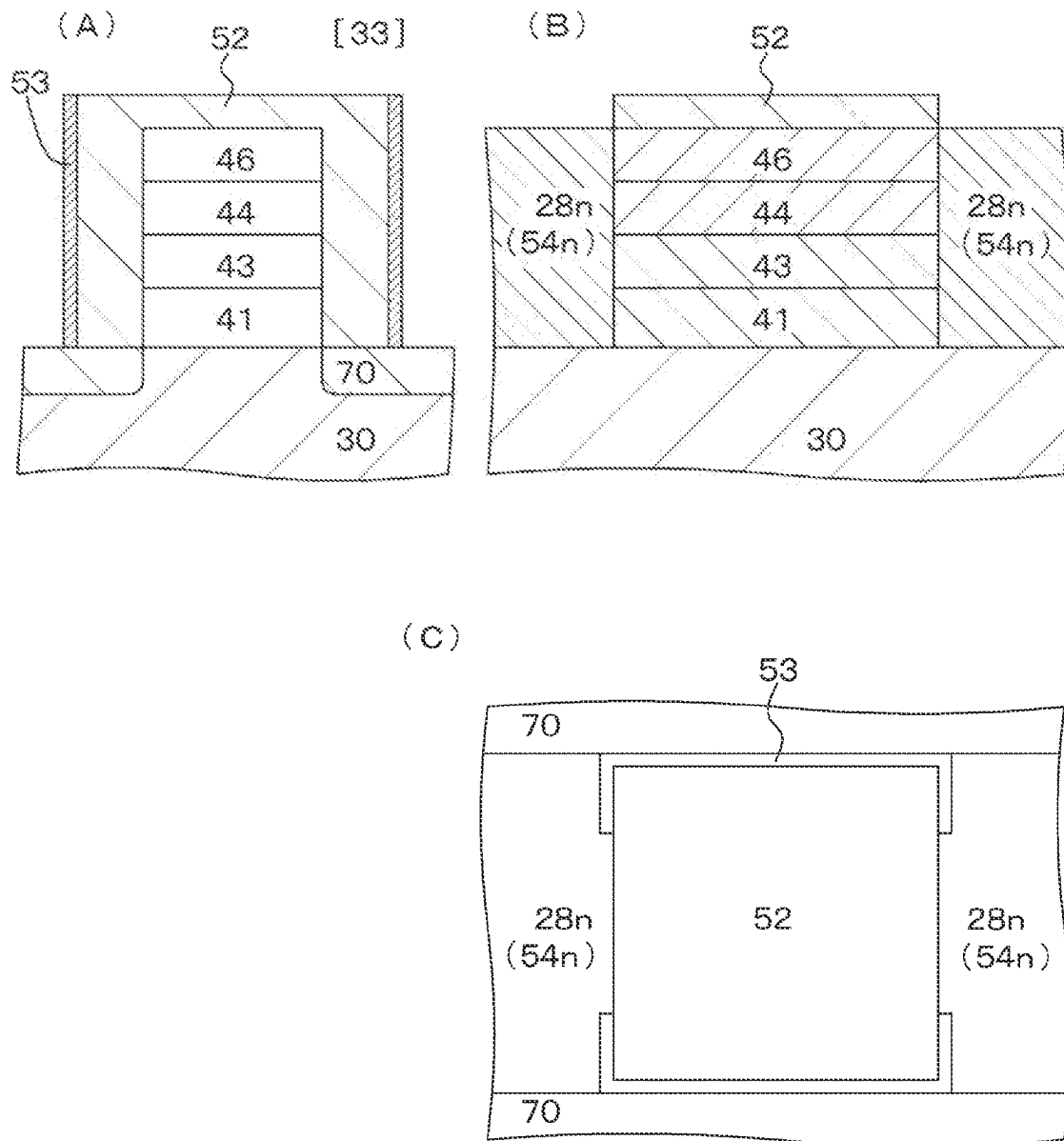


FIG. 27

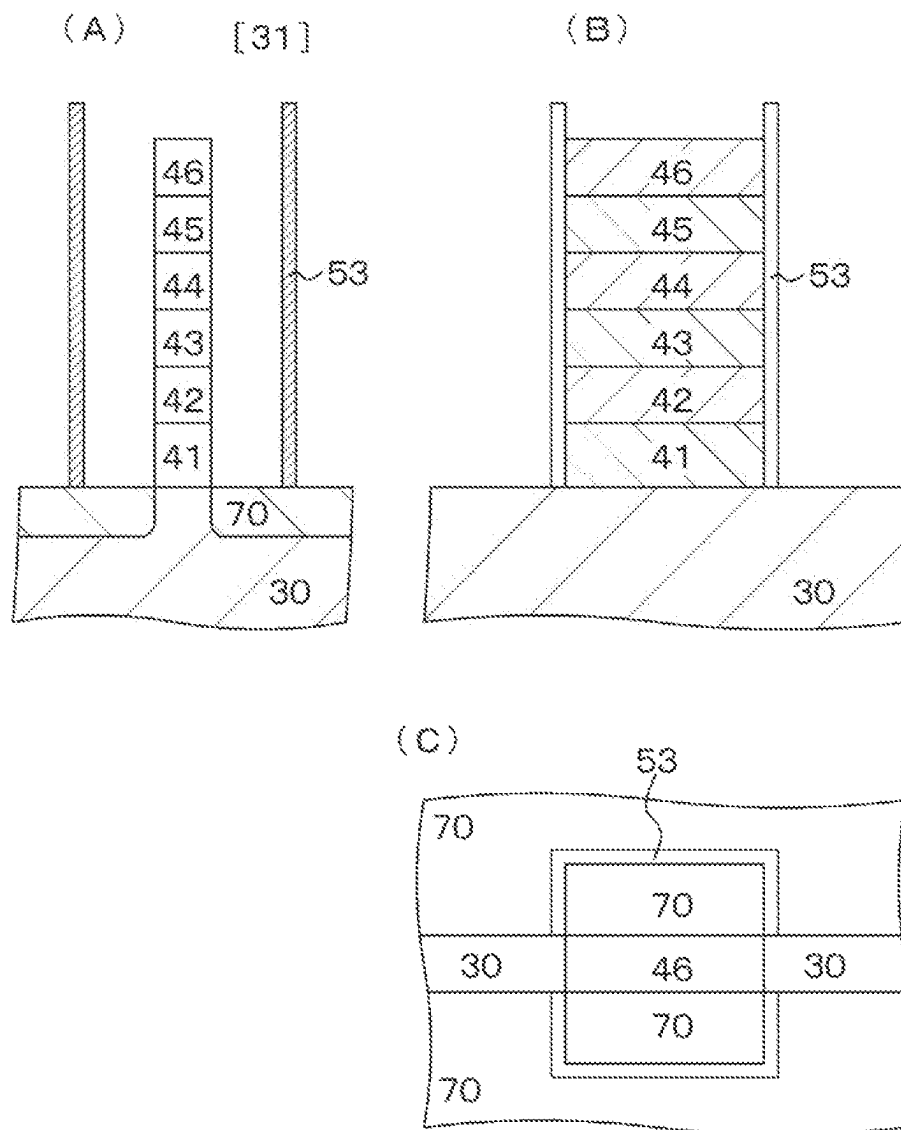


FIG. 28

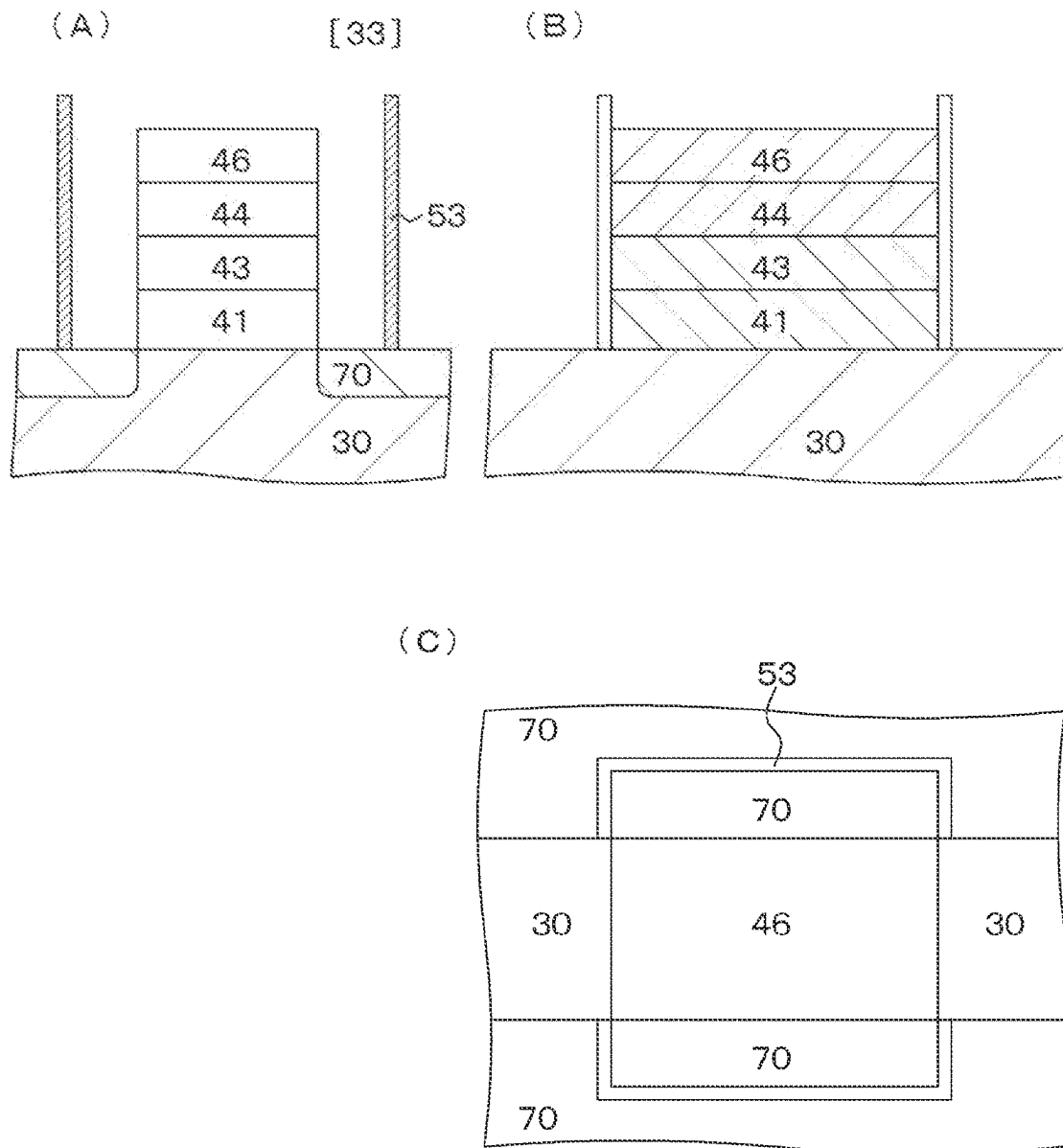


FIG. 29

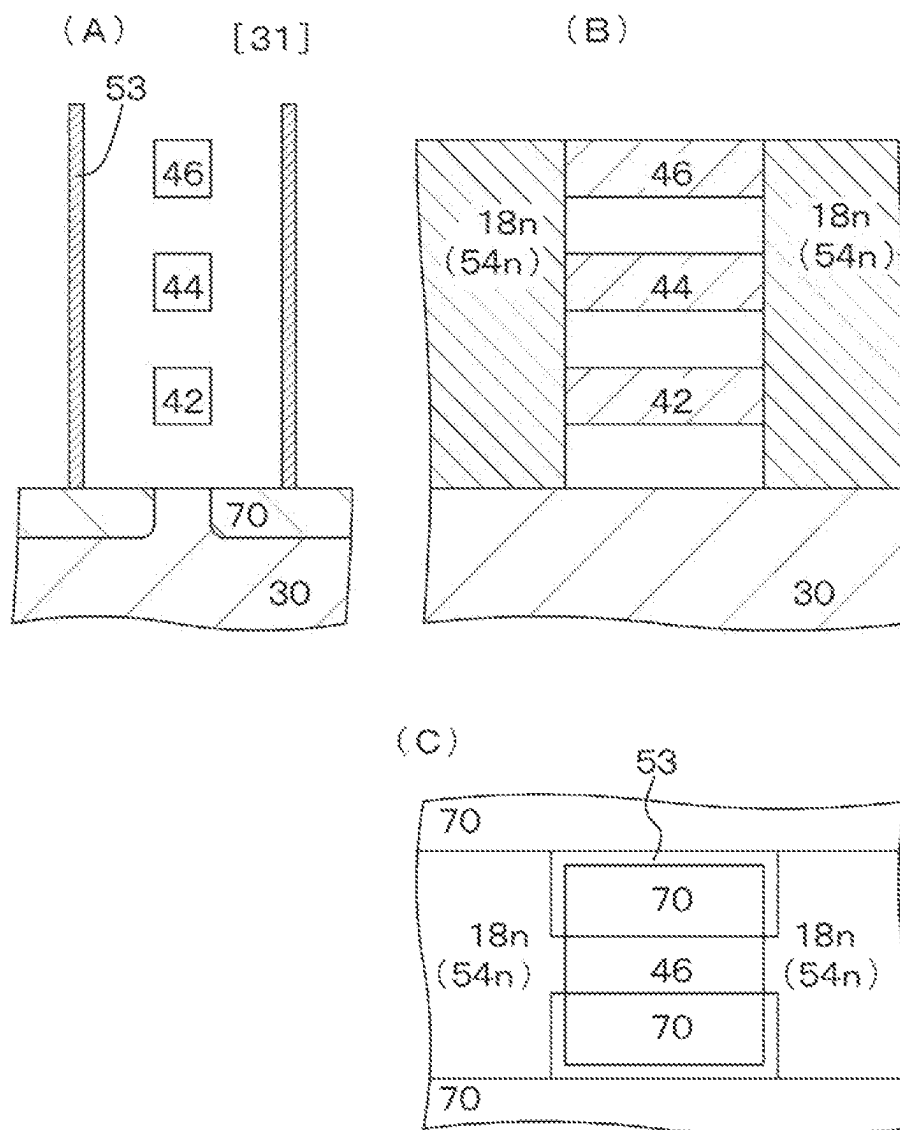


FIG. 30

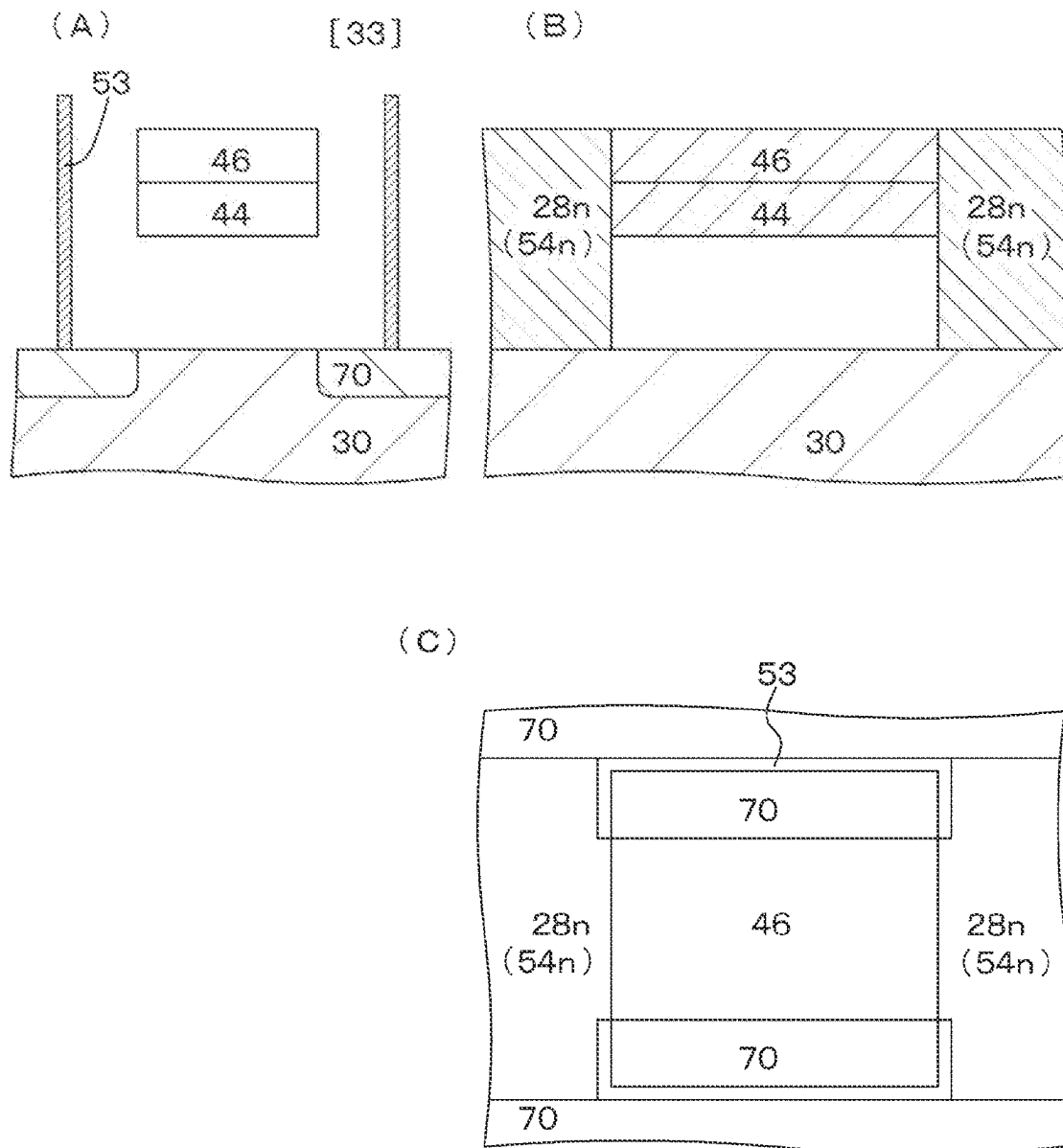


FIG. 31

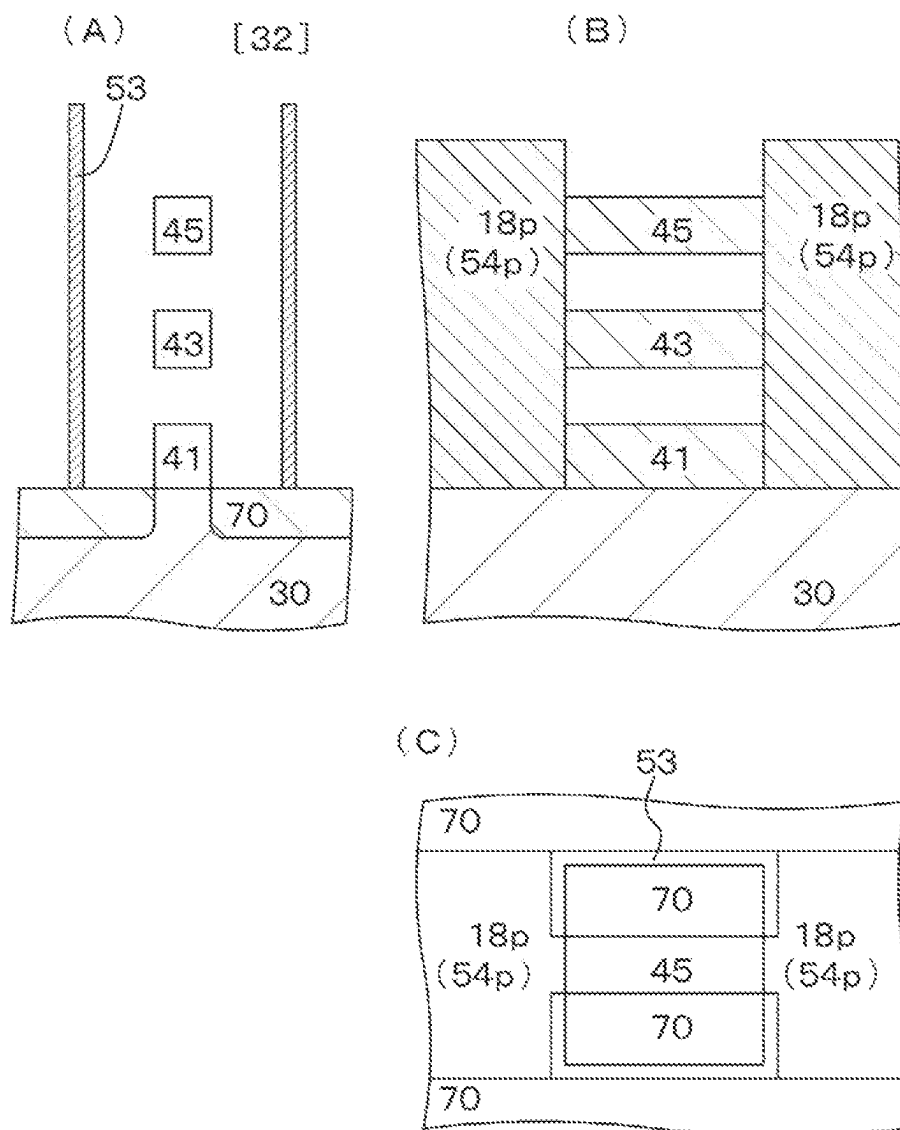


FIG. 32

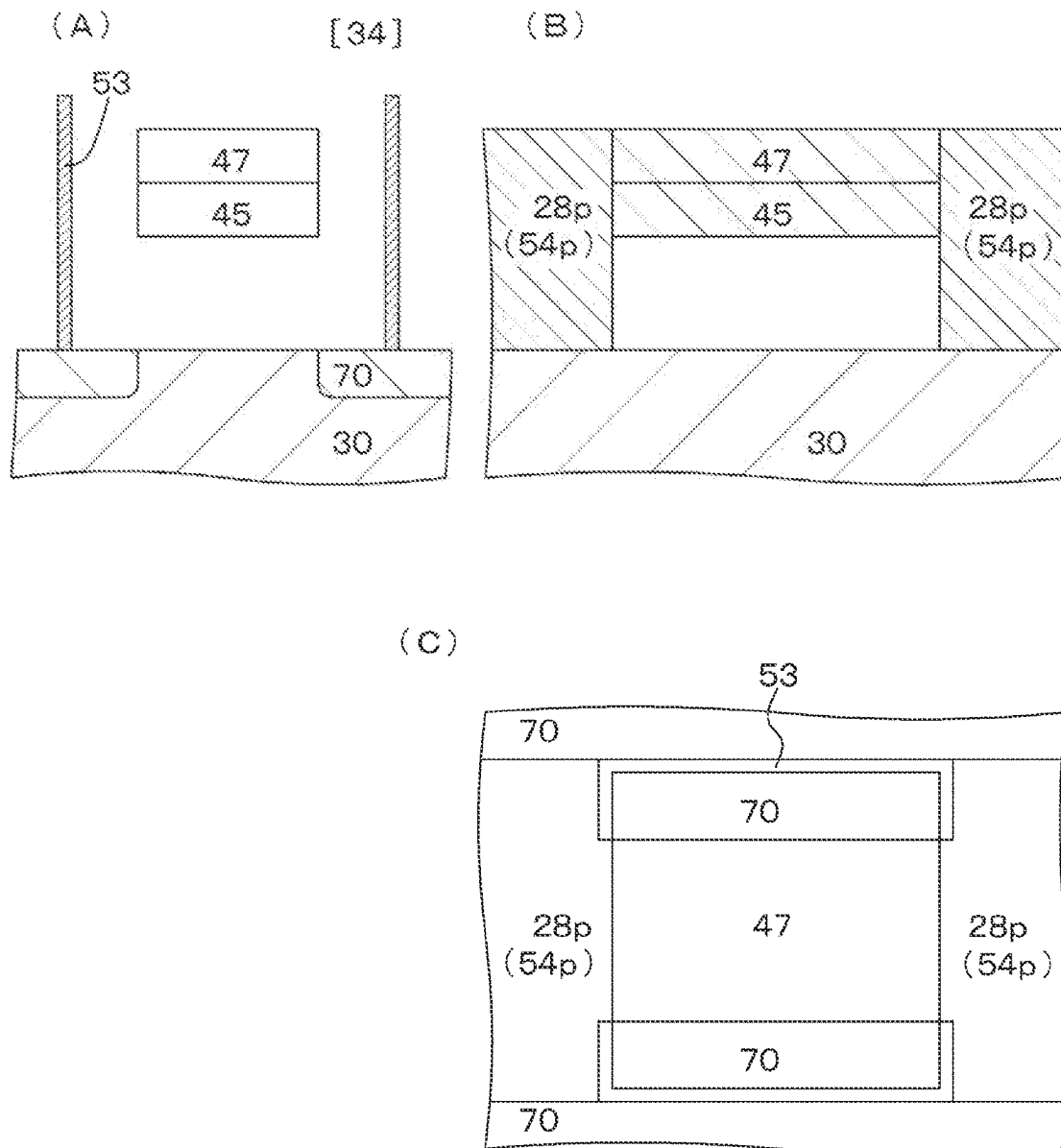


FIG. 33

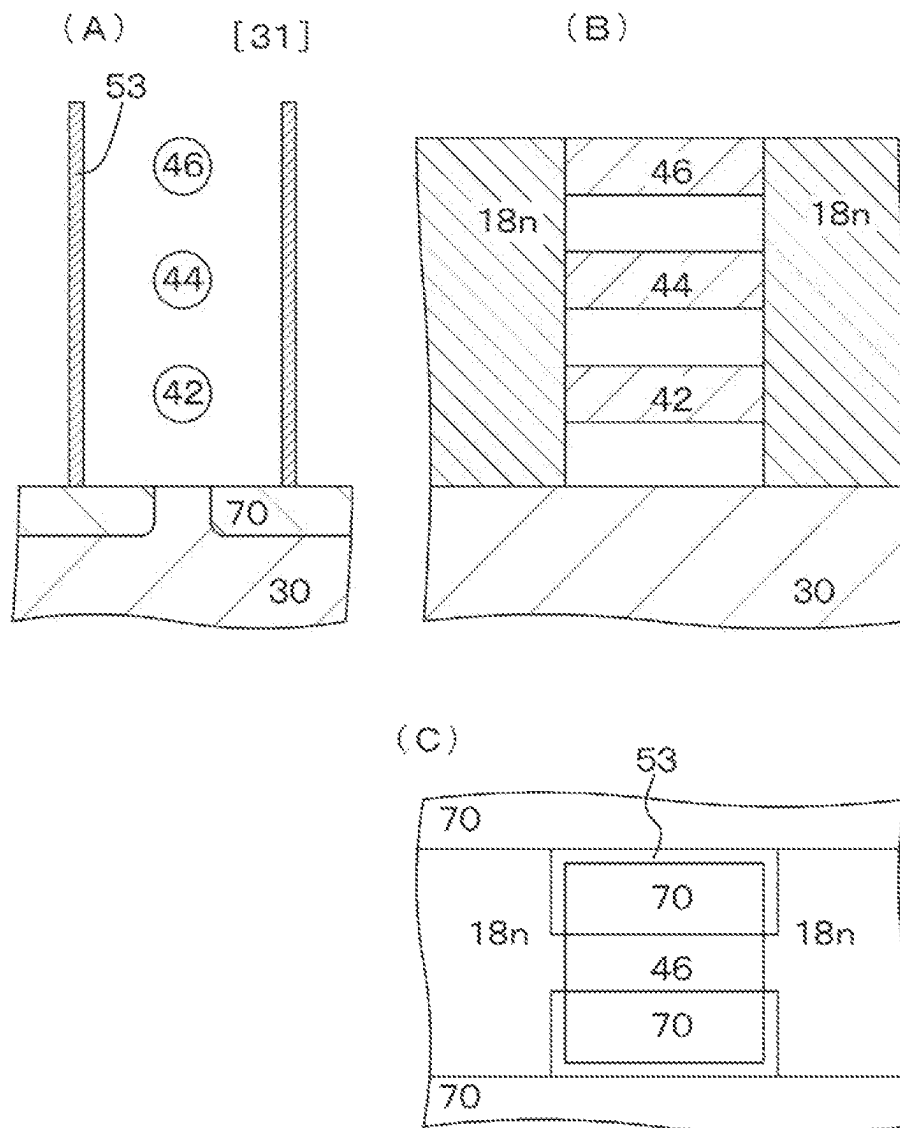


FIG. 34

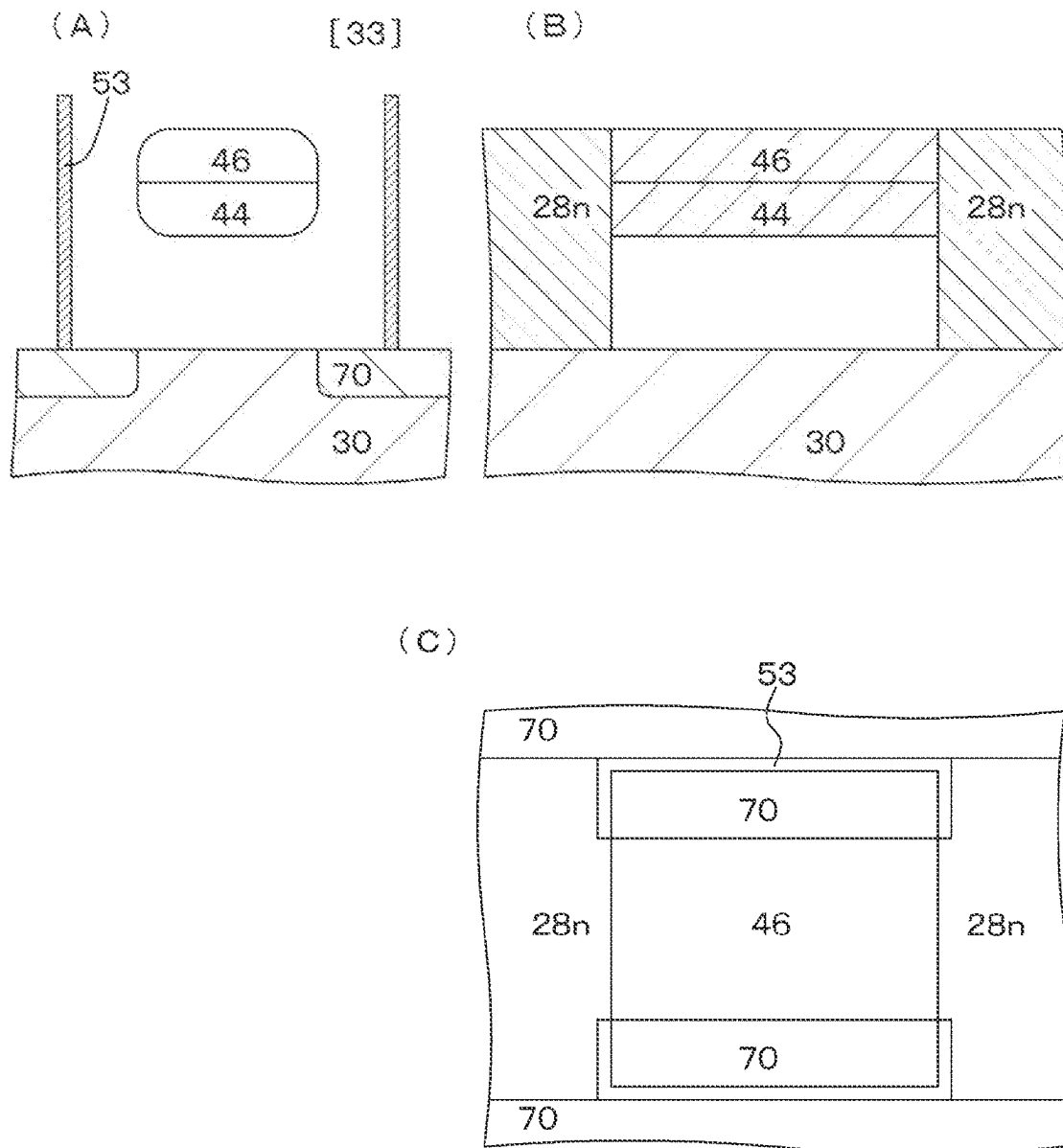


FIG. 35

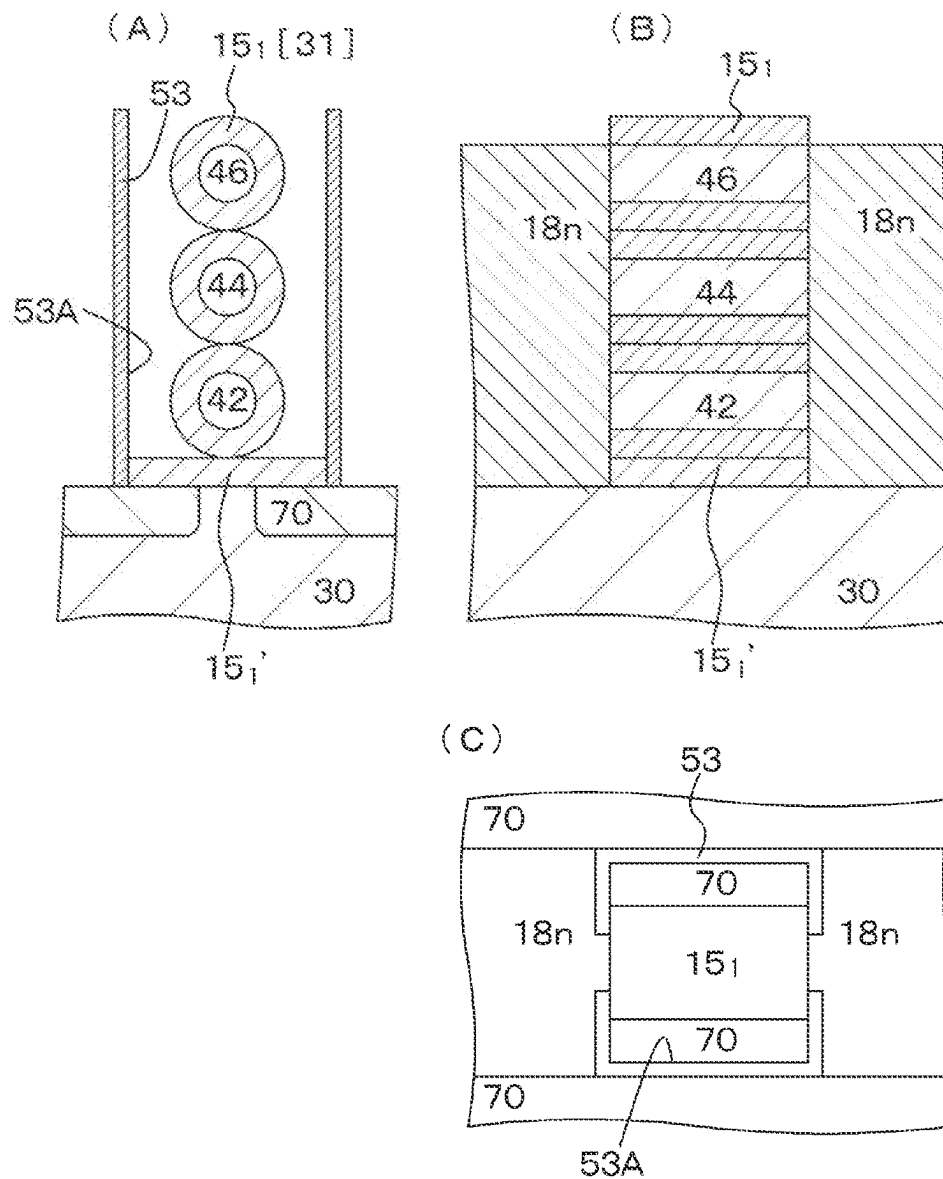


FIG. 36

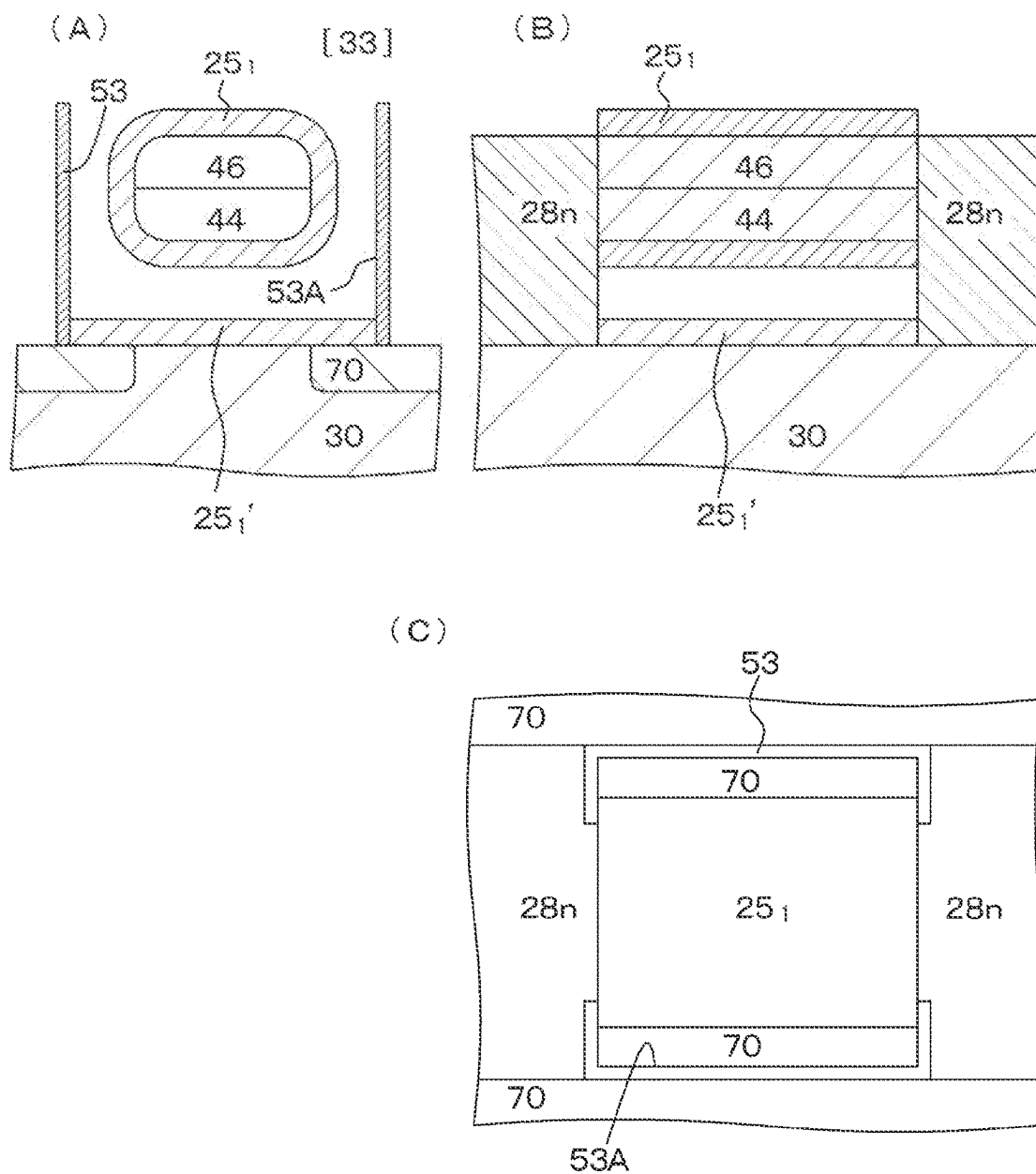


FIG. 37

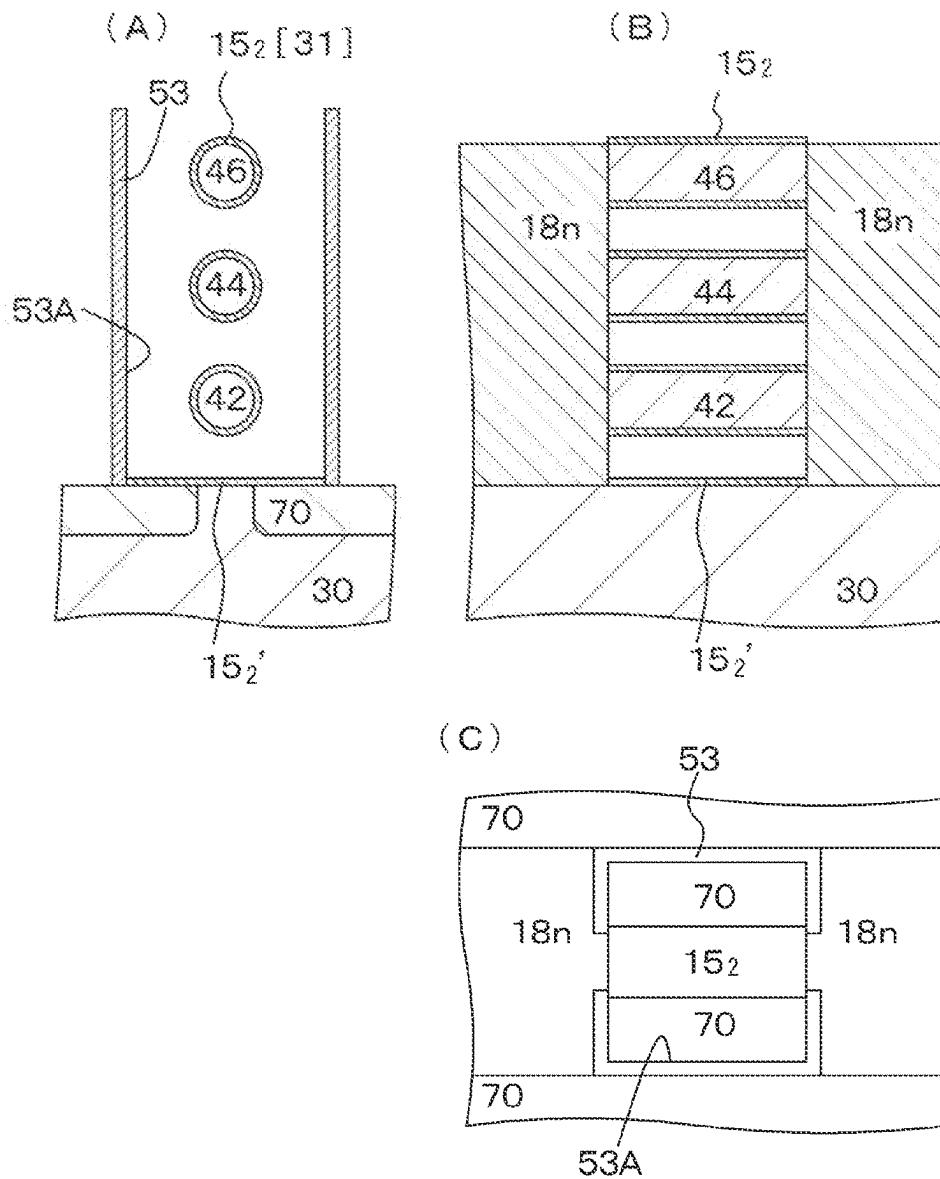


FIG. 38

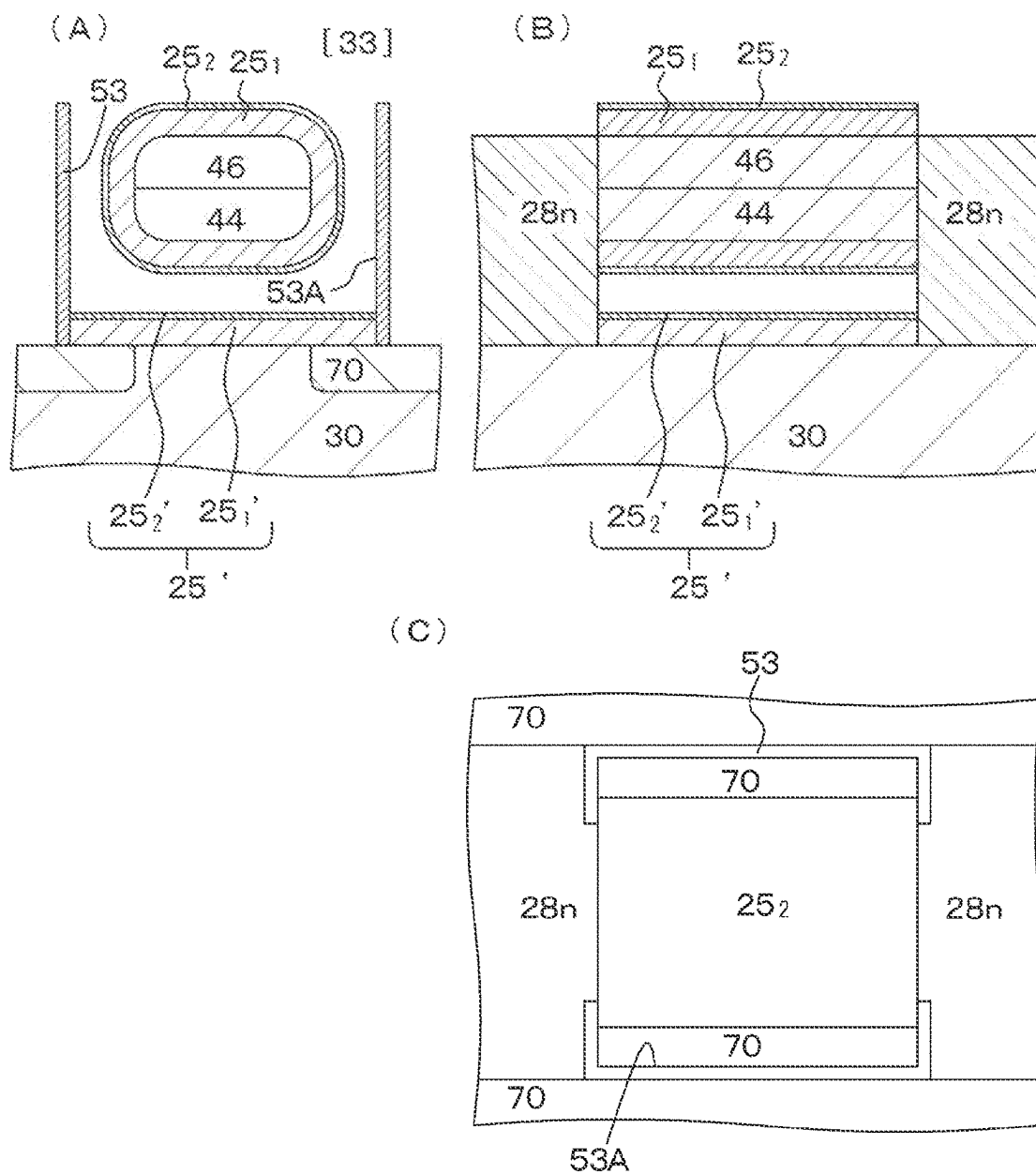


FIG. 39

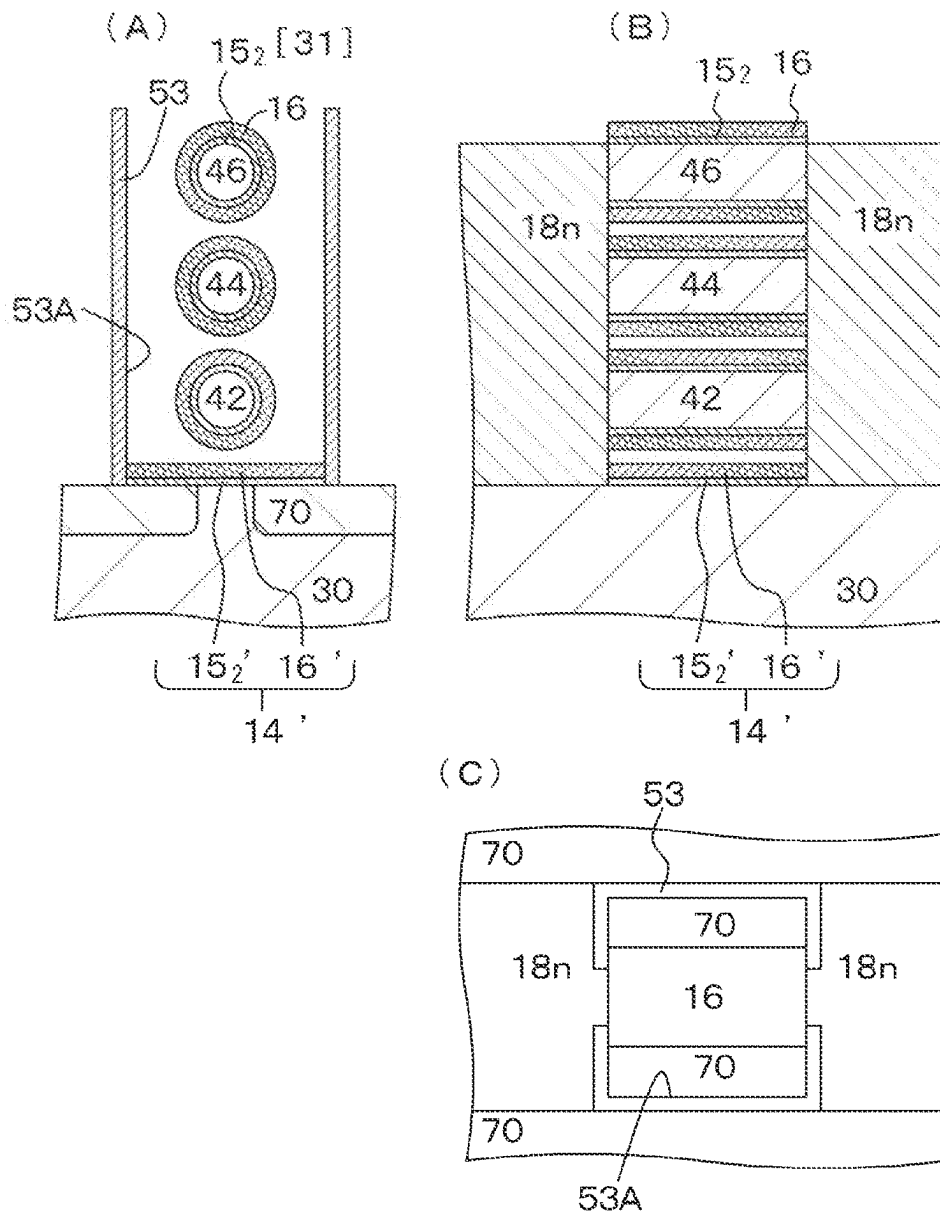


FIG. 40

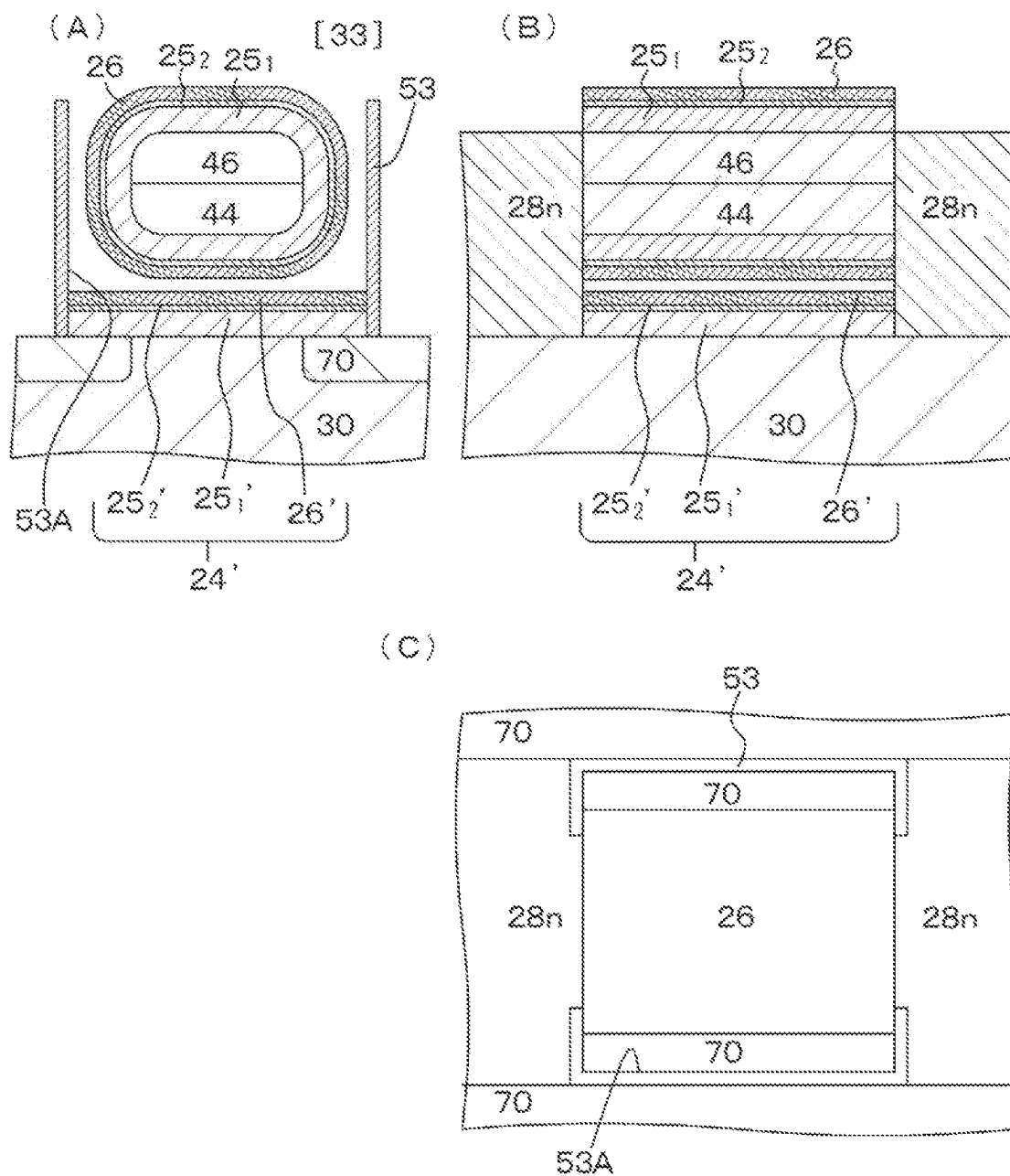


FIG. 41

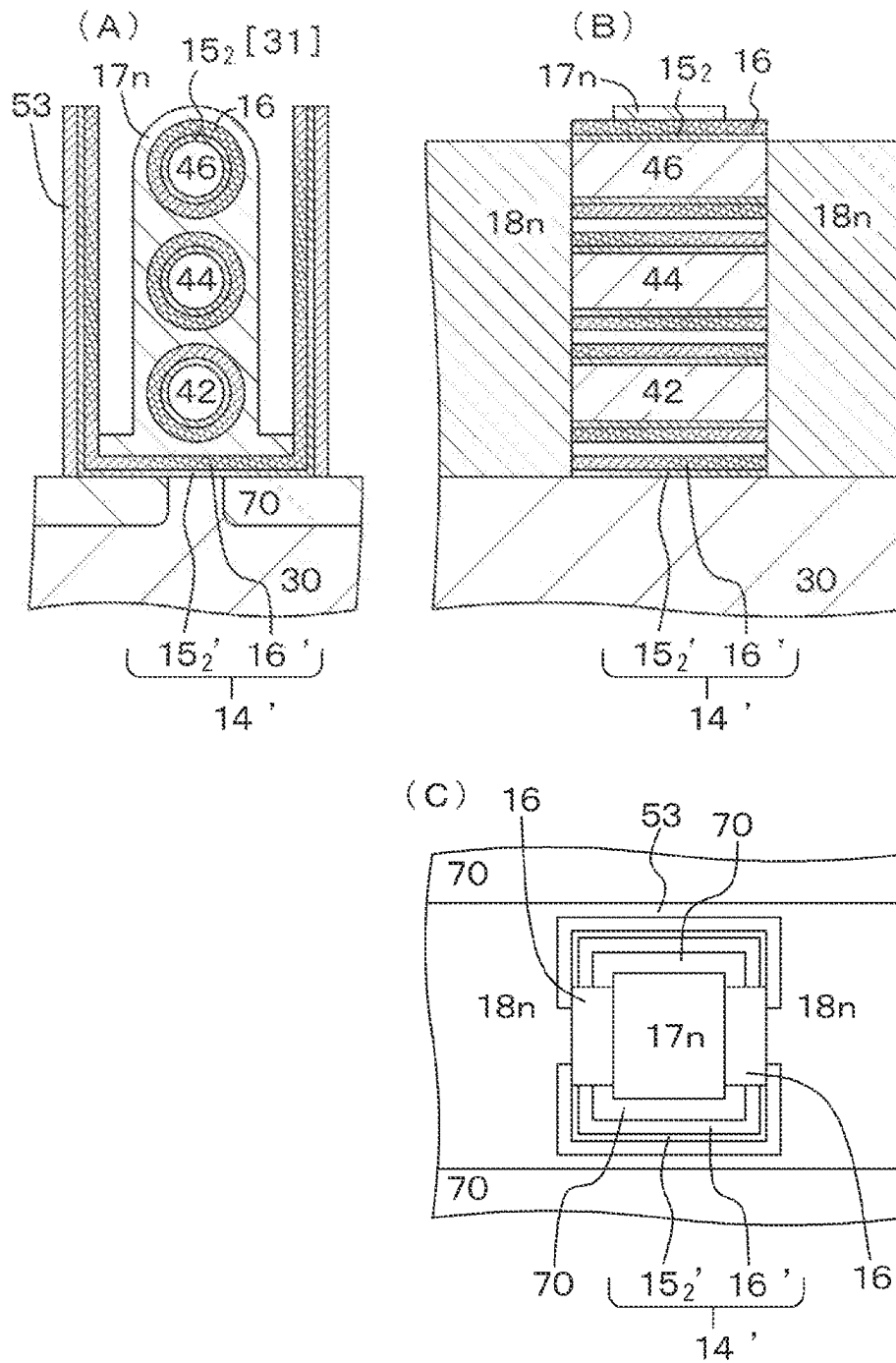


FIG. 42

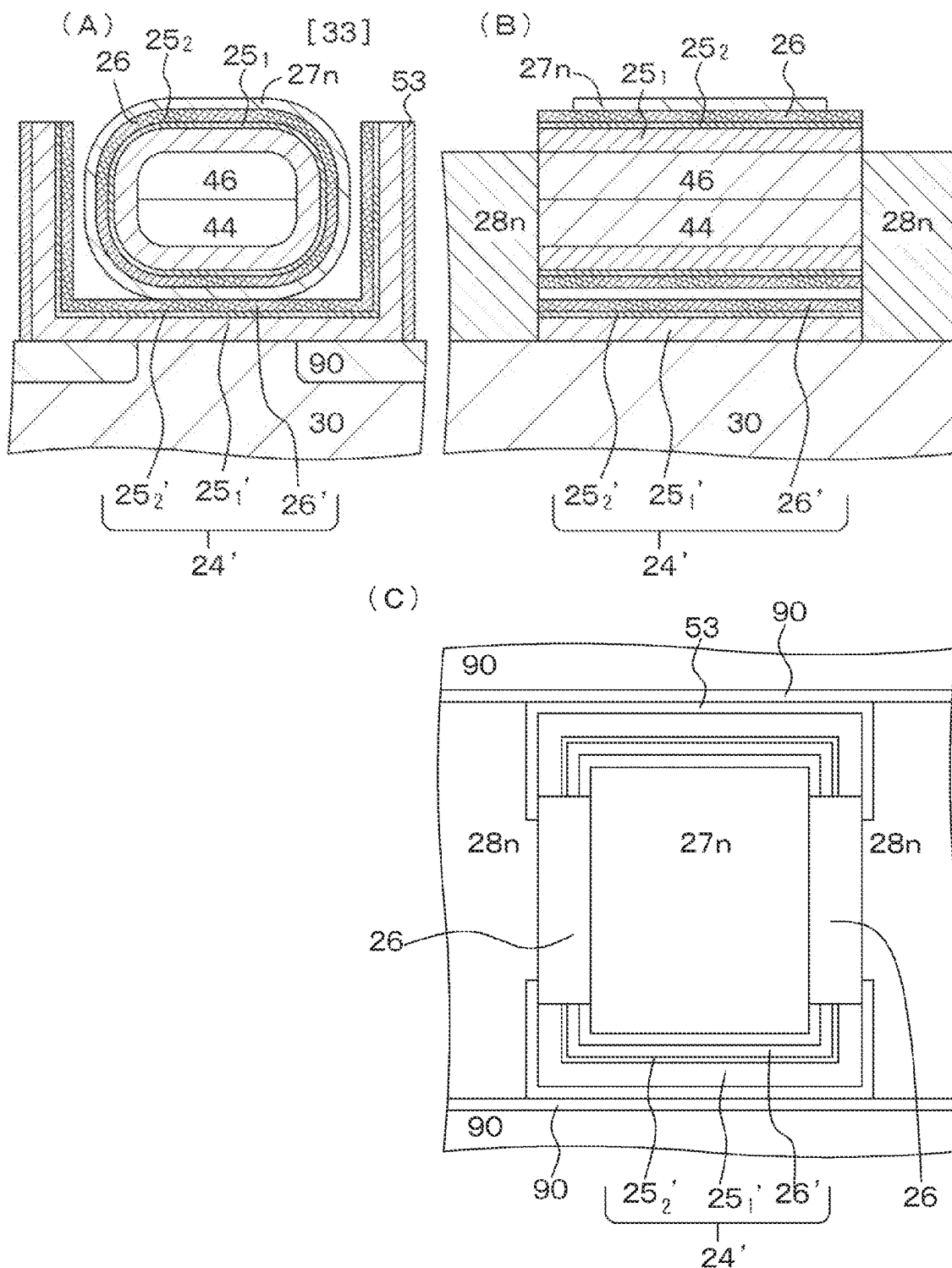


FIG. 43

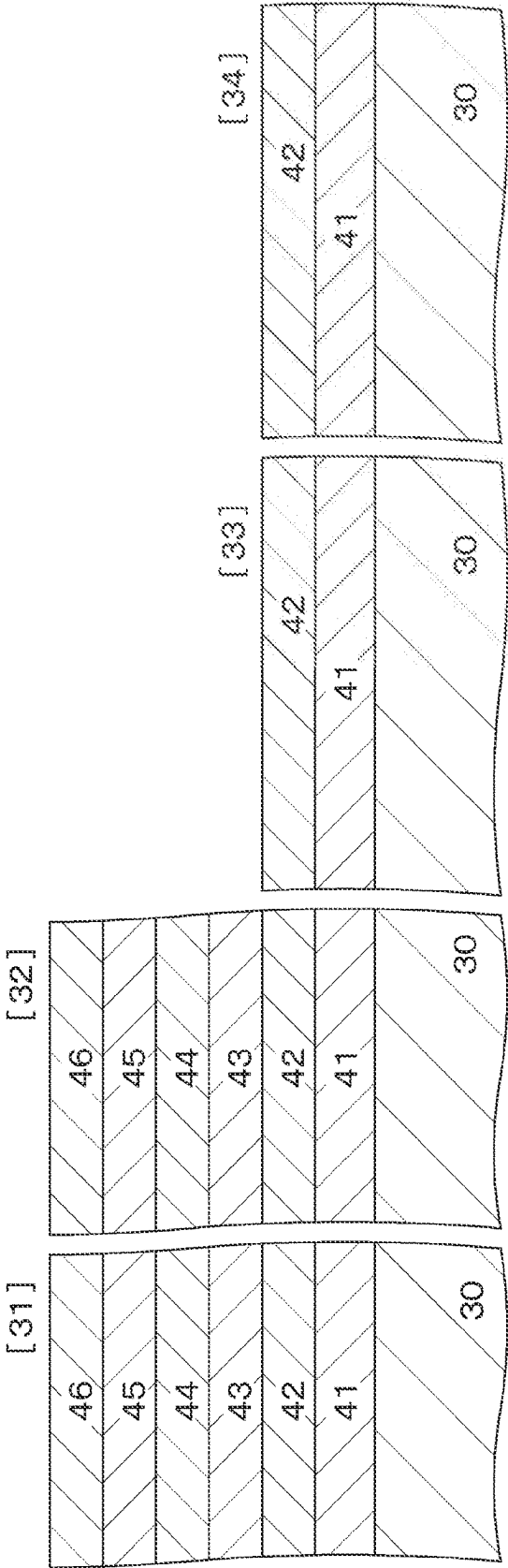


FIG. 44

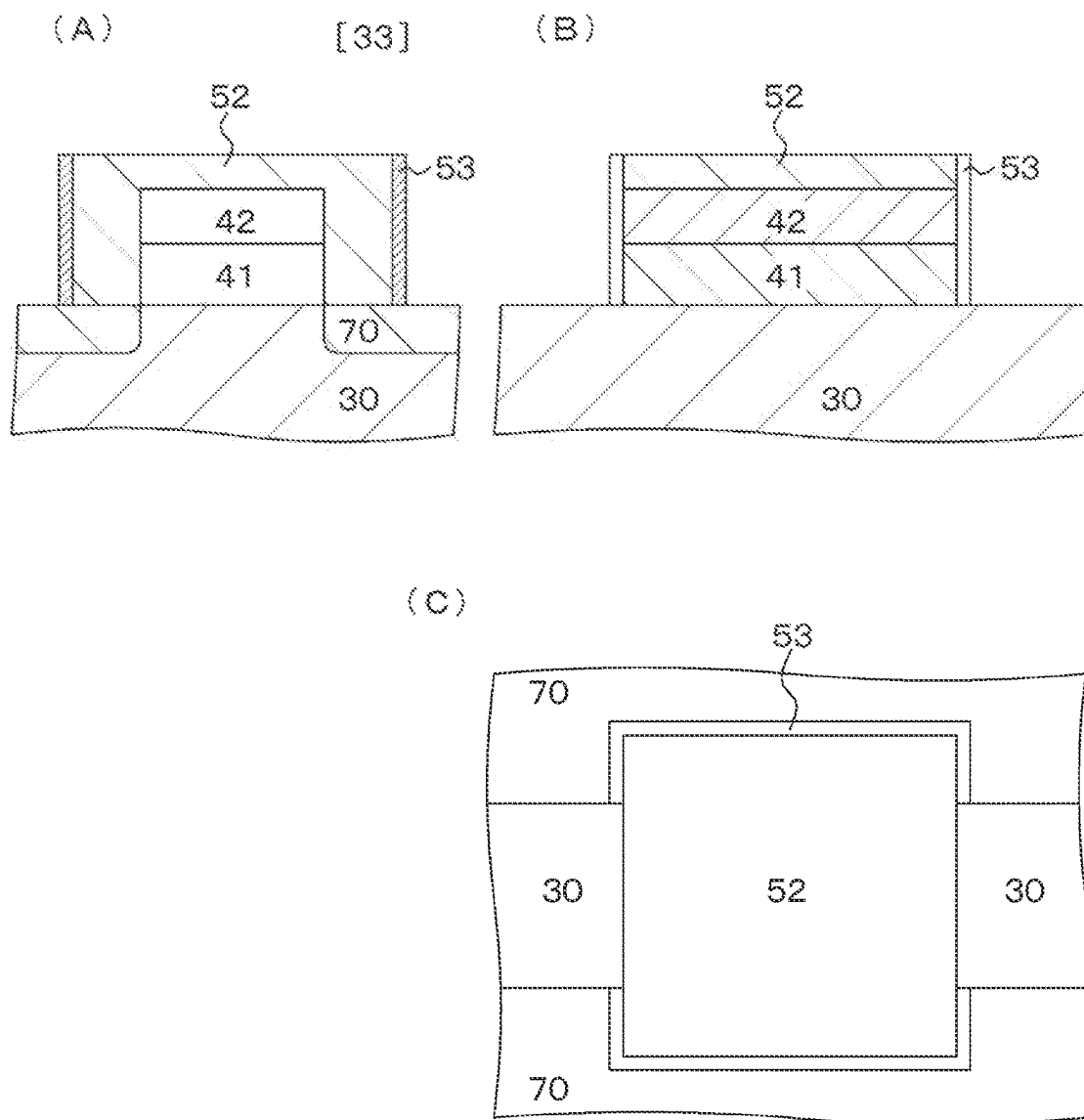


FIG. 45

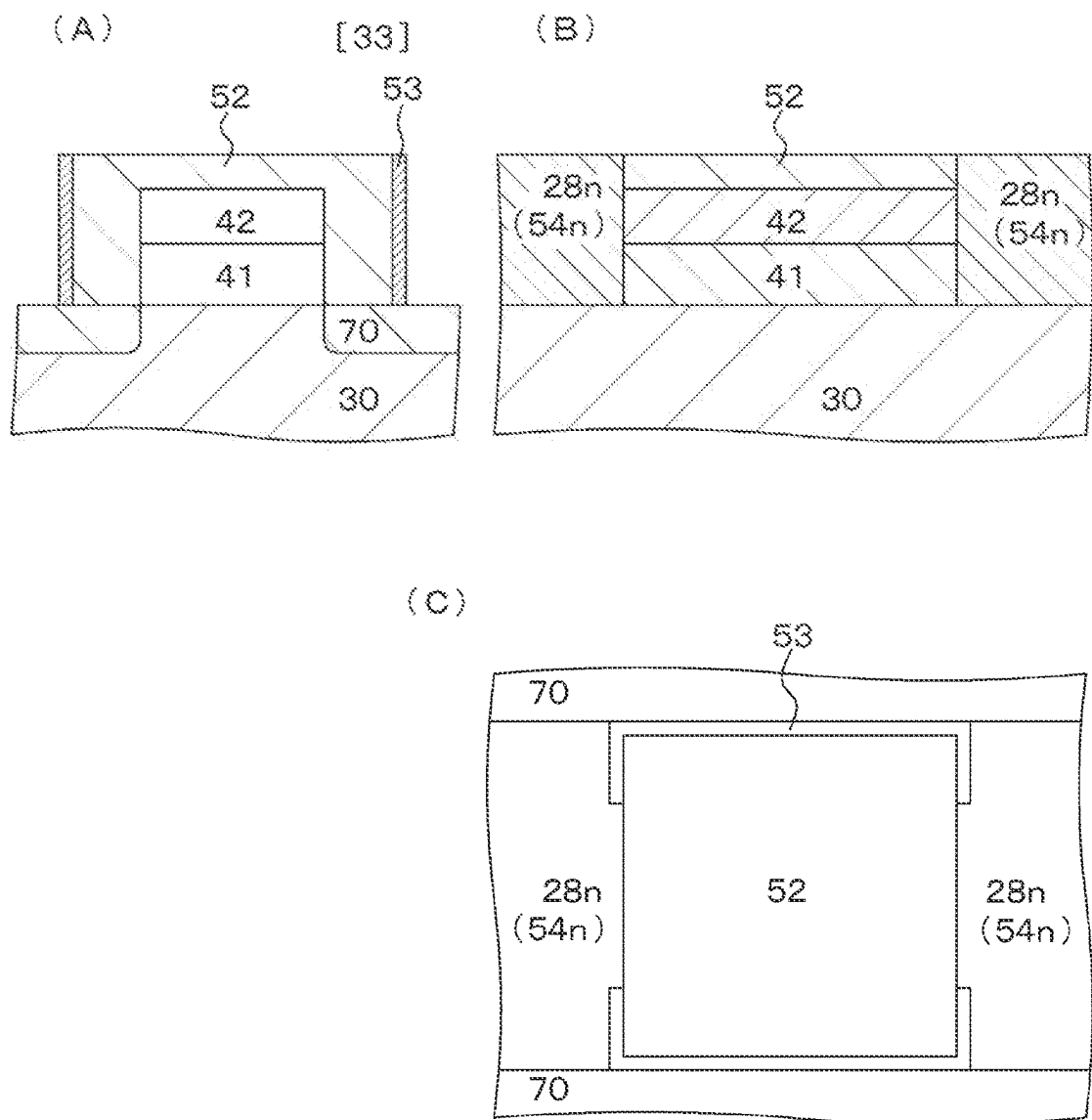


FIG. 46

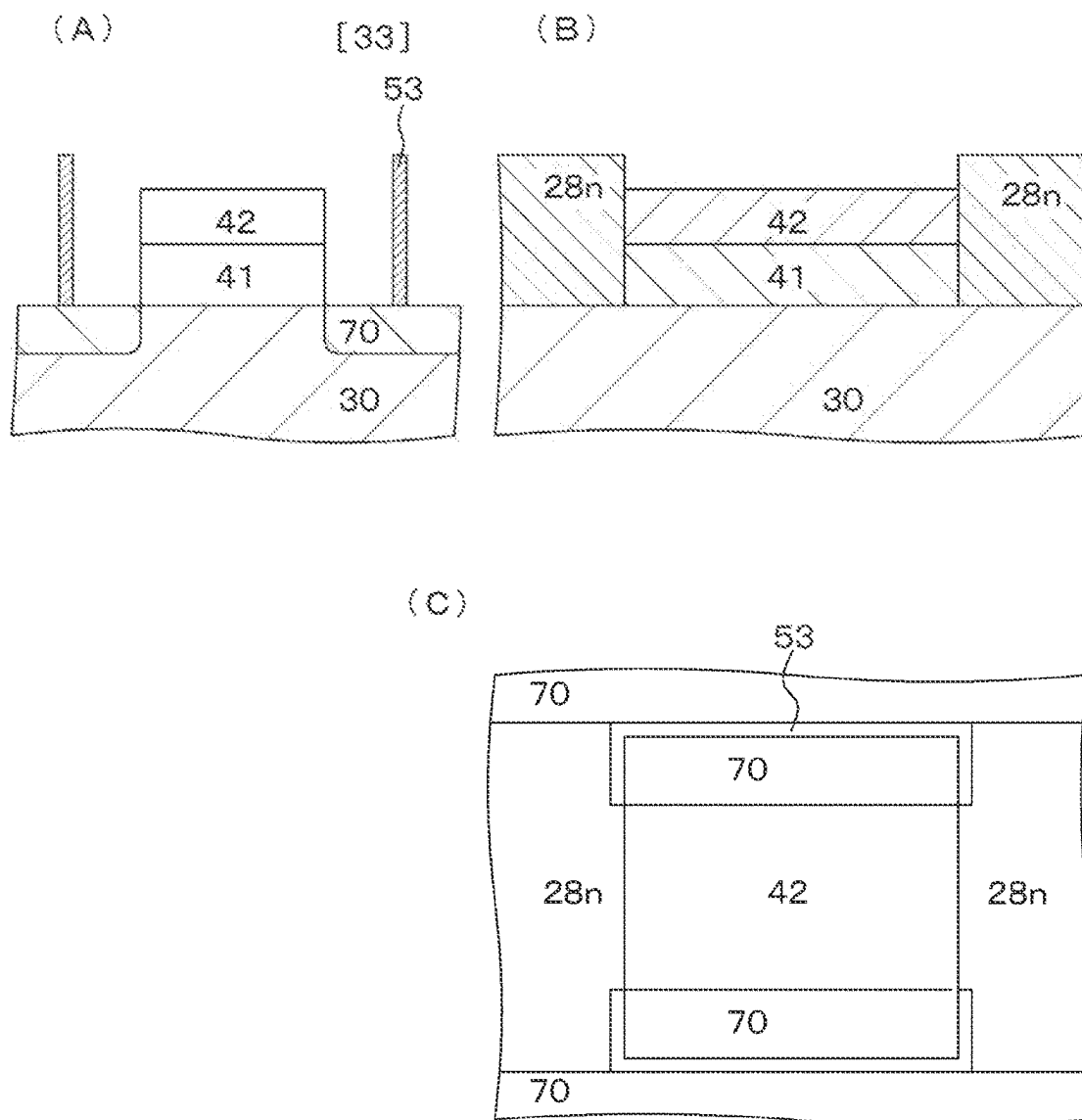


FIG. 47

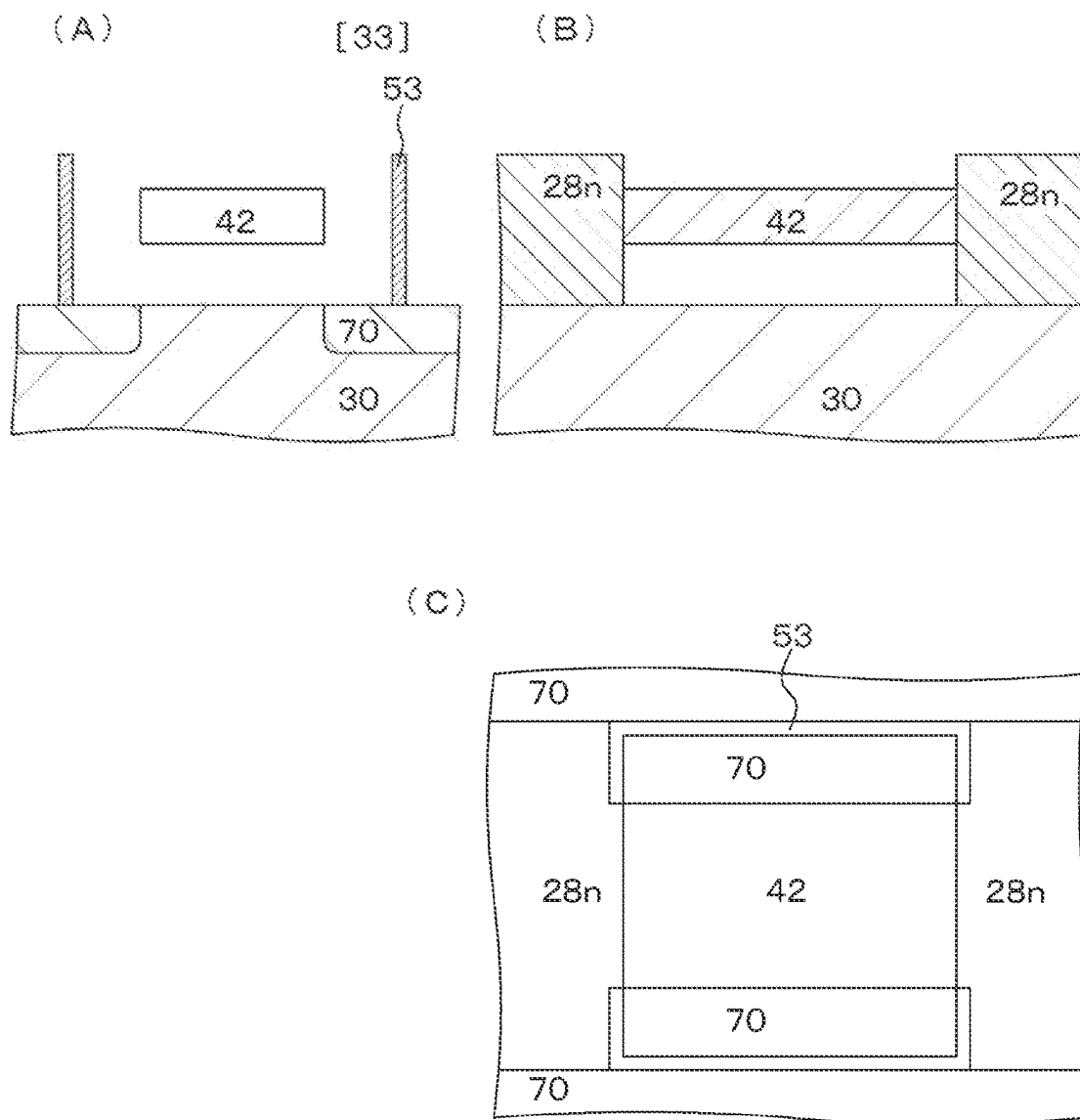


FIG. 48

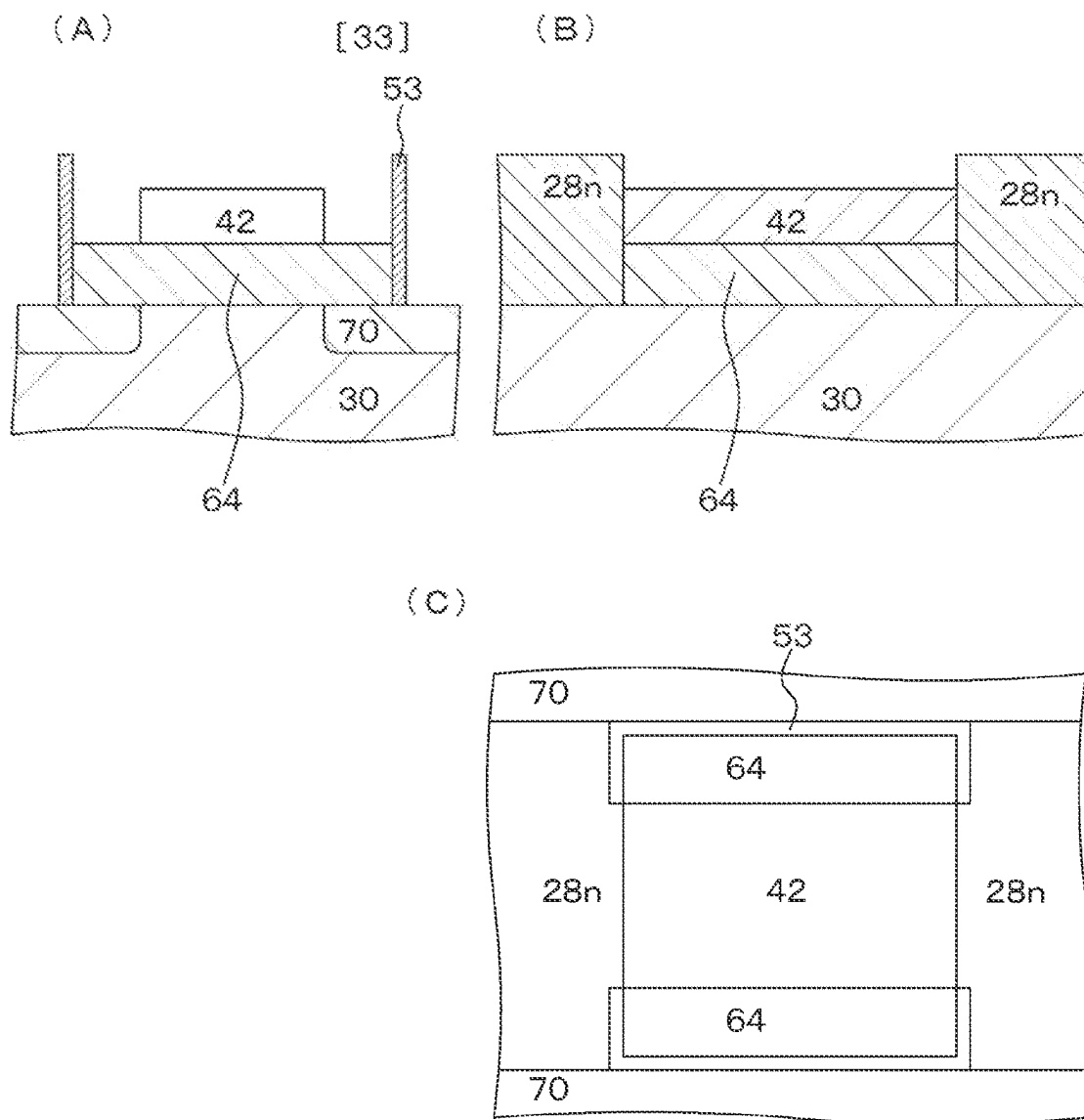


FIG. 49

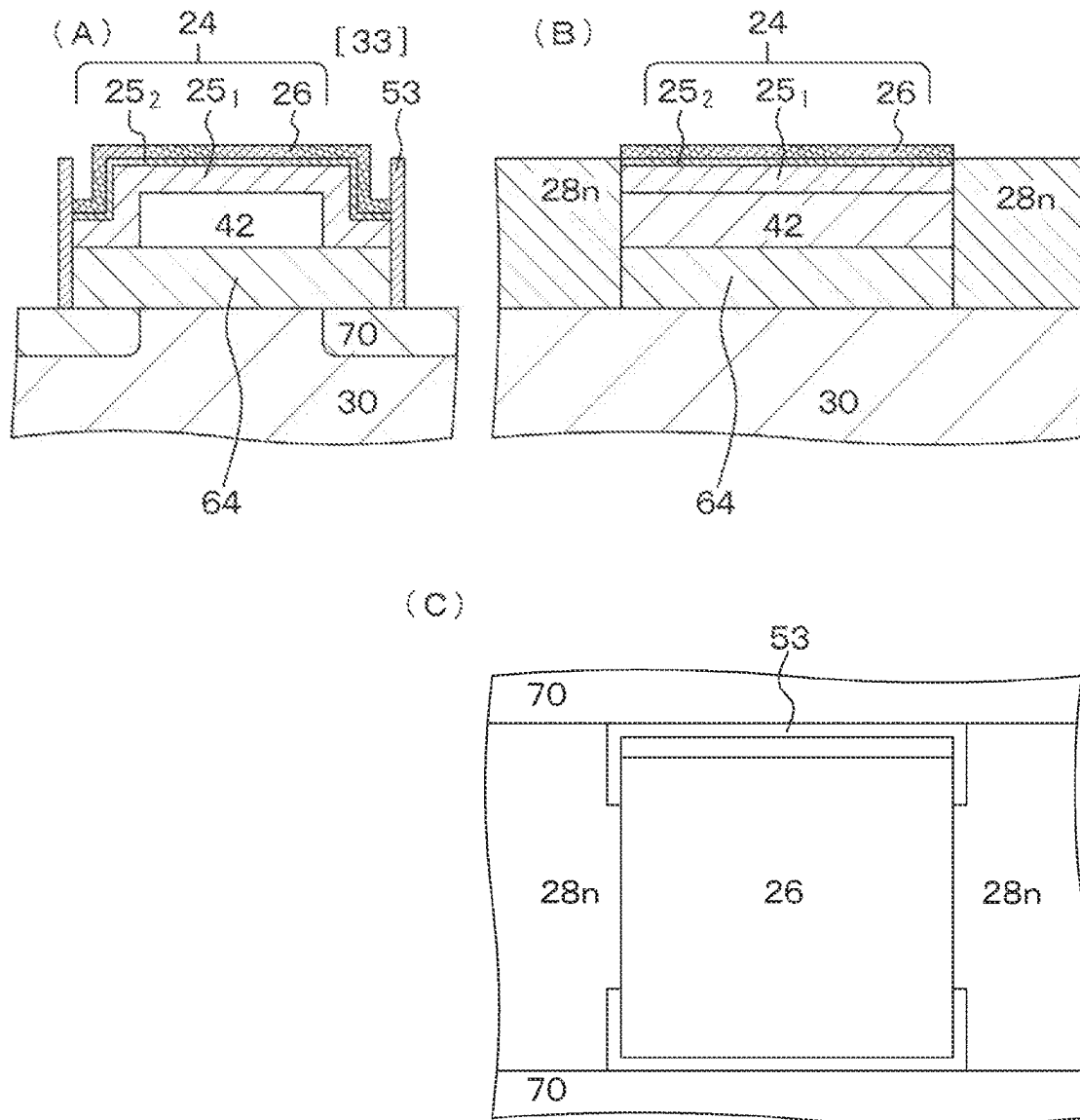
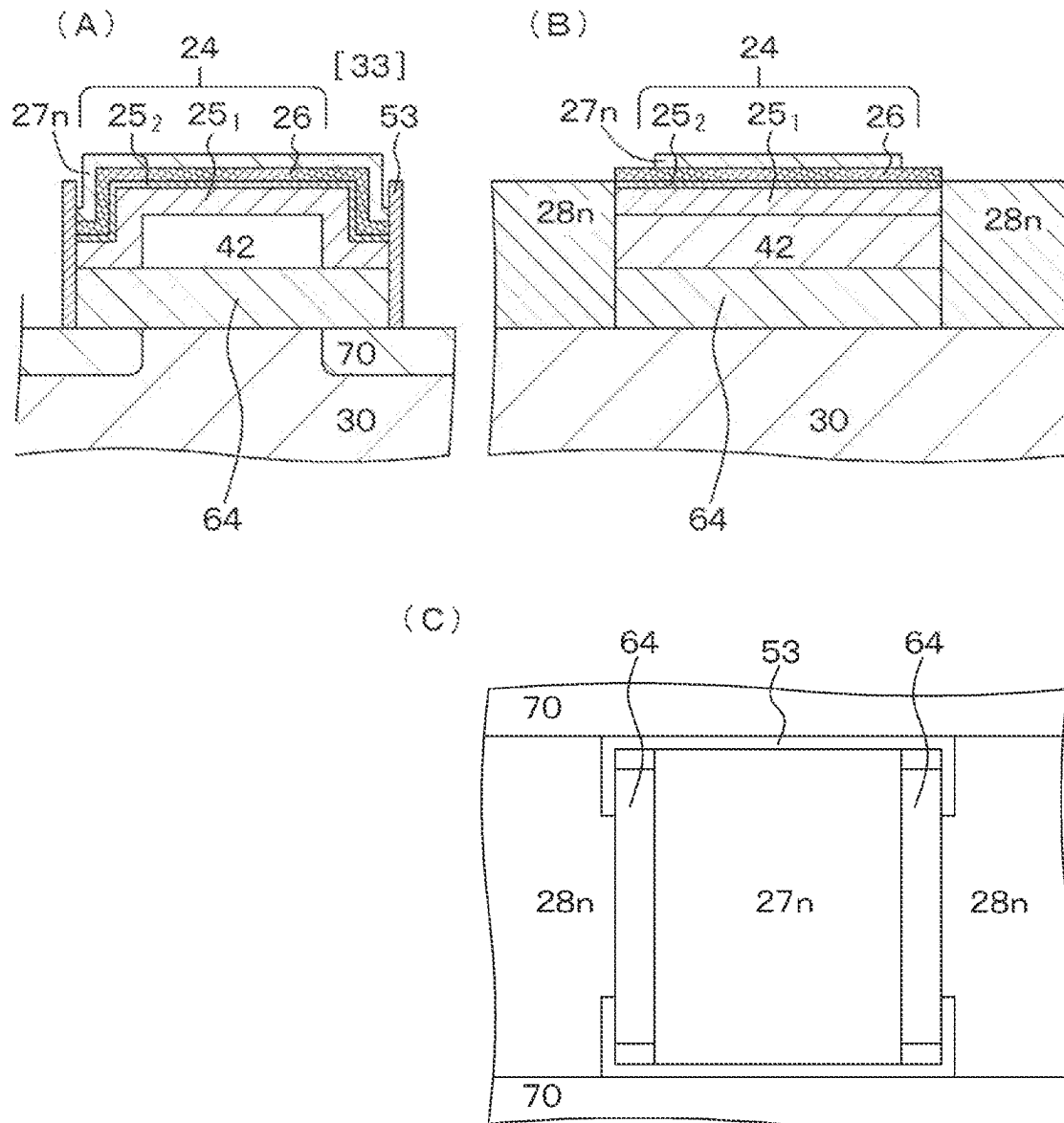


FIG. 50



1

**SEMICONDUCTOR DEVICE INCLUDING A
FIELD EFFECT TRANSISTOR WITH
NANOSTRUCTURE BASED LAMINATED
CHANNEL STRUCTURE AND A FIELD
EFFECT TRANSISTOR WITH A SINGLE
CHANNEL STRUCTURE**

**CROSS REFERENCE TO RELATED
APPLICATIONS**

This application is a continuation of U.S. patent application Ser. No. 17/943,337, filed Sep. 13, 2022, which is a continuation of U.S. patent application Ser. No. 17/261,226, filed Jan. 19, 2021, now U.S. Pat. No. 11,476,329, which is a national stage application under 35 U.S.C. 371 and claims the benefit of PCT Application No. PCT/JP2019/024075, having an international filing date of Jun. 18, 2019, which designated the United States, which PCT application claimed the benefit of Japanese Patent Application No. 2018-140217, filed Jul. 26, 2018, the entire disclosures of each of which are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to a semiconductor device, and specifically to a semiconductor device which includes a field effect transistor having a nanowire structure or a nanosheet structure.

BACKGROUND ART

Regarding scaling trends of advanced MOS transistors after 2012, a MOSFET having a bulk-planar structure was dominant until 20 nm generation. After 14 nm generation, however, a FET having a Fin structure (referred to as a “Fin-FET” for convenience) or a FET having an FD-SOI (Fully Depleted-Silicon On Insulator) structure (referred to as an “FD-SOI-FET” for convenience) has been thoroughly adopted. Meanwhile, a thickness of a silicon layer having a close relation with scaling of a gate length, i.e., a thickness of the Fin structure in a Fin-FET, and a thickness of a silicon layer in an FD-SOI-FET are important factors for size reduction of an FET. It is considered that a minimum thickness of the silicon layer is limited to 5 nm.

As a technology for eliminating this limitation on the thickness of the silicon layer constituting a channel forming region of a FET, a field effect transistor (referred to as a “nanowire-FET” for convenience) having a nanowire structure in a channel forming region has been studied (for example, see Japanese Patent Laid-open No. 2015-195405). The nanowire-FET has at least two nanowire structures. In addition, the nanowire-FET thus configured is driven in a range from 0.5 to 0.8 volts, for example.

On the other hand, a semiconductor device is often required to have not only the nanowire-FET, but also a field effect transistor driven in a range from 1.5 to 3.3 volts (referred to as a “second FET” for convenience), for example.

CITATION LIST

Patent Literature

[PTL 1]
Japanese Patent Laid-open No. 2015-195405

2

SUMMARY

Technical Problem

Meanwhile, the nanowire-FET and the second FET are simultaneously formed during manufacture of a typical semiconductor device. In this case, a space between nanowire structures of the nanowire-FET is so small that it is difficult to form a thick gate insulation film on the second FET. Moreover, there has been a strong demand for applying reverse bias to the second FET to control a threshold voltage V_{th} .

Accordingly, a first object of the present disclosure is to provide a semiconductor device which includes both a nanowire-FET and a second FET having a gate insulation film relatively thick with respect to the nanowire-FET. Moreover, a second object of the present disclosure is to provide a semiconductor device which includes both a nanowire-FET and a second FET having a configuration and a structure to which reverse bias is applicable.

Solution to Problem

A semiconductor device according to a first aspect of the present disclosure for achieving the above first object includes:

- a base,
- a first field effect transistor that includes at least two channel structure portions laminated, the channel structure portions each including a channel portion that has a nanowire structure or a nanosheet structure, a gate insulation film that surrounds the channel portion, and a gate electrode that surrounds at least a part of the gate insulation film, and
- a second field effect transistor that includes a channel forming layer, a gate insulation layer surrounding the channel forming layer, and a gate electrode surrounding at least a part of the gate insulation layer, in which the first field effect transistor and the second field effect transistor are provided above the base, the channel portions of the first field effect transistor are disposed apart from each other in a laminating direction of the channel structure portions, and assuming that each of a distance between the channel portions of the first field effect transistor is a distance L_1 and that a thickness of the gate insulation layer of the second field effect transistor is a thickness T_2 ,

$$T_2 \geq (L_1 / 2),$$

preferably,

$$T_2 \geq 1.1 \times (L_1 / 2),$$

more preferably,

$$T_2 \geq 1.2 \times (L_1 / 2),$$

is satisfied.

A semiconductor device according to a second aspect of the present disclosure for achieving the above second object includes

a base,
 a first field effect transistor that includes at least two channel structure portions laminated, the channel structure portions each including a channel portion that has a nanowire structure or a nanosheet structure, a gate insulation film that surrounds the channel portion, and a gate electrode that surrounds at least a part of the gate insulation film, and
 a second field effect transistor that includes a channel forming layer, a gate insulation layer formed on a top surface and a side surface of the channel forming layer, and a gate electrode formed on at least a top surface of the gate insulation layer, in which
 the first field effect transistor and the second field effect transistor are provided above the base,
 the channel portions of the first field effect transistor are disposed apart from each other in a laminating direction of the channel structure portions, and
 an insulation material layer is formed between a surface of the base and a bottom surface of the channel forming layer constituting the second field effect transistor.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a schematic partial cross-sectional diagram of a semiconductor device of Embodiment 1.

FIGS. 2A and 2B are schematic partial cross-sectional diagrams of a first field effect transistor included in the semiconductor device of Embodiment 1.

FIG. 3A is a diagram which depicts a schematic arrangement state of channel portions, a channel forming layer, and source/drain regions of the first field effect transistor included in the semiconductor device of the present disclosure, and FIG. 3B is a diagram which depicts a schematic arrangement of channel portions, a channel forming layer, and source/drain regions of a second field effect transistor included in the semiconductor device of the present disclosure.

FIG. 4 is a schematic partial cross-sectional diagram of a semiconductor device of a modification (Modification 1) of Embodiment 1.

FIG. 5 is a schematic partial cross-sectional diagram of a semiconductor device of a modification (Modification 4) of Embodiment 1.

FIG. 6 is a schematic partial cross-sectional diagram of a semiconductor device of Embodiment 2.

FIG. 7 is a schematic partial cross-sectional diagram of a semiconductor device of a modification of Embodiment 2.

FIGS. 8A, 8B, and 8C are schematic partial cross-sectional diagrams of a base and the like for explaining a manufacturing method of the semiconductor device of Embodiment 1, the diagrams being similar to the cross-sectional diagrams taken along the arrows A-A in FIG. 3A and the arrows B-B in FIG. 3B.

FIGS. 9A and 9B are schematic partial cross-sectional diagrams of the base and the like continuing from FIG. 8C, for explaining the manufacturing method of the semiconductor device of Embodiment 1, the diagrams being similar to the cross-sectional diagrams taken along the arrows A-A in FIG. 3A and the arrows B-B in FIG. 3B.

FIGS. 10A and 10B are schematic partial cross-sectional diagrams of the base and the like continuing from FIG. 9B, for explaining the manufacturing method of the semiconductor device of Embodiment 1, the diagrams being similar to the cross-sectional diagrams taken along the arrows A-A in FIG. 3A and the arrows B-B in FIG. 3B.

FIGS. 11A and 11B are schematic partial cross-sectional diagrams of the base and the like continuing from FIG. 10B, for explaining the manufacturing method of the semiconductor device of Embodiment 1, the diagrams being similar to the cross-sectional diagrams taken along the arrows A-A in FIG. 3A and the arrows B-B in FIG. 3B.

FIG. 12 is a schematic partial cross-sectional diagram of the base and the like continuing from FIG. 11B, for explaining the manufacturing method of the semiconductor device of Embodiment 1, the diagrams being similar to the cross-sectional diagrams taken along the arrows A-A in FIG. 3A and the arrows B-B in FIG. 3B.

FIG. 13 is a schematic partial cross-sectional diagram of the base and the like continuing from FIG. 12, for explaining the manufacturing method of the semiconductor device of Embodiment 1, the diagrams being similar to the cross-sectional diagrams taken along the arrows A-A in FIG. 3A and the arrows B-B in FIG. 3B.

FIG. 14 is a schematic partial cross-sectional diagram of the base and the like continuing from FIG. 13, for explaining the manufacturing method of the semiconductor device of Embodiment 1, the diagrams being similar to the cross-sectional diagrams taken along the arrows A-A in FIG. 3A and the arrows B-B in FIG. 3B.

FIG. 15 is a schematic partial cross-sectional diagram of the base and the like continuing from FIG. 14, for explaining the manufacturing method of the semiconductor device of Embodiment 1, the diagrams being similar to the cross-sectional diagrams taken along the arrows A-A in FIG. 3A and the arrows B-B in FIG. 3B.

FIG. 16 is a schematic partial cross-sectional diagram of the base and the like continuing from FIG. 15, for explaining the manufacturing method of the semiconductor device of Embodiment 1, the diagrams being similar to the cross-sectional diagrams taken along the arrows A-A in FIG. 3A and the arrows B-B in FIG. 3B.

FIG. 17 is a schematic partial cross-sectional diagram of the base and the like continuing from FIG. 16, for explaining the manufacturing method of the semiconductor device of Embodiment 1, the diagrams being similar to the cross-sectional diagrams taken along the arrows A-A in FIG. 3A and the arrows B-B in FIG. 3B.

FIG. 18 is a schematic partial cross-sectional diagram of the base and the like continuing from FIG. 17, for explaining the manufacturing method of the semiconductor device of Embodiment 1, the diagrams being similar to the cross-sectional diagrams taken along the arrows A-A in FIG. 3A and the arrows B-B in FIG. 3B.

FIG. 19 (A), (B), and (C) of FIG. 19 are schematic partial cross-sectional diagrams and a schematic partial plan diagram of the first field effect transistor included in the semiconductor device of Embodiment 1 in a midway of a manufacturing step of the semiconductor device of Embodiment 1.

FIG. 20 (A), (B), and (C) of FIG. 20 are schematic partial cross-sectional diagrams and a schematic partial plan diagram of the second field effect transistor included in the semiconductor device of Embodiment 1 in a midway of a manufacturing step of the semiconductor device of Embodiment 1.

FIG. 21 (A), (B), and (C) of FIG. 21 are schematic partial cross-sectional diagrams and a schematic partial plan diagram continuing from (A), (B), and (C) of FIG. 19 and depicting the first field effect transistor included in the semiconductor device of Embodiment 1 in a midway of a manufacturing step of the semiconductor device of Embodiment 1.

FIG. 31 (A), (B), and (C) of FIG. 31 are schematic partial cross-sectional diagrams and a schematic partial plan diagram continuing from (A), (B), and (C) of FIG. 27 and depicting the first field effect transistor included in the

FIG. 40 (A), (B), and (C) of FIG. 40 are schematic partial cross-sectional diagrams and a schematic partial plan diagram continuing from (A), (B), and (C) of FIG. 38 and depicting the second field effect transistor included in the semiconductor device of Embodiment 1 in a midway of a manufacturing step of the semiconductor device of Embodiment 1.

FIG. 41 (A), (B), and (C) of FIG. 41 are schematic partial cross-sectional diagrams and a schematic partial plan diagram continuing from (A), (B), and (C) of FIG. 39 and depicting the first field effect transistor included in the semiconductor device of Embodiment 1 in a midway of a manufacturing step of the semiconductor device of Embodiment 1.

FIG. 42 (A), (B), and (C) of FIG. 42 are schematic partial cross-sectional diagrams and a schematic partial plan diagram continuing from (A), (B), and (C) of FIG. 40 and depicting the second field effect transistor included in the semiconductor device of Embodiment 1 in a midway of a manufacturing step of the semiconductor device of Embodiment 1.

FIG. 43 is a schematic partial cross-sectional diagram of a base and the like for explaining a manufacturing method of a semiconductor device of Embodiment 2, the diagrams being similar to the cross-sectional diagrams taken along the arrows A-A in FIG. 3A and the arrows B-B in FIG. 3B.

FIG. 44 (A), (B), and (C) of FIG. 44 are schematic partial cross-sectional diagrams and a schematic partial plan diagram continuing from FIG. 43 and depicting the second field effect transistor included in the semiconductor device of Embodiment 2 in a midway of a manufacturing step of the semiconductor device of Embodiment 2.

FIGS. 45(A), (B), and (C) of FIG. 45 are schematic partial cross-sectional diagrams and a schematic partial plan diagram continuing from (A), (B), and (C) of FIG. 44 and depicting the second field effect transistor included in the semiconductor device of Embodiment 2 in a midway of a manufacturing step of the semiconductor device of Embodiment 2.

FIG. 46 (A), (B), and (C) of FIG. 46 are schematic partial cross-sectional diagrams and a schematic partial plan diagram continuing from (A), (B), and (C) of FIG. 45 and depicting the second field effect transistor included in the semiconductor device of Embodiment 2 in a midway of a manufacturing step of the semiconductor device of Embodiment 2.

FIG. 47 (A), (B), and (C) of FIG. 47 are schematic partial cross-sectional diagrams and a schematic partial plan diagram continuing from (A), (B), and (C) of FIG. 46 and depicting the second field effect transistor included in the semiconductor device of Embodiment 2 in a midway of a manufacturing step of the semiconductor device of Embodiment 2.

FIG. 48 (A), (B), and (C) of FIG. 48 are schematic partial cross-sectional diagrams and a schematic partial plan diagram continuing from (A), (B), and (C) of FIG. 47 and depicting the second field effect transistor included in the semiconductor device of Embodiment 2 in a midway of a manufacturing step of the semiconductor device of Embodiment 2.

FIG. 49 (A), (B), and (C) of FIG. 49 are schematic partial cross-sectional diagrams and a schematic partial plan diagram continuing from (A), (B), and (C) of FIG. 48 and depicting the second field effect transistor included in the semiconductor device of Embodiment 2 in a midway of a manufacturing step of the semiconductor device of Embodiment 2.

FIG. 50 (A), (B), and (C) of FIG. 50 are schematic partial cross-sectional diagrams and a schematic partial plan diagram continuing from (A), (B), and (C) of FIG. 49 and depicting the second field effect transistor included in the semiconductor device of Embodiment 2 in a midway of a manufacturing step of the semiconductor device of Embodiment 2.

DESCRIPTION OF EMBODIMENTS

The present disclosure will hereinafter be described with reference to the drawings while presenting embodiments. However, the present disclosure is not limited to the embodiments presented herein, and various numerical values and materials included in the embodiments are given only by way of example. Note that the description will proceed in the following order.

1. General description of semiconductor device according to first and second aspects of present disclosure
2. Embodiment 1 (semiconductor device according to first aspect of present disclosure)
3. Embodiment 2 (semiconductor device according to second aspect of present disclosure)
4. Others

General Description of Semiconductor Device According to First and Second Aspects of Present Disclosure

In a semiconductor device of a first aspect of the present disclosure, assuming that a distance between a surface of a base and a channel forming layer of a second field effect transistor is a distance L_2 , a mode satisfying the following relation may be adopted.

$$L_2 \geq L_1$$

$$L_2 \geq T_2$$

In this case, a preferable mode satisfying the following relation may be adopted.

$$L_2 \geq 2 \times L_1$$

In the semiconductor device of the first aspect of the present disclosure including the above preferable mode, assuming that a thickness of a gate insulation film of a first field effect transistor is T_1 , a mode satisfying the following relation may be adopted.

$$T_2 \geq 2 \times T_1$$

A preferable mode satisfying the following relation may be adopted.

$$T_2 \geq 3 \times T_1$$

By adopting such mode, the second field effect transistor having a gate insulation film relatively thick with respect to the first field effect transistor can be obtained reliably.

Moreover, in the semiconductor device of the first aspect of the present disclosure including the respective preferable modes described above, assuming that a thickness of a channel portion is T_{1-CH} and that a thickness of a channel forming layer is T_{2-CH} , a mode satisfying the following relation may be adopted.

$$T_{2-CH} \geq 2 \times T_{1-CH}$$

A preferable mode satisfying the following relation may be adopted.

$$T_{2-CH} \geq 3 \times T_{1-CH}$$

By adopting such mode, lowering of resistance of the channel forming layer of the second field effect transistor, raising of transconductance g_m , and reduction of parasitic capacitance are achievable.

Further, in the semiconductor device of the first aspect of the present disclosure including the respective preferable modes described above, at least a part of a channel portion in a lowermost layer constituting the first field effect transistor is surrounded by a first gate electrode, and a channel portion other than this channel portion is surrounded by a second gate electrode in an adoptable mode. In a case where the channel portion in the lowermost layer constituting the first field effect transistor is surrounded by the first gate electrode, an insulation layer (referred to as a “first insulation layer” in some cases for convenience) is formed between the first gate electrode and the surface of the base.

A structure formed such that at least a part of the channel portion in the lowermost layer constituting the first field effect transistor is surrounded by the first gate electrode and that the channel portion other than this channel portion is surrounded by the second gate electrode is applicable to a first field effect transistor of a semiconductor device according to a second aspect of the present disclosure.

As described above, the gate insulation film is formed between the first gate electrode and the channel portion of the first field effect transistors and between the second gate electrode and the channel portion of the first field effect transistor. Specifically, in the first field effect transistor, a gate insulation film (i.e., a gate insulation film formed on the outer peripheral portion of the channel portion) surrounding the channel portion located below and a gate insulation film (i.e., a gate insulation film formed on the outer peripheral portion of the channel portion) surrounding the channel portion located above are formed between the channel portions. Moreover, a gate electrode is provided between the respective gate insulation films. In such manner, a space between the channel portions is filled with the gate insulation film and the gate electrode. The total height of the channel portion is the sum total of diameters of materials (e.g., Si, SiGe, Ge, and InGaAs) constituting a nanowire structure forming the channel portion except for the gate insulation film and the gate electrode or the sum total of thicknesses of materials (e.g., Si, SiGe, Ge, and InGaAs) constituting a nanosheet structure except for the gate insulation film and the gate electrode. The foregoing discussion is applicable to the first field effect transistor of the semiconductor device according to the second aspect of the present disclosure.

In the following description, a gate electrode constituting the second field effect transistor will be referred to as a “third gate electrode” in some cases for convenience. In addition, in the second field effect transistor of the semiconductor device of the first aspect of the present disclosure, the third gate electrode surrounds at least a part of the gate insulation layer. In this case, the third gate electrode surrounds the gate insulation layer in one mode or surrounds a part of the gate

insulation layer in another mode. In the former case, the third gate electrode is formed between the surface of the base and the gate insulation layer via an insulation layer (referred to as a “second insulation layer” in some cases for convenience). A thickness of the second insulation layer is larger than a thickness of the first insulation layer described above. On the other hand, in the latter case, the third gate electrode is formed on a top surface and a side surface of the channel forming layer via the gate insulation layer, but not between the surface of the base and the gate insulation layer in a state of $L_2=T_2$.

Further, in the semiconductor device according to the first aspect of the present disclosure including the respective preferable modes and configurations described above or the semiconductor device according to the second aspect of the present disclosure, the following mode may be adopted.

The second field effect transistor includes an n-channel type field effect transistor and a p-channel type field effect transistor.

A channel forming layer of the n-channel type field effect transistor is made of silicon (Si).

A channel forming layer of the p-channel type field effect transistor is made of silicon (Si) or silicon-germanium (SiGe).

Further, in the semiconductor device according to the first aspect of the present disclosure including the respective preferable modes and configurations described above or the semiconductor device according to the second aspect of the present disclosure including the preferable mode described above, the following mode may be adopted.

The first field effect transistor includes an n-channel type field effect transistor and a p-channel type field effect transistor.

A channel portion of the n-channel type field effect transistor is made of silicon (Si).

A channel portion of the p-channel type field effect transistor is made of silicon-germanium (SiGe), germanium (Ge), or InGaAs.

However, this mode is not required to be adopted, and the following mode may be adopted.

The channel portion of the n-channel type field effect transistor is made of silicon-germanium (SiGe).

The channel portion of the p-channel type field effect transistor is made of silicon (Si), germanium (Ge), or InGaAs.

Another mode is adoptable. Specifically, the channel portion of the n-channel type field effect transistor is made of germanium (Ge).

The channel portion of the p-channel type field effect transistor is made of silicon (Si), silicon-germanium (SiGe), or InGaAs.

A further mode is adoptable. Specifically, the channel portion of the n-channel type field effect transistor is made of InGaAs.

The channel portion of the p-channel type field effect transistor is made of silicon (Si), silicon-germanium (SiGe), or germanium (Ge).

In the semiconductor device according to the second aspect of the present disclosure including the respective preferable modes described above, the following mode may be adopted. Reverse bias is applied to the base at a portion facing the bottom surface of the channel forming layer via the insulation material layer.

According to the semiconductor device of the second aspect of the present disclosure including the respective preferable modes described above, assuming that a thickness of the channel portion is T_{1-CH} and that a thickness of the

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insulation material layer is T_{ins} , a mode satisfying the following relation may be adopted.

$$0.2 \leq T_{1-CH}/T_{ins} \leq 2$$

preferably,

$$0.5 \leq T_{1-CH}/T_{ins} \leq 1$$

Further, in the semiconductor device according to the second aspect of the present disclosure including the preferable modes described above, the following configuration may be adopted. At least one semiconductor layer is formed between the channel forming layer and the insulation material layer in the second field effect transistor. In addition, in this case, adoptable is a configuration where an interlayer insulation layer is formed between the channel forming layer and the semiconductor layer and between the semiconductor layers. In this manner, reverse bias is applicable by applying appropriate potential to the semiconductor layer. Further, in these cases, the semiconductor layer may have a conductivity type opposite to a conductivity type of the channel forming layer. According to a MOSFET having a GAA (Gate-All-Around) structure where an entire outer periphery of a channel forming region is surrounded by a gate electrode, a threshold voltage V_{th} is normally determined only by a work function of a material constituting the gate electrode. However, the threshold voltage V_{th} is controllable by applying reverse bias, and thus, a desired threshold voltage V_{th} optimum for a circuit is allowed to be set. For example, a circuit speed can be raised by lowering the threshold voltage V_{th} , and a low leak current can be achieved by increasing the threshold voltage V_{th} .

According to the semiconductor device of the first and second aspects of the present disclosure including the preferred modes and configurations described above (hereinafter collectively referred to as the “semiconductor device of the present disclosure” in some cases), the following configuration may be adopted. The first field effect transistor is a low withstand voltage field effect transistor, while the second field effect transistor is a high withstand voltage field effect transistor. In addition, in this case, the following configuration may be adopted. Voltage applied to the gate electrode of the first field effect transistor is in a range from 0.5 to 0.8 volts, for example, while voltage applied to the gate electrode of the second field effect transistor is in a range from 1.5 to 3.3 volts, for example. Note here that whether a field effect transistor is suited for a low withstand voltage field effect transistor or for a high withstand voltage field effect transistor is greatly dependent on the sum total of cross-sectional areas of a channel portion and a channel forming layer and on each thickness of a gate insulation film and a gate insulation layer.

Whether the first field effect transistor is of an n-channel type or of a p-channel type is determined by a work function of a material constituting the gate electrode, for example. Whether the second field effect transistor is of an n-channel type or of a p-channel type is also determined by a work function of a material constituting the gate electrode, for example.

In a case where the channel portion or the channel forming layer is made of Si in the semiconductor device of the present disclosure, the gate electrode is made of such

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material as TiN, TaN, Al, TiAl, and W to form the respective field effect transistors of the n-channel type. On the other hand, in a case where the channel portion or the channel forming layer is made of SiGe, the gate electrode is made of material such as TiN and W to form the respective field effect transistors of the p-channel type.

In addition, the gate insulation film and the gate insulation layer may be made of such material as SiON and SiO₂ or such high dielectric constant material (what is generally called a High-k material) as HfO₂, HfAlON, and Y₂O₃.

The base included in the semiconductor device of the present disclosure may be constituted by a silicon semiconductor substrate or an SOI substrate, for example. The channel portion and the channel forming layer preferably have a crystalline nature, but may have a polycrystalline configuration, or may have an amorphous configuration depending on cases. The semiconductor layer may be made of the material constituting the channel portion or the channel forming layer, specifically, silicon (Si), silicon-germanium (SiGe), germanium (Ge), and InGaAs. The channel portion, the channel forming layer, and the semiconductor layer may be formed by such a forming method as epitaxial CVD, plasma CVD, and atomic layer CVD.

According to the semiconductor device of the present disclosure, it is sufficient if the number of the channel structure portions provided in a thickness direction of the first field effect transistor is two or more. In addition, it is sufficient if the number of the channel structure portions provided in the direction perpendicular to the thickness direction of the first field effect transistor is one or two or more. The nanowire structure constituting the semiconductor device of the present disclosure is formed such that both ends of a wire structure that is made of Si, SiGe or the like, for example, and that has a diameter ranging from 5 to 10 nm, for example, are supported by source/drain regions constituting the first field effect transistor. In addition, the nanosheet structure constituting the semiconductor device of the present disclosure is formed such that both ends of a material that has a substantially rectangular cross-sectional shape, is made of Si, SiGe or the like, for example, and has a size of a width by a thickness of (10 to 50 nm) by (5 to 10 nm), for example, are supported by source/drain regions constituting the first field effect transistor. Whether the nanowire structure is adopted or the nanosheet structure is adopted is dependent on a thickness and a width of a material constituting the nanowire or nanosheet structure. In addition, the channel forming layer constituting the second field effect transistor is supported by a source/drain region constituting the second field effect transistor.

An arrangement of the first field effect transistor and the second field effect transistor in the semiconductor device is dependent on required specifications of the semiconductor device, and is thus difficult to specify here. Arrangement examples include a configuration which has a first field effect transistor constituting such a digital circuit as a logic circuit, a SRAM circuit, and a CMOS circuit, for example, and a second field effect transistor constituting a transistor for transmitting and receiving signals or the like to and from the outside; a configuration which has a second field effect transistor for controlling an imaging element (light receiving element) of an imaging device such as a transistor constituting an analog-digital converter and a first field effect transistor constituting a logic circuit controlling the imaging device or a driving circuit of the imaging element (light receiving element) constituting the imaging device; and a configuration which has a first field effect transistor constituting a CPU, a GPU, or the like and a second field effect

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transistor constituting a transistor for transmitting and receiving signals or the like to and from the outside. However, other configurations may be adopted.

Embodiment 1

Embodiment 1 relates to a semiconductor device according to a first aspect of the present disclosure.

FIGS. 1, 2A, and 2B each depict a schematic partial cross-sectional diagram of a semiconductor device according to Embodiment 1, FIG. 3A depicts a schematic arrangement state of channel portions, a channel forming layer, and source/drain regions of a first field effect transistor in the semiconductor device of Embodiment 1, and FIG. 3B depicts a schematic arrangement of channel portions, a channel forming layer, and source/drain regions of a second field effect transistor in the semiconductor device of Embodiment 1. FIG. 1 is a schematic partial cross-sectional diagram taken along arrows A-A in FIG. 3A and arrows B-B in FIG. 3B, while FIGS. 2A and 2B are schematic partial cross-sectional diagrams taken along the arrows A-A in FIG. 3A. Note that FIG. 1 includes a schematic partial cross-sectional diagram of one channel structure portion in an n-channel type first field effect transistor, a schematic partial cross-sectional diagram of one channel structure portion in a p-channel type first field effect transistor, and schematic partial cross-sectional diagrams of an n-channel type second field effect transistor and a p-channel type second field effect transistor. In addition, FIG. 2A is a schematic partial cross-sectional diagram of the n-channel type first field effect transistor, while FIG. 2B is a schematic partial cross-sectional diagram of the p-channel type first field effect transistor. FIG. 1 is a cross-sectional diagram in which hatching lines are omitted.

The semiconductor device of Embodiment 1 includes a base 30, first field effect transistors 10n and 10p where at least two channel structure portions 11n and two channel structure portions 11p (three for each in a laminating direction of the channel structure portions 11n and 11p in the depicted example) are laminated, each of the channel structure portions 11n and each of the channel structure portions 11p including a channel portion 13n and a channel portion 13p, respectively, the channel portion 13n and the channel portion 13p each having a nanowire structure or a nanosheet structure (having a nanowire structure 12n and a nanowire structure 12p, respectively, in the depicted example), a gate insulation film 14 surrounding the channel portion 13n and a gate insulation film 14 surrounding the channel portion 13p, respectively, and a gate electrode 17n and a gate electrode 17p, respectively, the gate electrode 17n and the gate electrode 17p each surrounding at least a part of the gate insulation film 14, and second field effect transistors 20n and 20p which include a channel forming layer 23n and a channel forming layer 23p, respectively, a gate insulation layer 24 surrounding the channel forming layer 23n and a gate insulation layer 24v surrounding the channel forming layer 23p, respectively, and a gate electrode 27n and a gate electrode 27p, respectively, each surrounding at least a part of the gate insulation layer 24 (surrounding the gate insulation layer 24 in the depicted figure).

The first field effect transistors 10n and 10p and the second field effect transistors 20n and 20p are provided above the base 30.

The channel portions 13n of the first field effect transistors 10n are disposed apart from each other in the laminating direction of the channel structure portions 11n, and the channel portions 13p of the first field effect transistors 10p

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are disposed apart from each other in the laminating direction of the channel structure portions 11p.

In addition, assuming that each of a distance between the channel portions 13n of the first field effect transistors 10n and a distance between the channel portions 13p of the first field effect transistor 10p is a distance L_1 and that a thickness of the gate insulation layer 24 of each of the second field effect transistors 20n and 20p is a distance T_2 , the following relation is satisfied.

$$T_2 \geq (L_1/2)$$

preferably,

$$T_2 \geq 1.1 \times (L_1/2)$$

more preferably,

$$T_2 \geq 1.2 \times (L_1/2)$$

According to an example depicted in FIGS. 2A and 2B, the three channel structure portions 11n and the three channel structure portions 11p are provided in a width direction. Meanwhile, FIG. 1 depicts one of the channel structure portions 11n and one of the channel structure portion 11p. FIG. 1 depicts a cross-sectional shape of the one channel structure portion of the first field effect transistor. Accordingly, FIG. 1 is different from each of FIGS. 2A and 2B. This is also applicable to Embodiment 2 described below.

In addition, assuming that each of a distance between a surface of the base 30 and the channel forming layer 23n of the second field effect transistor 20n and a distance between the surface of the base 30 and the channel forming layer 23p of the second field effect transistor 20p is a distance L_2 , the following relations are satisfied.

$$L_2 \geq L_1$$

$$L_2 \geq T_2$$

preferably,

$$L_2 \geq 2 \times L_1$$

Moreover, assuming that a thickness of the gate insulation film 14 of each of the first field effect transistors 10n and 10p is T_1 , the following relation is satisfied.

$$T_2 \geq 2 \times T_1$$

preferably,

$$T_2 \geq 3 \times T_1$$

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Further, assuming that a thickness of each of the channel portions **13n** and **13p** is T_{1-CH} and that a thickness of the channel forming layers **23n** and **23p** is T_{2-CH} , the following relation is satisfied.

$$T_{2-CH} \geq 2 \times T_{1-CH}$$

preferably,

$$T_{2-CH} \geq 3 \times T_{1-CH}$$

Specifically, the following values are set.

$$T_2/(L_1/2) = 1.5$$

$$L_2/L_1 = 2.0$$

$$L_2/T_2 = 2.7$$

$$T_2/T_1 = 3.0$$

$$T_{2-CH}/T_{1-CH} = 2.0$$

However, these values are not required to be set. Here, the following values are designated.

$$T_{1-CH} = 8 \text{ nm}$$

$$T_{2-CH} = 16 \text{ nm}$$

$$T_1 = 2 \text{ nm}$$

$$T_2 = 6 \text{ nm}$$

$$L_1 = 8 \text{ nm}$$

$$L_2 = 16 \text{ nm}$$

Moreover, at least a part of a channel portion in a lowermost layer constituting the first field effect transistor is surrounded by a first gate electrode, while channel portions other than this channel portion are each surrounded by a second gate electrode. Specifically, in the depicted example, the channel portion **13n** in the lowermost layer constituting the first field effect transistor **10n** is surrounded by a gate electrode **17n**, while the channel portions **13n** other than this channel portion (channel portions **13n** in the second layer and the uppermost layer) are also surrounded by the gate electrode **17n**. The first field effect transistors **10n** and **10p** included in a semiconductor device of Embodiment 2 describe below may be configured similarly. On the other hand, a part of a channel portion **13p** in a lowermost layer constituting the first field effect transistor **10p** is surrounded by a gate electrode **17p**, while the channel portions **13p** other than this channel portion (channel portions **13p** in the second layer and the uppermost layer) are surrounded by the gate electrode **17p**. Furthermore, each of gate electrodes (third gate electrodes) **27n** and **27p** constituting the second field effect transistor **20n** and **20p**, respectively, surrounds at least a part of the gate insulation layer **24** (in the depicted example, an entire outer peripheral portion of the gate insulation layer **24**).

The channel portion **13n** in the lowermost layer constituting the first field effect transistor **10n** is surrounded by the

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first gate electrode **17n**, and a first insulation layer **14'** is formed between the first gate electrode **17n** and the surface of the base **30**. Moreover, gate electrodes **27n** and **27p** are provided between the surface of the base **30** and the gate insulation layers **24** constituting the second field effect transistors **20n** and **20p**, respectively, via second insulation layers **24'**. A thickness of each of the second insulation layers **24'** is larger than a thickness of each of the first insulation layers **14'**. For example, the following relation may be adopted as a relation between a thickness T_2' of the second insulation layer **24'** and a thickness T_1' of the first insulation layer **14'**

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$$T_2'/T_1' \geq 3$$

The gate insulation film **14** is formed between the channel portion **13n** of the first field effect transistor **10n** and the first gate electrode **17n** and between the channel portion **13p** of the first field effect transistor **10p** and the first gate electrode **17p**, and further between the channel portion **13n** of the first field effect transistor **10n** and the second gate electrode **17n** and between the channel portion **13p** of the first field effect transistor **10p** and the second gate electrode **17p**. Specifically, in the first field effect transistors **10n** and **10p**, the gate insulation films (i.e., gate insulation films formed on the outer peripheral portions of the channel portions **13n** and **13p**) **14** surrounding the channel portions located below and the gate insulation films (i.e., gate insulation films formed on the outer peripheral portions of the channel portions) **14** surrounding the channel portions **13n** and **13p** located above are formed between the channel portions **13n** and between the channel portion **13p**, respectively. Moreover, the gate electrodes **17n** and **17p** are provided between the respective gate insulation films **14**. In such manner, spaces between the channel portions **13n** and between the channel portions **13p** are filled with the gate insulation films **14** and the gate electrodes **17n** and **17p**. The total heights (thicknesses) of the channel portions **13n** and **13p** are the sum totals of materials constituting the nanowire structures **12n** and **12p** forming the channel portions **13n** and **13p** except for the gate insulation films **14** and the gate electrodes **17n** and **17p**, respectively. This is also applicable to Embodiment 2 described below.

Each of the third gate electrodes **27n** and **27p** of the second field effect transistors **20n** and **20p** surrounds at least a part of the gate insulation layer **24**. In this case, as depicted in the figure, the third gate electrodes **27n** and **27p** may be configured to surround the gate insulation layers **24**, or the third gate electrodes **27n** and **27p** may be configured to surround a part of the gate insulation layers **24** as described below. According to the depicted example, the third gate electrodes **27n** and **27p** are provided between the surface of the base **30** and the gate insulation layers **24**.

Further, according to the semiconductor device of Embodiment 1, the second field effect transistor includes the n-channel type field effect transistor **20n** and the p-channel type field effect transistor **20p**. The channel forming layer **23n** of the n-channel type field effect transistor **20n** is made of silicon (Si), while the channel forming layer **23p** of the p-channel type field effect transistor **20p** is made of silicon-germanium (SiGe). The first field effect transistor includes the n-channel type field effect transistor **10n** and the p-channel type field effect transistor **10p**. The channel portion **13n** of the n-channel type field effect transistor **10n** is made of silicon (Si), while the channel portion **13p** of the p-channel

type field effect transistor **10p** is made of silicon-germanium (SiGe). Such configuration of the first field effect transistor is also applicable to Embodiment 2 described below.

Moreover, examples of the material forming the gate electrodes **17n**, **17p**, **27n**, and **27p** include TiN, TaN, Al, TiAl, and W. Specifically, the gate electrodes **17n**, **17p**, **27n**, and **27p** are made of TiN, for example. Gate insulation films **15** and gate insulation layers **25** constituting a part (lower layers) of the gate insulation films **14** and the gate insulation layers **24** are made of SiO₂, while gate insulation films **16** and gate insulation layers **26** constituting a remaining part (upper layers) of the gate insulation films **14** and the gate insulation layers **24** are made of a high dielectric constant material, specifically HfO₂, for example. The base **30** is constituted by a silicon semiconductor substrate. The base **30** includes an element separation region **70** made of an insulation material **71**. This is also applicable to Embodiment 2 described below.

Both ends of the channel portions **13n** and **13p** are supported by source/drain regions **18n** and **18p** constituting the first field effect transistors **10n** and **10p**, respectively, while both ends of the channel forming layers **23n** and **23p** are supported by source/drain regions **28n** and **28p** constituting the second field effect transistors **20n** and **20p**. This is also applicable to Embodiment 2 described below.

According to the semiconductor device of Embodiment 1, each of the first field effect transistors **10n** and **10p** is a low withstand voltage field effect transistor, while each of the second field effect transistors **20n** and **20p** is a high withstand voltage field effect transistor. Voltage applied to the gate electrodes **17n** and **17p** of the first field effect transistors **10n** and **10p** is in a range from 0.5 to 0.8 volts, while voltage applied to the gate electrodes **27n** and **27p** of the second field effect transistors **20n** and **20p** is in a range from 1.5 to 3 volts. This is also applicable to Embodiment 2 described below.

A manufacturing method of the semiconductor device of Embodiment 1 will hereinafter be described with reference to FIGS. **8A**, **8B**, **8C**, **9A**, **9B**, **10A**, **10B**, **11A**, **11B**, **12**, **13**, **14**, **15**, **16**, **17**, and **18** each of which is a schematic partial cross-sectional diagram similar to the cross-sectional diagram taken along the arrows A-A in FIG. **3A** or the arrows B-B in FIG. **3B**, FIGS. **19**, **21**, **23**, **25**, **27**, **29**, **31**, **33**, **35**, **37**, **39**, and **41** each of which includes a schematic partial cross-sectional diagram of the first field effect transistor similar to the cross-sectional diagram taken along the arrows A-A in FIG. **3A** (indicated by (A) in each of the figures; see arrows A-A in (C) of FIG. **19** as well), a schematic partial cross-sectional diagram of the first field effect transistor similar to the cross-sectional diagram taken along the arrows A'-A' in FIG. **3** (indicated by (B) in each of the figures; see arrows B-B in (C) of FIG. **19** as well), and a schematic partial plan diagram of the first field effect transistor (indicated by (C) in each of the figures), and FIGS. **20**, **22**, **24**, **26**, **28**, **30**, **32**, **34**, **36**, **38**, **40**, and **42** each of which includes a schematic partial cross-sectional diagram of the second field effect transistor similar to the cross-sectional diagram taken along the arrows B-B in FIG. **3B** (indicated by (A) in each of the figures; see arrows A-A in (C) of FIG. **20** as well), a schematic partial cross-sectional diagram of the second field effect transistor similar to the cross-sectional diagram taken along the arrows B'-B' in FIG. **3B** (indicated by (B) in each of the figures; see arrows B-B in (C) of FIG. **20** as well), and a schematic partial plan diagram of the second field effect transistor (indicated by (C) in each of the figures). Note that, in the following description, a region where the first field effect transistor **10n** is to be formed in

the base **30** will be referred to as a first region **31**, a region where the first field effect transistor **10p** is to be formed in the base **30** will be referred to as a second region **32**, a region where the second field effect transistor **20n** is to be formed in the base **30** will be referred to as a third region **33**, and a region where the second field effect transistor **20p** is to be formed in the base **30** will be referred to as a fourth region **34** in some cases. In addition, in the figures, the first region **31**, the second region **32**, the third region **33**, and the fourth region **34** are indicated by reference numbers **[31]**, **[32]**, **[33]**, and **[34]**, respectively. Moreover, hatching lines in a first Si—Ge layer **41**, a first Si layer **42**, a second Si—Ge layer **43**, a second Si layer **44**, a third Si—Ge layer **45**, a third Si layer **46**, and a fourth Si—Ge layer **47** are not depicted in (A) of each of FIGS. **19**, **20**, **21**, **22**, **23**, **24**, **25**, **26**, **27**, **28**, **29**, **30**, **31**, **32**, **33**, **34**, **35**, **36**, **37**, **38**, **39**, **40**, **41**, and **42**. Furthermore, in a case where respective types of processing or the like in a certain region have an undesired effect on another region, it is sufficient if a mask layer or the like is provided in the affected other region as needed. Description of such a mask layer is omitted.

[Step-100A]

First, the first Si—Ge layer **41** is formed by a known method in each of the first region **31**, the second region **32**, the third region **33**, and the fourth region **34** of the base **30** (see FIG. **8A**), and then the first Si—Ge layer **41** on the fourth region **34** is removed by a known method. As a result, a structure depicted in FIG. **8B** is obtained.

[Step-100B]

Next, the first Si layer **42** is formed on the entire surface by a known method (see FIG. **8C**), and then the first Si layer **42** on the third region **33** is removed by a known method. As a result, a structure depicted in FIG. **9A** is obtained.

[Step-100C]

Subsequently, the second Si—Ge layer **43** is formed on the entire surface by a known method (see FIG. **9B**), and then the second Si-layer **43** on the fourth region **34** is removed by a known method. As a result, a structure depicted in FIG. **10A** is obtained.

[Step-100D]

Then, the second Si layer **44** is formed on the entire surface by a known method. As a result, a structure depicted in FIG. **10B** is obtained.

[Step-100E]

Next, the third Si—Ge layer **45** is formed on the entire surface by a known method (see FIG. **11A**), and then the third Si-layer **45** on the third region **33** is removed by a known method. As a result, a structure depicted in FIG. **11B** is obtained.

[Step-100F]

Then, the third Si layer **46** is formed on the entire surface by a known method (see FIG. **12**), and then the third Si layer **46** on the fourth region **34** is removed by a known method. As a result, a structure depicted in FIG. **13** is obtained.

[Step-100G]

Subsequently, the fourth Si—Ge layer **47** is formed on the entire surface by a known method (see FIG. **14**), and then the fourth Si—Ge layer **47** on each of the first region **31**, the second region **32**, and the third region **33** is removed by a known method. As a result, a structure depicted in FIG. **15** is obtained.

In such manner, a laminated structure including the first Si—Ge layer **41**, the first Si layer **42**, the second Si—Ge layer **43**, the second Si layer **44**, the third Si—Ge layer **45**, and the third Si layer **46** is formed on each of the first region **31** and the second region **32**, a laminated structure including the first Si—Ge layer **41**, the second Si—Ge layer **43**, the

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second Si layer 44, and the third Si layer 46 is formed on the third region 33, and a laminated structure including the first Si layer 42, the second Si layer 44, the third Si—Ge layer 45, and the fourth Si—Ge layer 47 is formed on the fourth region 34.

[Step-110]

Thereafter, to form the channel structure portions 11n and 11p, a mask layer 51 made of SiN is formed by a known method on the third Si layer 46 of each of the first region 31 and the second region 32, on the third Si layer 46 of the third region 33, and on the fourth Si—Ge layer 47 of the fourth region 34 (see FIG. 16), the laminated structure in each of the first region 31, the second region 32, the third region 33, and the fourth region 34 is etched using the mask layer 51 as an etching mask, and further, a part of the exposed base 30 in a thickness direction is etched. As a result, a structure depicted in FIG. 17 is obtained. A groove for forming the element separation region 70 having a shallow-trench structure is formed in the base 30. Subsequently, a film of the insulation material 71 made of SiO₂ is formed on the entire surface, and a top surface is smoothed using CMP. Following this, the insulation material 71 is etched and left in the groove to form the element separation region 70 having the shallow-trench structure (see FIG. 18). Note that formation of the element separation region 70 in the first field effect transistors 10n and 10p and formation of the element separation region 70 in the second field effect transistors 20n and 20p may be performed either simultaneously or independently.

[Step-120]

Subsequently, the mask layer 51 is removed (see (A), (B), and (C) in FIGS. 19 and (A), (B), and (C) in FIG. 20), and then thermal oxidation treatment is performed to form an unillustrated dummy oxide layer on a surface of the laminated structure of each of the first region 31, the second region 32, the third region 33, and the fourth region 34. Thereafter, a dummy gate portion covering the laminated structure of the first region 31, a dummy gate portion covering the laminated structure of the second region 32, a dummy gate portion covering the laminated structure of the third region 33, and a dummy gate portion covering the laminated structure of the fourth region 34 are formed by a known method. Each of the dummy gate portions is indicated by a reference number 52. Each of the dummy gate portions 52 is made of polysilicon. Subsequently, the laminated structure on each of the first region 31, the second region 32, the third region 33, and the fourth region 34 each in an exposed state is etched using the dummy gate portion 52 as an etching mask. As a result, a structure depicted in each of (A), (B), and (C) in FIGS. 21 and (A), (B), and (C) in FIG. 22 is obtained. A similar structure is also providable in each of the second region 32 and the fourth region 34.

[Step-130]

Thereafter, a SiN layer is formed on the entire surface. This SiN layer is etched back to form a side wall 53 made of SiN on a side surface of the dummy gate portion 52. However, the side wall 53 is not formed on portions corresponding to both ends of the channel portions 13n and 13p and both ends of the channel forming layers 23n and 23p. As a result, a structure depicted in each of (A), (B), and (C) in FIGS. 23 and (A), (B), and (C) in FIG. 24 is obtained. A similar structure is also providable in each of the second region 32 and the fourth region 34.

[Step-140]

Thereafter, a phosphor-doped SiC layer 54n for forming the source/drain region 18n constituting the first field effect transistor 10n and the source/drain region 28n constituting

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the second field effect transistor 20n is epitaxially grown from the surface of the exposed base 30 by a known method, and then patterned by a known method. As a result, the source/drain region 18n constituting the first field effect transistor 10n and the source/drain region 28n constituting the second field effect transistor 20n are obtained. Similarly, a boron-doped Si—Ge layer 54p for forming the source/drain region 18p constituting the first field effect transistor 10p and the source/drain region 28p constituting the second field effect transistor 20p is epitaxially grown from the surface of the exposed base 30 by a known method, and then patterned by a known method. In this manner, the source/drain region 18p constituting the first field effect transistor 10p and the source/drain region 28p constituting the second field effect transistor 20p are obtained.

As a result, a structure depicted in each of (A), (B), and (C) in FIGS. 25 and (A), (B), and (C) in FIG. 26 is obtained. A similar structure is also providable in each of the second region 32 and the fourth region 34. A part of the source/drain region 18n is in contact with the side surface of the laminated structure in the first region 31, while the remaining part is in contact with the side wall 53. Similarly, a part of the source/drain region 18p is in contact with the side surface of the laminated structure in the second region 32, while the remaining part is in contact with the side wall. A part of the source/drain region 28n is in contact with the side surface of the laminated structure in the third region 33, while the remaining part is in contact with the side wall 53. A part of the source/drain region 28p is in contact with the side surface of the laminated structure in the fourth region 34, while the remaining part is in contact with the side wall.

[Step-150]

Subsequently, the dummy gate portion 52 is removed by a known method [see (A), (B), and (C) of FIGS. 27 and (A), (B), and (C) of FIG. 28], and an unillustrated dummy oxide layer is further removed by a known method. Then, the third Si—Ge layer 45, the second Si—Ge layer 43, and the first Si—Ge layer 41 in the laminated structure of the first region 31 are selectively removed, and the second Si—Ge layer 43 and the first Si—Ge layer 41 in the laminated structure of the third region 33 are selectively removed. Moreover, the third Si layer 46, the second Si layer 44, and the first Si layer 42 in the laminated structure of the second region 32 are selectively removed, and the second Si layer 44 and the first Si layer 42 in the laminated structure of the fourth region 34 are selectively removed. As a result, a structure depicted in each of (A), (B), and (C) in FIGS. 29, (A), (B), and (C) in FIGS. 30, (A), (B), and (C) in FIG. 31, and (A), (B), and (C) in FIG. 32 is obtained.

[Step-160]

Subsequently, outer peripheries of the third Si layer 46, the second Si layer 44, and the first Si layer 42 each exposed in the first region 31, the third Si—Ge layer 45, the second Si—Ge layer 43, and the first Si—Ge layer 41 each exposed in the second region 32, the second Si layer 44 and the first Si layer 42 each exposed in the third region 33, and the second Si—Ge layer 43 and the first Si—Ge layer 41 each exposed in the fourth region 34 are thermally oxidized to form an oxide film. After this thermal oxidation treatment, a cross-sectional shape of each of the nanowire structures 12n and 12p of the first field effect transistors 10n and 10p having the nanowire structure becomes circular. Note that these oxide films are not depicted in the figures. As a result, a structure depicted in each of (A), (B), and (C) in FIGS. 33 and (A), (B), and (C) in FIG. 34 is obtained. A similar structure is also providable in each of the second region 32 and the fourth region 34.

[Step-170A]

Thereafter, a gate insulation film 15_1 (made of SiO_2) constituting a lower layer of the gate insulation film 14 is formed on the formed oxide film by atomic layer deposition (ALD), and also a gate insulation layer 25_1 (made of SiO_2) constituting a lower layer of the gate insulation layer 24 is formed on the formed oxide film. As a result, a structure depicted in each of (A), (B), and (C) in FIGS. 35 and (A), (B), and (C) in FIG. 36 is obtained. A similar structure is also providable in each of the second region 32 and the fourth region 34 . Note that an insulation film similar to the gate insulation film and an insulation layer similar to the gate insulation layer are similarly accumulated on a side surface $53A$ inside the side wall 53 . However, these film and layer are, in principle, not depicted in the figures.

[Step-170B]

Subsequently, the gate insulation film 15_1 and a first insulation lower layer $15'$ of each of the first field effect transistors $10n$ and $10p$ are removed by a known method. Thereafter, again by ALD, a gate insulation film 15_2 (made of SiO_2) constituting a lower layer of the gate insulation film 14 is formed on the formed oxide film, and also a gate insulation layer 25_2 (made of SiO_2) constituting a lower layer of the gate insulation layer 24 is formed on the gate insulation layer 25_1 . The gate insulation film of each of the first field effect transistors $10n$ and $10p$ is constituted by the gate insulation film 15_2 . On the other hand, the gate insulation layer of each of the second field effect transistors $20n$ and $20p$ is constituted by a laminated structure of the gate insulation layer 25_1 and the gate insulation layer 25_2 . In addition, a second insulation lower layer $25'$ (second insulation lower layers $25'1$ and $25'2$) having a thickness larger than a thickness of the first insulation lower layer $15'$ may be formed on the surface of the base 30 . In this case, the first insulation lower layer $15'$ is formed simultaneously with the gate insulation film 15_2 , while the second insulation lower layer $25'$ is formed simultaneously with the gate insulation layer 25_1 and the gate insulation layer 25_2 . As a result, a structure depicted in each of (A), (B), and (C) in FIGS. 37 and (A), (B), and (C) in FIG. 38 is obtained. A similar structure is also providable in each of the second region 32 and the fourth region 34 .

Alternatively, the gate insulation layer 25_2 may be formed on each of the second field effect transistors $20n$ and $20p$ without forming the gate insulation film 15_2 on each of the first field effect transistors $10n$ and $10p$. In this case, the gate insulation film of each of the first field effect transistors $10n$ and $10p$ is constituted by the gate insulation film 15_1 . On the other hand, the gate insulation layer of each of the second field effect transistors $20n$ and $20p$ is constituted by a laminated structure of the gate insulation layer 25_1 and the gate insulation layer 25_2 . In this case as well, the second insulation lower layer $25'$ having a thickness larger than the thickness of the first insulation lower layer $15'$ may be formed on the surface of the base 30 . The first insulation lower layer $15'$ is formed simultaneously with the gate insulation film 15_1 , while the second insulation lower layer $25'$ is formed simultaneously with the gate insulation layer 25_1 and the gate insulation layer 25_2 .

Alternatively, formation of the gate insulation film 15 on each of the first field effect transistors $10n$ and $10p$ and formation of the gate insulation layer 25 on each of the second field effect transistors $20n$ and $20p$ may be performed independently. In this case, the gate insulation film of each of the first field effect transistors $10n$ and $10p$ is constituted by the gate insulation film 15 . On the other hand, the gate insulation layer of each of the second field effect transistors

$20n$ and $20p$ is constituted by the gate insulation layer 25 . In this case as well, the second insulation lower layer $25'$ having a thickness larger than the thickness of the first insulation lower layer $15'$ may be formed on the surface of the base 30 . In this case, the first insulation lower layer $15'$ is formed simultaneously with the gate insulation film 15 , while the second insulation lower layer $25'$ is formed simultaneously with the gate insulation layer 25 .

The gate insulation film 15 constituting each of the first field effect transistors $10n$ and $10p$ is formed by either the first ALD or the second ALD, while the gate insulation layer 25 constituting each of the second field effect transistors $20n$ and $20p$ is formed by the first and second ALD. Accordingly, the gate insulation film 15 and the gate insulation layer 25 each having a desired thickness can be obtained by controlling a thickness of an SiO_2 film or an SiO_2 layer formed by the first ALD and a thickness of an SiO_2 film or an SiO_2 layer formed by the second ALD. Alternatively, the gate insulation film 15 and the gate insulation layer 25 each having a desired thickness may be obtained by forming a single layer of the gate insulation film 15 on each of the first field effect transistors $10n$ and $10p$ and forming a plurality of the gate insulation layers 25 on each of the second field effect transistors $20n$ and $20p$, or by independently performing formation of the gate insulation film 15 on each of the first field effect transistors $10n$ and $10p$ and formation of the gate insulation layer 25 on each of the second field effect transistors $20n$ and $20p$.

[Step-170C]

Subsequently, the gate insulation film 16 and the gate insulation layer 26 constituting upper layers of the gate insulation film 14 and the gate insulation layer 24 are formed by ALD on the gate insulation film 15_2 and the gate insulation layer 25_2 . Each of the gate insulation film 16 and the gate insulation layer 26 is made of HfO_2 . Moreover, a first insulation upper layer $16'$ and a second insulation upper layer $26'$ may be formed on the surface of the base 30 . As a result, a structure depicted in each of (A), (B), and (C) in FIGS. 39 and (A), (B), and (C) in FIG. 40 is obtained. A similar structure is also providable in each of the second region 32 and the fourth region 34 .

By performing the processes described above, the first insulation layer $14'$ (having a configuration similar to the laminated configuration of the gate insulation film 15_2 and the gate insulation film 16) is formed on the surface of the base 30 between the first gate electrode $17n$ in the lowermost layer and the surface of the base 30 , and the second insulation layer $24'$ (having a configuration similar to the laminated configuration of the gate insulation layer 25_1 , the gate insulation layer 25_2 , and the gate insulation layer 26) is formed on the surface of the base 30 between the gate insulation layer 24 and the surface of the base 30 . An insulation layer having a laminated configuration similar to that of the first insulation layer $14'$ and the second insulation layer $24'$ is also formed on the side surface of the side wall 53 . However, this insulation layer is not depicted in the figures other except for FIGS. 41 and 42.

[Step-180]

Thereafter, the gate electrodes $17n$, $17p$, $27n$, and $27p$ made of TiN are provided in regions inside the side wall 53 by a known method. As a result, a structure depicted in each of (A), (B), and (C) in FIGS. 41 and (A), (B), and (C) in FIG. 42 is obtained. A similar structure is also providable in each of the second region 32 and the fourth region 34 . Furthermore, conductive material layers 19 and 29 each made of tungsten (W) are formed on the gate electrodes $17n$, $17p$, $27n$, and $27p$ by a known method. As a result, a structure

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depicted in a schematic partial cross-sectional diagram of FIGS. 1, 2A, and 2B is obtained.

According to the semiconductor device of Embodiment 1, the relation $T_2 \geq (L_1/2)$ is satisfied. Accordingly, such a semiconductor device which includes both the second field effect transistor having a relatively thick gate insulation film with respect to the first field effect transistor and the first field effect transistor having a nanowire structure or the like is providable. In this case, driving of the first field effect transistor at a low voltage and driving of the second field effect transistor at a high voltage are achievable. Moreover, lowering of resistance of the channel forming layer of the second field effect transistor, raising of transconductance g_m , and reduction of parasitic capacitance are achievable. Note that parasitic resistance can be further reduced by enlarging each width of the channel forming layers 23n and 23p of the second field effect transistors 20n and 20p.

According to Modification 1 of Embodiment 1, the gate insulation layer 25 having a large thickness is formed on each of the second field effect transistors 20n and 20p in [Step-170A] described above, and the gate insulation layer 26 is formed between the gate insulation layer 25 and the surface of the base 30 in [Step-170C] described above, i.e., a space between the surface of the base 30 and each of the channel forming layers 23n and 23p is filled with the gate insulation layer 24. As a result, obtained is such a structure where the top surface and the side surface of the gate insulation layer 24 are covered with the gate electrodes (third gate electrodes) 17n and 17p constituting the second field effect transistors 20n and 20p as depicted in a schematic partial cross-sectional diagram in FIG. 4.

Depending on cases, the following structure may be adopted in Modification 2 of Embodiment 1.

A laminated structure including the first Si—Ge layer 41, the first Si layer 42, the second Si—Ge layer 43, the second Si layer 44, the third Si—Ge layer 45, and the third Si layer 46 is formed in each of the first region 31 and the second region 32.

A laminated structure including the first Si—Ge layer 41 (or the second Si—Ge layer 43), the second Si layer 44, and the third Si layer 46 is formed in the third region 33.

A laminated structure including the first Si layer 42 (or the first Si layer 44), the third Si—Ge layer 45, and the fourth Si—Ge layer 47 is formed in the fourth region 34.

Alternatively, depending on cases, the following structure may be adopted in Modification 3 of Embodiment 1.

A laminated structure including the first Si—Ge layer 41, the first Si layer 42, the second Si—Ge layer 43, the second Si layer 44, the third Si—Ge layer 45, and the third Si layer 46 is formed in each of the first region 31 and the second region 32.

A laminated structure including the first Si—Ge layer 41 (or the second Si—Ge layer 43) and the second Si layer 44 (or the third Si layer 46) is formed in the third region 33.

A laminated structure including the first Si layer 42 (or the first Si layer 44) and the third Si—Ge layer 45 (or the fourth Si—Ge layer 47) is formed in the fourth region 34.

In these laminated structures of Modification 2 or Modification 3 of Embodiment 1, the thickness T_{2-CH} of the channel forming layer of the second field effect transistor, the thickness T_2 of the gate insulation layer of the second field effect transistor, or the distance L_2 from the surface of the base to the channel forming layer of the second field effect transistor is different. However, a structure similar to the structure described in Embodiment 1 can be obtained.

In addition, in Modification 4 of Embodiment 1, depending on cases, the channel forming layers 23n and 23p of the

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second field effect transistors 20n and 20p may be made of silicon (Si), for example. In this case, it is sufficient if different materials are used to form the gate electrodes 27n and 27p. Specifically, examples of the material forming the gate electrode 27n include Ti, V, Cr, Zr, Nb, Mo, Hf, Ta, W, and a compound containing these metals, while examples of the material forming the gate electrode 27p include Fe, Co, Ni, Cu, Ru, Rh, Pd, Ag, Os, Ir, Pt, Au, and a compound containing these metals. FIG. 5 is a schematic partial cross-sectional diagram of a modification (Modification 4) of the semiconductor device of Embodiment 1.

Embodiment 2

Embodiment 2 relates to a semiconductor device according to a second aspect of the present disclosure.

FIG. 6 is a schematic partial cross-sectional diagram of the semiconductor device of Embodiment 2. FIG. 6 is a schematic partial cross-sectional diagram similar to the cross-sectional diagram taken along the arrows A-A in FIG. 3A and the arrows B-B in FIG. 3B. FIG. 6 includes a schematic partial cross-sectional diagram of one channel structure portion in an n-channel type first field effect transistor, a schematic partial cross-sectional diagram of one channel structure portion in a p-channel type first field effect transistor, and schematic partial cross-sectional diagrams of an n-channel type second field effect transistor and a p-channel type second field effect transistor. Note that the schematic partial cross-sectional diagram of the n-channel type first field effect transistor and the p-channel type first field effect transistor is similar to the cross-sectional diagrams of FIGS. 1, 2A, and 2B. FIG. 6 is a cross-sectional diagram in which hatching lines are omitted.

The semiconductor device of Embodiment 2 includes the base 30,

the first field effect transistors 10n and 10p where at least the two channel structure portions 11n and the two channel structure portions 11p (three for each in a laminating direction of the channel structure portions 11n and 11p in the depicted example) are laminated, respectively, each of the channel structure portions 11n and each of the channel structure portions 11p including the channel portion 13n and the channel portion 13p, respectively, the channel portion 13n and the channel portion 13p each having a nanowire structure or a nanosheet structure (having the nanowire structure 12n and the nanowire structure 12p, respectively, in the depicted example), the gate insulation film 14 surrounding the channel portion 13n and the gate insulation film 14 surrounding the channel portion 13p, respectively, and the gate electrode 17n and the gate electrode 17p, respectively, the gate electrode 17n and the gate electrode 17p surrounding at least a part of the gate insulation film 14, and

the second field effect transistors 20n and 20p which include the channel forming layers 23n and 23p, the gate insulation layers 24 formed on the top surfaces and the side surfaces of the channel forming layers 23n and 23p, and the gate electrodes 27n and 27p each formed on at least the top surface of the gate insulation layer 24 (formed on the top surface and the side surface of the gate insulation layer 24 in the depicted figure).

The first field effect transistors 10n and 10p and the second field effect transistors 20n and 20p are provided above the base 30.

The channel portions 13n of the first field effect transistors 10n are disposed apart from each other in the laminating

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direction of the channel structure portions **11n**, and the channel portions **13p** of the first field effect transistors **10p** are disposed apart from each other in the laminating direction of the channel structure portions **11p**.

An insulation material layer **64** is formed between the surface of the base **30** and a bottom surface of the channel forming layer **23n** constituting the second field effect transistor **20n** and between the surface of the base **30** and a bottom surface of the channel forming layer **23p** constituting the second field effect transistor **20p**.

In addition, reverse bias is applicable to the base **30** at portions facing the bottom surfaces of the channel forming layers **23n** and **23p** via the insulation material layers **64**. Specifically, an n-type well or a p-type well is formed by ion-implantation, for example, at each of these portions of the base **30** inside the base **30** constituted by a silicon semiconductor substrate, and wiring (not depicted) through which voltage is applicable to the well is connected to the well. By applying bias to the well, reverse bias is applicable to each of the first field effect transistors **10n** and **10p** and the second field effect transistors **20n** and **20p**.

In addition, assuming that a thickness of the channel portion is T_{1-CH} and that a thickness of the insulation material layer is T_{Ins} , the following relation is satisfied.

$$0.2 \leq T_{1-CH}/T_{Ins} \leq 2$$

Specifically, the following value is set.

$$T_{1-CH}/T_{Ins} = 8 \text{ nm}/8 \text{ nm}$$

However, this value is not required to be set.

A manufacturing method of the semiconductor device of Embodiment 2 will be described with reference to FIG. **43** which is a schematic partial cross-sectional diagram similar to the cross-sectional diagram taken along the arrows A-A in FIG. **3A** and the arrows B-B in FIG. **3B**, FIGS. **44**, **45**, **46**, **47**, **48**, **49**, and **50** each of which includes a schematic partial cross-sectional diagram of the second field effect transistor similar to the cross-sectional diagram taken along the arrows B-B in FIG. **3B** (indicated by (A) in each of the figures; see arrows A-A in (C) of FIG. **20** as well), a schematic partial cross-sectional diagram of the second field effect transistor similar to the cross-sectional diagram taken along the arrows B'-B' in FIG. **3B** (indicated by (B) in each of the figures; see arrows B-B in (C) of FIG. **20** as well), and a schematic partial plan diagram (indicated by (C) in each of the figures), together with FIGS. **23**, **25**, **29**, **31**, **33**, **35**, **37**, **39**, and **41** each of which relates to the first field effect transistor described in Embodiment 1. Note that hatching lines in the first Si—Ge layer **41** and the first Si layer **42** are not omitted in (A) of each of FIGS. **44**, **45**, **46**, **47**, **48**, **49**, and **50**. [Step-200A]

First, the first Si—Ge layer **41** is formed by a known method in each of the first region **31**, the second region **32**, the third region **33**, and the fourth region **34** of the base **30**. [Step-200B]

Next, the first Si layer **42** is formed on the entire surface by a known method. [Step-200C]

Subsequently, the second Si—Ge layer **43** is formed on the entire surface by a known method, and then the second

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Si-layer **43** on each of the third region **33** and the fourth region **34** is removed by a known method.

[Step-200D]

Then, the second Si layer **44** is formed on the entire surface by a known method, and then the second Si layer **44** on each of the third region **33** and the fourth region **34** is removed by a known method.

[Step-200E]

Subsequently, the third Si—Ge layer **45** is formed on the entire surface by a known method, and then the third Si—Ge layer **45** on each of the third region **33** and the fourth region **34** is removed by a known method.

[Step-200F]

Subsequently, the third Si layer **46** is formed on the entire surface by a known method, and then the third Si layer **46** on each of the third region **33** and the fourth region **34** is removed by a known method. As a result, a structure depicted in FIG. **43** is obtained.

In this manner, a laminated structure including the first Si—Ge layer **41**, the first Si layer **42**, the second Si—Ge layer **43**, the second Si layer **44**, the third Si—Ge layer **45**, and the third Si layer **46** is formed on each of the first region **31** and the second region **32**, and a laminated structure including the first Si—Ge layer **41** and the first Si layer **42** is formed on the third region **33** and the fourth region **34**. [Step-210]

Thereafter, to form the channel structure portions **11n** and **11p**, a mask layer **51** made of SiN is formed by a known method on the third Si layer **46** of each of the first region **31** and the second region **32** and on the first Si layer **42** of each of the third region **33** and the fourth region **34**, the laminated structure in each of the first region **31**, the second region **32**, the third region **33**, and the fourth region **34** is etched using the mask layer **51** as an etching mask, and further a part of the exposed base **30** in a thickness direction is etched. A groove for forming the element separation region **70** having a shallow-trench structure is formed in the base **30**. Subsequently, a film of the insulation material **71** made of SiO₂ is formed on the entire surface, and a top surface is smoothed by CMP. Thereafter, the insulation material **71** is etched and left in the groove to form the element separation region **70** having the shallow-trench structure. Note that formation of the element separation regions **70** in the first field effect transistors **10n** and **10p** and formation of the element separation regions **70** in the second field effect transistors **20n** and **20p** may be performed either simultaneously or independently.

[Step-220]

Subsequently, the mask layer **51** is removed, and then thermal oxidation treatment is performed to form an unilustrated dummy oxide layer on a surface of the laminated structure on each of the first region **31**, the second region **32**, the third region **33**, and the fourth region **34**. Thereafter, a dummy gate portion covering the laminated structure of the first region **31**, a dummy gate portion covering the laminated structure of the second region **32**, a dummy gate portion covering the laminated structure of the third region **33**, and a dummy gate portion covering the laminated structure of the fourth region **34** are formed by a known method. Each of the dummy gate portions is indicated by a reference number **52**. Each of the dummy gate portions **52** is made of polysilicon. Subsequently, the laminated structure on each of the first region **31**, the second region **32**, the third region **33**, and the fourth region **34** each in an exposed state is etched using the dummy gate portion **52** as an etching mask.

[Step-230]

Thereafter, an SiN layer is formed on the entire surface. This SiN layer is etched back to form a side wall 53 made of SiN on a side surface of the dummy gate portion 52. However, the side wall 53 is not formed on portions corresponding to both ends of the channel portions 13n and 13p and both ends of the channel forming layers 23n and 23p. As a result, a structure depicted in each of (A), (B), and (C) in FIGS. 23 and (A), (B), and (C) in FIG. 44 is obtained.

[Step-240]

Thereafter, a phosphor-doped SiC layer 54n for forming the source/drain region 18n constituting the first field effect transistor 10n and the source/drain region 28n constituting the second field effect transistor 20n is epitaxially grown from the surface of the exposed base 30 by a known method, and then patterned by a known method. As a result, the source/drain region 18n constituting the first field effect transistor 10n and the source/drain region 28n constituting the second field effect transistor 20n are obtained. Similarly, a boron-doped Si—Ge layer 54p for forming the source/drain region 18p constituting the first field effect transistor 10p and the source/drain region 28p constituting the second field effect transistor 20p is epitaxially grown from the surface of the exposed base 30 by a known method, and then patterned by a known method. In this manner, the source/drain region 18p constituting the first field effect transistor 10p and the source/drain region 28p constituting the second field effect transistor 20p are obtained.

As a result, a structure depicted in each of (A), (B), and (C) in FIGS. 25 and (A), (B), and (C) in FIG. 45 is obtained. A similar structure is also providable in each of the second region 32 and the fourth region 34. A part of the source/drain region 18n is in contact with the side surface of the laminated structure in the first region 31, while the remaining part is in contact with the side wall 53. Similarly, a part of the source/drain region 18p is in contact with the side surface of the laminated structure in the second region 32, while the remaining part is in contact with the side wall. A part of the source/drain region 28n is in contact with the side surface of the laminated structure in the third region 33, while the remaining part is in contact with the side wall 53. A part of the source/drain region 28p is in contact with the side surface of the laminated structure in the fourth region 34, while the remaining part is in contact with the side wall.

[Step-250]

Thereafter, the dummy gate portion 52 on each of the third region 33 and the fourth region 34 is removed by a known method (see (A), (B), and (C) of FIG. 46). The dummy gate portion 52 on each of the first region 31 and the second region 32 is left. Subsequently, the unillustrated dummy oxide layer on each of the third region 33 and the fourth region 34 is removed by a known method, and then the first Si—Ge layer 41 of the laminated structure in each of the third region 33 and the fourth region 34 is selectively removed (see (A), (B), and (C) of FIG. 47). Thereafter, the insulation material layer 64 is formed on the entire surface and then etched back. As a result, the insulation material layer 64 is formed between the surface of the base 30 and each of the first Si layers 42 of the laminated structures on the third region 33 and the fourth region 34. The insulation material layer 64 extends from a portion below each of the first Si layers 42 on the third region 33 and the fourth region 34 toward the surface of the base 30. As a result, a structure depicted in each of (A), (B), and (C) in FIG. 48 is obtained.

[Step-260]

Subsequently, the dummy gate portion 52 on each of the first region 31 and the second region 32 is removed by a

known method, and the unillustrated dummy oxide layer is further removed by a known method. Then, the third Si—Ge layer 45, the second Si—Ge layer 43, and the first Si—Ge layer 41 in the laminated structure of the first region 31 are selectively removed (see (A), (B), and (C) of FIG. 29). Moreover, the third Si layer 46, the second Si layer 44, and the first Si layer 42 in the laminated structure of the second region 32 are selectively removed (see (A), (B), and (C) of FIG. 31). In addition, outer peripheries of the third Si layer 46, the second Si layer 44, and the first Si layer 42 exposed in the first region 31, the third Si—Ge layer 45, the second Si—Ge layer 43, and the first Si—Ge layer 41 exposed in the second region 32, the first Si layer 42 exposed in the third region 33, and the first Si layer 42 exposed in the fourth region 34 are thermally oxidized to form an oxide film. After this thermal oxidation treatment, a cross-sectional shape of each of the nanowire structures 12n and 12p of the first field effect transistors 10n and 10p having the nanowire structure becomes circular. Note that these oxide films are not depicted in the figures. As a result, a structure depicted in each of (A), (B), and (C) in FIG. 33 is obtained. A similar structure is also obtained in the second region 32.

[Step-270A]

Thereafter, the gate insulation film 15₁ (made of SiO₂) constituting a lower layer of the gate insulation film 14 is formed on the formed oxide film by atomic layer deposition (ALD), and also the gate insulation layer 25₁ (made of SiO₂) constituting a lower layer of the gate insulation layer 24 is formed on the formed oxide film. As a result, a structure depicted in each of (A), (B), and (C) in FIG. 35 is obtained. A similar structure is also providable in the second region 32.

[Step-270B]

Subsequently, the gate insulation film 15₁ and the first insulation lower layer 15' of each of the first field effect transistors 10n and 10p are removed by a known method. Thereafter, again by using ALD, the gate insulation film 15₂ (made of SiO₂) constituting a lower layer of the gate insulation film 14 is formed on the formed oxide film, and also the gate insulation layer 25₂ (made of SiO₂) constituting a lower layer of the gate insulation layer 24 is formed on the gate insulation layer 25₁. The gate insulation film of each of the first field effect transistors 10n and 10p is constituted by the gate insulation film 15₂. On the other hand, the gate insulation layer of each of the second field effect transistors 20n and 20p is constituted by a laminated structure of the gate insulation layer 25₁ and the gate insulation layer 25₂. The first insulation lower layer 15' is formed simultaneously with the gate insulation film 15₂. As a result, a structure depicted in each of (A), (B), and (C) in FIG. 37 is obtained. A similar structure is also providable in the second region 32.

Alternatively, the gate insulation layer 25₂ may be formed on each of the second field effect transistors 20n and 20p without forming the gate insulation film 15₂ on each of the first field effect transistors 10n and 10p. In this case, the gate insulation film of each of the first field effect transistors 10n and 10p is constituted by the gate insulation film 15₁. On the other hand, the gate insulation layer of each of the second field effect transistors 20n and 20p is constituted by a laminated structure of the gate insulation layer 25₁ and the gate insulation layer 25₂. In this case, the first insulation lower layer 15' is also formed simultaneously with the gate insulation film 15₁.

Alternatively, formation of the gate insulation film 15 on each of the first field effect transistors 10n and 10p and formation of the gate insulation layer 25 on each of the

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second field effect transistors **20n** and **20p** may be performed independently. In this case, the gate insulation film of each of the first field effect transistors **10n** and **10p** is constituted by the gate insulation film **15**. On the other hand, the gate insulation layer of each of the first field effect transistors **20n** and **20p** is constituted by the gate insulation layer **25**. In this case, the first insulation lower layer **15'** is also formed simultaneously with the gate insulation film **15**.

The gate insulation film **15** constituting each of the first field effect transistors **10n** and **10p** is formed by either the first ALD or the second ALD, while the gate insulation layer **25** constituting each of the second field effect transistors **20n** and **20p** is formed by the first and second ALD. Accordingly, the gate insulation film **15** and the gate insulation layer **25** each having a desired thickness can be obtained by controlling a thickness of an SiO₂ film or an SiO₂ layer formed by the first ALD and a thickness of an SiO₂ film or an SiO₂ layer formed by the second ALD. Alternatively, the gate insulation film **15** and the gate insulation layer **25** each having a desired thickness may be obtained by forming a single layer of the gate insulation film **15** on each of the first field effect transistors **10n** and **10p** and forming a plurality of the gate insulation layers **25** on each of the second field effect transistors **20n** and **20p**, or by independently performing formation of the gate insulation film **15** on each of the first field effect transistors **10n** and **10p** and formation of the gate insulation layer **25** on each of the second field effect transistors **20n** and **20p**.

[Step-270C]

Subsequently, the gate insulation film **16** and the gate insulation layer **26** constituting upper layers of the gate insulation film **14** and the gate insulation layer **24** are formed by ALD on the gate insulation film **15₂** and the gate insulation layer **25₂**. Each of the gate insulation film **16** and the gate insulation layer **26** is made of HfO₂. Moreover, the first insulation upper layer **16'** may be formed on the surface of the base **30**. As a result, a structure depicted in each of (A), (B), and (C) in FIGS. **39** and (A), (B), and (C) in FIG. **49** is obtained. A similar structure is also providable in each of the second region **32** and the fourth region **34**.

By performing the processes described above, the first insulation layer **14'** (having a configuration similar to the laminated configuration of the gate insulation film **15₂** and the gate insulation film **16**) is formed on the surface of the base **30** between the first gate electrode **17n** in the lowermost layer and the surface of the base **30**. An insulation layer having a laminated configuration similar to that of the first insulation layer **14'** and the second insulation layer **24'** is also formed on the side surface of the side wall **53**. However, this insulation layer is not depicted in the figures except for FIG. **41**.

[Step-280]

Thereafter, the gate electrodes **17n**, **17p**, **27n**, and **27p** made of TiN are provided in regions inside the side wall **53** by a known method. As a result, a structure depicted in each of (A), (B), and (C) in FIGS. **41** and (A), (B), and (C) in FIG. **50** is obtained. A similar structure is also providable in each of the second region **32** and the fourth region **34**. Furthermore, conductive material layers **19** and **29** each made of tungsten (W) are formed on the gate electrodes **17n**, **17p**, **27n**, and **27p** by a known method. As a result, a structure depicted in a schematic partial cross-sectional diagram of FIG. **6** is obtained.

According to the semiconductor device of Embodiment 2, the gate electrode constituting the second field effect transistor is provided at least on the top surface of each of the channel forming layers. In addition, the insulation material

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layer having a small thickness (e.g., a thickness identical or similar to the thickness of the channel portion) is formed between the surface of the base and the bottom surface of each of the channel forming layers constituting the second field effect transistor. Accordingly, a semiconductor device which includes both the second field effect transistor capable of controlling the threshold voltage V_{th} by application of reverse bias and the first field effect transistor having a nanowire structure or the like is providable.

Depending on cases, at least one semiconductor layer **61** may be formed between the insulation material layer **64** and each of the channel forming layers **23n** and **23p** as depicted in a schematic partial cross-sectional diagram of the second field effect transistors **20n** and **20p** in FIG. **7**. According to the depicted example, two layers of the semiconductor layer **61** are formed. Reverse bias is applicable to the semiconductor layers **61** by connecting the semiconductor layers **61** to wiring layers (not depicted) formed below the source/drain regions **28n** and **28p**. An interlayer insulation layer **65** is formed between the semiconductor layers **61** and each of the channel forming layers **23n** and **23p** and between the semiconductor layers **61**.

Such structure can be manufactured by the following method. Specifically, a laminated structure including the first Si—Ge layer **41**, the first Si layer **42**, the second Si—Ge layer **43**, the second Si layer **44**, the third Si—Ge layer **45**, and the third Si layer **46** is formed on each of the third region **33** and the fourth region **34**, and the third Si—Ge layer **45**, the second Si—Ge layer **43**, and the first Si—Ge layer **41** are selectively removed by a step similar to [Step-250]. In addition, the insulation material layer **64** is formed on the entire surface and then etched back. As a result, the insulation material layer **64** is formed between the surface of the base **30** and each of the first Si layers **42** of the laminated structures in the third region **33** and the fourth region **34**, and the interlayer insulation layer **65** is formed between the semiconductor layers **61**.

Moreover, depending on cases, the semiconductor layer **61** in the third region **33** may have a conductivity type (p-type) opposite to the conductivity type (n-type) of the channel forming layer **23n**, while the semiconductor layer **61** in the fourth region **34** may have a conductivity type (n-type) opposite to the conductivity type (p-type) of the channel forming layer **23p**. It is sufficient if the semiconductor layer **61** thus configured is formed by introducing appropriate impurities into the semiconductor layer **61** by ion implantation, or by collectively implanting ions into the semiconductor layer **61**. Alternatively, formation of the semiconductor layer **61** and execution of ion implantation may be performed repeatedly by the number of layers of the semiconductor layer.

Alternatively, depending on cases, the following structure may be adopted.

A laminated structure including the first Si—Ge layer **41**, the first Si layer **42**, the second Si—Ge layer **43**, the second Si layer **44**, the third Si—Ge layer **45**, and the third Si layer **46** is formed in each of the first region **31** and the second region **32**.

A laminated structure including two or a larger number of an Si—Ge layer or an Si layer may be formed in each of the third region **33** and the fourth region.

While the present disclosure has been described on the basis of the preferred embodiments, the configuration and the structure of the semiconductor device, the material constituting the semiconductor device, and the manufacturing method of the semiconductor device described in the

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embodiments are presented by way of example, and may be modified in appropriate manners.

Moreover, the order of steps in the manufacturing method of the semiconductor device in each of the embodiments may appropriately be changed according to demands. While the channel structure portion exclusively has the nanowire structure in the embodiments described above, the channel structure portion may have a nanosheet structure. The base may be constituted by an SOI substrate instead of the silicon semiconductor substrate. Depending on cases, the element separation region may be formed in an initial step of the manufacturing method of the semiconductor device.

According to the embodiments, the laminated structure includes the first Si—Ge layer **41**, the first Si layer **42**, the second Si—Ge layer **43**, the second Si layer **44**, the third Si—Ge layer **45**, the third Si layer **46**, and the fourth Si—Ge layer **47**. Alternatively, the laminated structure may include the first Si layer **42**, the first Si—Ge layer **41**, the second Si layer **44**, the second Si—Ge layer **43**, the third Si layer **46**, the third Si—Ge layer **45**, and the fourth Si layer **47**. In addition, it is sufficient if the number of the laminated channel structure portions is two or larger.

The SiGe layer may be obtained by processes of forming an upper layer made of SiGe on a lower layer made of Si and performing oxidation treatment to convert the upper layer made of SiGe into an SiO₂ layer and the lower layer made of Si into an SiGe layer.

Note that the present disclosure may also have following configurations.

[A01]<<Semiconductor Device: First Aspect>>

A semiconductor device including:

a base;

a first field effect transistor that includes at least two channel structure portions laminated, the channel structure portions each including a channel portion that has a nanowire structure or a nanosheet structure, a gate insulation film that surrounds the channel portion, and a gate electrode that surrounds at least a part of the gate insulation film; and

a second field effect transistor that includes a channel forming layer, a gate insulation layer surrounding the channel forming layer, and a gate electrode surrounding at least a part of the gate insulation layer, in which the first field effect transistor and the second field effect transistor are provided above the base,

the channel portions of the first field effect transistor are disposed apart from each other in a laminating direction of the channel structure portions, and

assuming that each of a distance between the channel portions of the first field effect transistor is a distance L₁ and that a thickness of the gate insulation layer of the second field effect transistor is a thickness T₂,

$$T_2 \geq (L_1/2)$$

is satisfied.

[A02]

The semiconductor device according to [A01], in which T₂ ≥ 1.1 × (L₁/2), preferably T₂ ≥ 1.2 × (L₁/2), is satisfied.

[A03]

The semiconductor device according to [A01] or [A02], in which, assuming that a distance between a surface of the base and the channel forming layer of the second field effect transistor is a distance L₂,

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$$L_2 \geq L_1,$$

and

$$L_2 \geq T_2$$

are satisfied.

[A04]

The semiconductor device according to [A03], in which L₂ ≥ 2 × L₁ is satisfied.

[A05]

The semiconductor device according to any one of [A01] to [A04], in which, assuming that a thickness of each of the gate insulation films of the first field effect transistor is a thickness T₁,

$$T_2 \geq 2 \times T_1$$

is satisfied.

[A06]

The semiconductor device according to any one of [A01] to [A05], in which, assuming that a thickness of each of the channel portions is T_{1-CH} and that a thickness of the channel forming layer is T_{2-CH},

$$T_{2-CH} \geq 2 \times T_{1-CH}$$

is satisfied.

[A07]

The semiconductor device according to any one of [A01] to [A06], in which at least a part of a channel portion in a lowermost layer constituting the first field effect transistor is surrounded by a first gate electrode, and a channel portion other than the channel portion in the lowermost layer is surrounded by a second gate electrode.

[A08]

The semiconductor device according to any one of [A01] to [A07], in which the second field effect transistor includes an n-channel type field effect transistor and a p-channel type field effect transistor,

a channel forming layer of the n-channel type field effect transistor includes silicon, and

a channel forming layer of the p-channel type field effect transistor includes silicon or silicon-germanium.

[A09]

The semiconductor device according to any one of [A01] to [A08], in which

the first field effect transistor includes an n-channel type field effect transistor and a p-channel type field effect transistor,

a channel portion of the n-channel type field effect transistor includes silicon, and

a channel portion of the p-channel type field effect transistor includes silicon-germanium, germanium, or InGaAs.

[B01]<<Semiconductor Device: Second Aspect>>

A semiconductor device including:

a base;

a first field effect transistor that includes at least two channel structure portions laminated, the channel structure portions each including a channel portion that has a nanowire structure or a nanosheet structure, a gate

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insulation film that surrounds the channel portion, and a gate electrode that surrounds at least a part of the gate insulation film; and

a second field effect transistor that includes a channel forming layer, a gate insulation layer formed on a top surface and a side surface of the channel forming layer, and a gate electrode formed on at least a top surface of the gate insulation layer, in which

the first field effect transistor and the second field effect transistor are provided above the base,

the channel portions of the first field effect transistor are disposed apart from each other in a laminating direction of the channel structure portions, and

an insulation material layer is formed between a surface of the base and a bottom surface of the channel forming layer constituting the second field effect transistor. [B02]

The semiconductor device according to [B01], in which reverse bias is applied to the base at a portion facing the bottom surface of the channel forming layer via the insulation material layer.

[B03]

The semiconductor device according to [B01] or [B02], in which, assuming that a thickness of each of the channel portions is T_{1-CH} and that a thickness of the insulation material layer is T_{Ins} ,

$$0.2 \leq T_{1-CH}/T_{Ins} \leq 2$$

is satisfied.

[B04]

The semiconductor device according to any one of [B01] to [B03], in which at least one semiconductor layer is formed between the channel forming layer and the insulation material layer in the second field effect transistor.

[B05]

The semiconductor device according to [B04], in which an interlayer insulation layer is formed between the channel forming layer and the semiconductor layer and between the semiconductor layers.

[B06]

The semiconductor device according to [B04] or [B05], in which the semiconductor layer has a conductivity type opposite to a conductivity type of the channel forming layer.

REFERENCE SIGNS LIST

10n, 10p First field effect transistor, **11n, 11p** Channel structure portion, **12n, 12p** Nanowire structure, **13n, 13p** Channel portion, **14** Gate insulation film, **14'** First insulation layer, **15, 15₁, 15₂** Part of gate insulation film (lower layer of gate insulation film), **15'** First insulation lower layer, **16** Remaining part of gate insulation film (upper layer of gate insulation film), **16'** First insulation upper layer, **17n, 17p** Gate electrode, **18n 18p** Source/drain region, **19** Conductive material layer, **20n, 20p** Second field effect transistor, **23n, 23p** Channel forming layer, **24** Gate insulation layer, **24'** Second insulation layer, **25, 25₁, 25₂** Part of gate insulation layer (lower layer of gate insulation layer), **25'** Second insulation lower layer, **26** Remaining part of gate insulation layer (upper layer of gate insulation layer), **26'** Second insulation upper layer, **27n, 27p** Gate electrode, **28n, 28p** Source/drain region, **29** Conductive material layer, **30** Base, **31** First region, **32** Second region, **33** Third

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region, **34** Fourth region, **41** First Si—Ge layer, **42** First Si layer, **43** Second Si—Ge layer, **44** Second Si layer, **45** Third Si—Ge layer, **46** Third Si layer, **47** Fourth Si—Ge layer, **51** Mask layer, **52** Dummy gate portion, **53** Side wall, **54n** Phosphor-doped SiC layer, **54p** Boron-doped Si—Ge layer, **61** Semiconductor layer, **64** Insulation material layer, **65** Interlayer insulation layer, **70** Element separation region, **71** Insulation material

What is claimed is:

1. A semiconductor device, comprising:

a base;

a first field effect transistor that includes at least two laminated channel structure portions, each laminated channel structure portion including a channel portion; and

a second field effect transistor that includes a channel forming layer, a gate insulation layer formed on a top surface and a side surface of the channel forming layer, and a gate electrode formed on at least a top surface of the gate insulation layer,

wherein the first field effect transistor and the second field effect transistor are provided above the base,

wherein each channel portion of the first field effect transistor is disposed apart from one another in a laminating direction of the at least two laminated channel structure portions,

wherein an insulation material layer is formed between a surface of the base and a bottom surface of the channel forming layer constituting the second field effect transistor, and

wherein assuming that a thickness of each channel portion of the first field effect transistor is T_{1-CH} and that a thickness of the insulation material layer is T_{Ins} ,

$$0.5 \leq T_{1-CH}/T_{Ins} \leq 1$$

is satisfied.

2. The semiconductor device according to claim **1**, wherein at least one semiconductor layer is formed between the channel forming layer and the insulation material layer in the second field effect transistor.

3. The semiconductor device according to claim **2**, wherein an interlayer insulation layer is formed between the channel forming layer and the at least one semiconductor layer.

4. The semiconductor device according to claim **2**, wherein the at least one semiconductor layer has a conductivity type opposite to a conductivity type of the channel forming layer.

5. The semiconductor device according to claim **1**, wherein each channel portion of the first field effect transistor is at least partially surrounded by a corresponding gate insulation film.

6. The semiconductor device according to claim **1**, wherein each channel portion of the first field effect transistor has a nanowire structure.

7. The semiconductor device according to claim **1**, wherein each channel portion of the first field effect transistor is at least partially surrounded by a corresponding gate insulation film.

8. The semiconductor device according to claim **7**, wherein a gate electrode surrounds at least a part of each corresponding gate insulation film.

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9. A semiconductor device, comprising:
a base;

a first field effect transistor that includes at least two
laminated channel structure portions, each laminated
channel structure portion including a channel portion
that has a nanowire structure or a nanosheet structure,
a gate insulation film that surrounds the channel portion,
a gate electrode that surrounds at least a part
of the gate insulation film; and
a second field effect transistor that includes a channel
forming layer, a gate insulation layer surrounding the
channel forming layer, and a gate electrode surrounding
at least a part of the gate insulation layer,
wherein the first field effect transistor and the second field
effect transistor are provided above the base,
wherein each channel portion of the first field effect
transistor are disposed apart from each other in a
laminating direction of the at least two laminated
channel structure portions, and
wherein assuming that each of a distance between the
channel portions of the first field effect transistor is a
distance L1 and that a distance between a surface of the
base and the channel forming layer of the second field
effect transistor is a distance L2,

$$L2 \geq L1$$

is satisfied.

10. The semiconductor device according to claim 9,
wherein assuming that a thickness of the gate insulation
layer of the second field effect transistor is a thickness T2,

$$T2 \geq (L1/2)$$

is satisfied.

11. The semiconductor device according to claim 10,
wherein $T2 \geq 1.1 \times (L1/2)$ is satisfied.

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12. The semiconductor device according to claim 10,
wherein $L2 \geq T2$ is satisfied.

13. The semiconductor device according to claim 9,
wherein the second field effect transistor includes an
n-channel type field effect transistor and a p-channel
type field effect transistor,

wherein a channel forming layer of the n-channel type
field effect transistor includes silicon, and

wherein a channel forming layer of the p-channel type
field effect transistor includes silicon or silicon-germa-
nium.

14. The semiconductor device according to claim 9,
wherein the first field effect transistor includes an n-chan-
nel type field effect transistor and a p-channel type field
effect transistor,

wherein a channel portion of the n-channel type field
effect transistor includes silicon, and

wherein a channel portion of the p-channel type field
effect transistor includes silicon-germanium, germa-
nium, or InGaAs.

15. A semiconductor device, comprising:
a base;

a first field effect transistor including at least two lami-
nated channel structure portions, each laminated chan-
nel structure portion including a channel portion that
has a nanowire structure or a nanosheet structure, a gate
insulation film that surrounds the channel portion, and
a gate electrode that surrounds at least a part of the gate
insulation film; and

a second field effect transistor including a channel form-
ing layer, a gate insulation layer formed on a top
surface and side surfaces of the channel forming layer,
and a gate electrode formed on at least a top surface of
the gate insulation layer; and

an insulation material layer formed between the base and
a bottom surface of the channel forming layer of the
second field effect transistor, wherein reverse bias is
applied to the base at a portion facing the bottom
surface of the channel forming layer of the second field
effect transistor via the insulation material layer.

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